Introduction to Harmonic Analysis

Lecture notes, expanded

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Abstract

These notes follow the handwritten notebook page by page; the grey rules "notebook page N" mark where page N of the notebook begins. Everything said in the lectures is here, but in addition every notion that was used without definition is defined (in grey *Background* boxes), every "check this" and every exercise is worked out, and remarks are explained.

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1 Fourier analysis in Hilbert spaces

1.1 Inner products and norms

Background 1. (Vector spaces over $\mathbb{C}$)

A *linear space* (vector space) over $\mathbb{C}$ is a set H with an addition $H \times H \to H$ and a scalar multiplication $\mathbb{C} \times H \to H$ such that $(H, +)$ is an abelian group (with neutral element 0) and for all $\alpha, \beta \in \mathbb{C}$, $g, h \in H$:

$$\alpha(g+h) = \alpha g + \alpha h, \quad (\alpha+\beta)g = \alpha g + \beta g, \quad (\alpha\beta)g = \alpha(\beta g), \quad 1 \cdot g = g.$$

In particular $g, h \in H,\ \alpha, \beta \in \mathbb{C} \Rightarrow \alpha g + \beta h \in H$. For $z = a + ib \in \mathbb{C}$ we write $\overline{z} = a - ib$, $\operatorname{Re} z = a = \frac{z+\overline{z}}{2}$, $\operatorname{Im} z = b = \frac{z-\overline{z}}{2i}$ and $|z|^2 = z\overline{z}$.

Definition 1.2 (Inner product)**.** Let H be a linear space over $\mathbb{C}$. An *inner product* on H is a map $(\cdot,\cdot) : H \times H \to \mathbb{C}$ such that

(i) $(x, y) = \overline{(y, x)}$ for all $x, y \in H$ (Hermitian symmetry)

(ii) $(x, x) \geq 0$ for all $x \in H$, and $(x, x) = 0 \iff x = 0$ (positive definiteness)

(iii) $(\alpha x + \beta y, z) = \alpha\,(x, z) + \beta\,(y, z)$ for all $\alpha, \beta \in \mathbb{C}$, $x, y, z \in H$ (linearity in the first slot)

Note that (i) implies $(x, x) = \overline{(x, x)} \in \mathbb{R}$, so the inequality in (ii) makes sense.

Remark 1.3 (Conjugate linearity in the second slot). $(x, \alpha y + \beta z) = \overline{\alpha}\,(x, y) + \overline{\beta}\,(x, z)$. Indeed, by (i), (iii), (i): $(x, \alpha y + \beta z) = \overline{(\alpha y + \beta z, x)} = \overline{\alpha\,(y, x) + \beta\,(z, x)} = \overline{\alpha}\,\overline{(y, x)} + \overline{\beta}\,\overline{(z, x)} = \overline{\alpha}\,(x, y) + \overline{\beta}\,(x, z)$. (Physicists use the opposite convention: linear in the second slot.)

Definition 1.4. For $x \in H$ put $\|x\| := \sqrt{(x, x)}$.

Background 5. (Norm)

A *norm* on a linear space H is a map $\|\cdot\| : H \to [0, \infty)$ with

(N1) $\|x\| = 0 \iff x = 0$;

(N2) $\|\lambda x\| = |\lambda|\ \|x\|$ for $\lambda \in \mathbb{C}$;

(N3) $\|x + y\| \leq \|x\| + \|y\|$ (triangle inequality).

A norm induces a metric $d(x, y) = \|x - y\|$, hence notions of convergence ($x_n \to x$ iff $\|x_n - x\| \to 0$), open and closed sets, closure, etc.

Proposition 1.6. *$\|\cdot\|$ is a norm on H.*

The key step is the following inequality, which we will use constantly.

Theorem 1.7 (Cauchy–Schwarz inequality)**.** *For all $x, y \in H$,*

$$|(x, y)| \leq \|x\|\ \|y\|\,. \tag{1}$$

Equality holds if and only if x and y are linearly dependent.

Proof. If $(x, y) = 0$ there is nothing to prove; so let $(x, y) \neq 0$, in particular $x \neq 0 \neq y$. For every $\alpha \in \mathbb{C}$,

$$\begin{aligned} 0 \leq \|x + \alpha y\|^2 &= (x + \alpha y, x + \alpha y) = (x, x) + (x, \alpha y) + (\alpha y, x) + |\alpha|^2\,(y, y) \\ &= \|x\|^2 + |\alpha|^2\,\|y\|^2 + 2\operatorname{Re}(x, \alpha y)\,, \end{aligned}$$

because $(\alpha y, x) = \overline{(x, \alpha y)}$ and $w + \overline{w} = 2 \operatorname{Re} w$. Choose α with $|\alpha| = 1$ such that $(x, \alpha y) = \overline{\alpha}\,(x, y) = -\,|(x, y)|$, i.e.

$$\overline{\alpha} = -\frac{|(x,y)|}{(x,y)}, \qquad \alpha = -\frac{|(x,y)|}{\overline{(x,y)}} = -\frac{|(x,y)|}{(y,x)}.$$

Then $0 \le \|x\|^2 + \|y\|^2 - 2\,|(x,y)|$, that is

$$2\,|(x,y)| \le \|x\|^2 + \|y\|^2 \qquad \forall x, y \in H. \tag{2}$$

This is weaker than (1) but it is not homogeneous, and we can exploit that: apply (2) to x and ty, $t > 0$:

$$2t\,|(x,y)| \le \|x\|^2 + t^2\,\|y\|^2\,.$$

Choosing $t = \|x\| \,/\, \|y\|$ (so that $t^2\,\|y\|^2 = \|x\|^2$) gives $2t\,|(x,y)| \le 2\,\|x\|^2$, i.e. $|(x,y)| \le \frac{1}{t}\,\|x\|^2 = \frac{\|y\|}{\|x\|}\,\|x\|^2 = \|x\|\,\|y\|$.

Equality. If $y = \lambda x$ (or $x = 0$) equality is clear. Conversely, if $|(x,y)| = \|x\|\,\|y\|$ with $x, y \neq 0$, then for $t = \|x\| \,/\, \|y\|$ and the α above, $\|x + \alpha t y\|^2 = \|x\|^2 + t^2\,\|y\|^2 - 2t\,|(x,y)| = 0$, so $x = -\alpha t y$. □

Proof of Proposition 1.6. (N1) is (ii) of Definition 1.2. (N2): $\|\lambda x\|^2 = (\lambda x, \lambda x) = \lambda\overline{\lambda}\,(x,x) = |\lambda|^2\,\|x\|^2$. (N3): using $\operatorname{Re} w \le |w|$ and (1),

$$\|x+y\|^2 = \|x\|^2 + 2\operatorname{Re}(x,y) + \|y\|^2 \le \|x\|^2 + 2\,\|x\|\,\|y\| + \|y\|^2 = \big(\|x\| + \|y\|\big)^2. \qquad \square$$

Remark 1.8. Two identities that follow by simply expanding $\|x \pm y\|^2$ are worth remembering:

$$\|x+y\|^2 + \|x-y\|^2 = 2\,\|x\|^2 + 2\,\|y\|^2 \qquad \text{(parallelogram law)},$$
$$\|x+y\|^2 = \|x\|^2 + \|y\|^2 \quad \text{if } (x,y) = 0 \qquad \text{(Pythagoras)}.$$

The parallelogram law characterises norms coming from an inner product (Jordan–von Neumann theorem); e.g. $\max_i |x_i|$ on $\mathbb{C}^2$ does not satisfy it.

1.2 Hilbert spaces

Background 9. (Cauchy sequences and completeness)

A sequence (x_n) in a normed space is a *Cauchy sequence* if

$$\forall \varepsilon > 0\ \exists N_\varepsilon\ \forall m, n \ge N_\varepsilon :\ \|x_n - x_m\| < \varepsilon.$$

Every convergent sequence is Cauchy ($\|x_n - x_m\| \le \|x_n - x\| + \|x - x_m\|$). The space is *complete* if conversely every Cauchy sequence converges, i.e. there is x in the space with $\|x_n - x\| \to 0$. A complete normed space is called a *Banach space*. Example: $\mathbb{R}$ and $\mathbb{C}$ are complete, $\mathbb{Q}$ is not ($1,\ 1.4,\ 1.41,\ 1.414, \cdots \to \sqrt{2} \notin \mathbb{Q}$).

Definition 1.10 (Hilbert space)**.** A *Hilbert space* is a linear space H with an inner product $(\cdot,\cdot)$ such that H is complete as a normed space $(H, \|\cdot\|)$, $\|x\| = \sqrt{(x,x)}$: for every Cauchy sequence $(x_n) \subset H$ there exists $x \in H$ with $\|x_n - x\| \xrightarrow{n\to\infty} 0$.

Example 1.11 (An inner product space which is not complete)**.** On $C([0,1])$ consider $(f,g) = \int_0^1 f\overline{g}\,dx$. Let f_n be 0 on $[0, \frac{1}{2}]$, 1 on $[\frac{1}{2} + \frac{1}{n}, 1]$ and linear in between. Then $\|f_n - f_m\|^2 \le \frac{1}{\min(m,n)}$, so (f_n) is Cauchy, but its L^2-limit is the indicator of $(\frac{1}{2}, 1]$, which is not (a.e. equal to) a continuous function. So completeness is a genuine requirement; $L^2([0,1])$ below is the "completion" of this space.

Example 1.12. The following are Hilbert spaces.

(a) $\mathbb{C}^N := \left\{ a = \begin{pmatrix} a_1 \\ \vdots \\ a_N \end{pmatrix} : a_k \in \mathbb{C} \right\}$ with $(a,b)_{\mathbb{C}^N} = \sum_{k=1}^N a_k \overline{b_k}$. Completeness: a Cauchy sequence in $\mathbb{C}^N$ is Cauchy in each coordinate, and $\mathbb{C}$ is complete.

(b) $\ell^2(\mathbb{Z}) := \left\{ \{a_k\}_{k\in\mathbb{Z}} : \sum_{k\in\mathbb{Z}} |a_k|^2 < \infty \right\}$ with $(\{a_k\},\{b_k\})_{\ell^2(\mathbb{Z})} = \sum_{k\in\mathbb{Z}} a_k \overline{b_k}$.

Why this is well defined: $|a_k\overline{b_k}| \le \frac{1}{2}(|a_k|^2 + |b_k|^2)$, so the series converges absolutely; the same inequality shows $\ell^2(\mathbb{Z})$ is closed under addition. *Completeness (sketch):* if $(a^{(n)})$ is Cauchy in ℓ^2, then for each k the numbers $a_k^{(n)}$ form a Cauchy sequence in $\mathbb{C}$, with limit a_k. For $\varepsilon > 0$ and $m,n \ge N_\varepsilon$ and every finite K: $\sum_{|k|\le K} |a_k^{(n)} - a_k^{(m)}|^2 < \varepsilon^2$; let $m \to \infty$, then $K \to \infty$: $\left\| a^{(n)} - a \right\| \le \varepsilon$. Hence $a \in \ell^2$ and $a^{(n)} \to a$.

(c) $L^2(X,\mu)$, see Theorem 1.13 for the notions involved:

$$L^2(X,\mu) := \left\{ f\colon X \to \mathbb{C} \text{ measurable} : \int_X |f|^2 \, d\mu < \infty \right\} \Big/ \sim, \qquad (f,g) = \int_X f\,\overline{g}\,d\mu,$$

where $f \sim g$ if $f = g$ μ-almost everywhere on X. The quotient is needed for (ii) of Definition 1.2: $\int |f|^2 d\mu = 0$ only gives $f = 0$ a.e., not $f \equiv 0$. Completeness of L^2 is a theorem of measure theory (the *Riesz–Fischer theorem* in its original form).

Obviously $\ell^2(\mathbb{Z}) = L^2(\mathbb{Z},\#)$, where $\#$ is the *counting measure* on $\mathbb{Z}$ ($\#(A)$ = number of elements of A; the notebook writes $\delta_{\mathbb{Z}}$ – the "discrete measure" $\sum_{n\in\mathbb{Z}} \delta_n$). Then $\int_{\mathbb{Z}} f\, d\# = \sum_{k\in\mathbb{Z}} f(k)$, and only the empty set is a null set.

(d) $S_2(H)$, the Hilbert–Schmidt operators on a Hilbert space H (Theorem 1.14), with $(A,B) = \operatorname{trace}(AB^*) = \sum_n (Ae_n, Be_n)$. For $H = \mathbb{C}^N$ these are all $N \times N$ matrices and $(A,B) = \sum_{i,j} A_{ij}\overline{B_{ij}}$ – the matrix is just regarded as a vector in $\mathbb{C}^{N^2}$.

(e) The *Fock space* (Bargmann–Fock space)

$$\mathcal{F}_2 = \left\{ f \text{ entire} : \int_{\mathbb{C}} |f(z)|^2 e^{-|z|^2} \, d\lambda_2(z) < \infty \right\}, \qquad (f,g) = \frac{1}{\pi} \int_{\mathbb{C}} f(z)\,\overline{g(z)}\, e^{-|z|^2} \, d\lambda_2(z),$$

where λ_2 is Lebesgue measure on $\mathbb{C} = \mathbb{R}^2$ and a function is *entire* if it is holomorphic (complex differentiable) on all of $\mathbb{C}$. See Example 1.15.

Background 13. (Measure spaces, measurable functions, a.e.)

A *σ-algebra* on a set X is a family $\mathcal{F}$ of subsets of X containing X and closed under complements and countable unions. A *measure* is a map $\mu\colon \mathcal{F} \to [0,\infty]$ with $\mu(\emptyset) = 0$ and $\mu(\bigsqcup_n A_n) = \sum_n \mu(A_n)$ for disjoint $A_n \in \mathcal{F}$. $(X,\mathcal{F},\mu)$ is a *measure space*.
The *Borel σ-algebra* $\mathcal{B}(Y)$ of a metric (topological) space Y is the smallest σ-algebra containing all open sets. On $\mathbb{C}$ we use $\mathcal{B}(\mathbb{C}) = \mathcal{B}(\mathbb{R}^2)$. A measure on $\mathcal{B}(X)$ is a *Borel measure*.
$f\colon X \to \mathbb{C}$ is *measurable* if $f^{-1}(B) \in \mathcal{F}$ for every $B \in \mathcal{B}(\mathbb{C})$. A property holds *$\mu$-almost everywhere* (a.e.) if it fails only on a set $N \in \mathcal{F}$ with $\mu(N) = 0$.

Background 14. (Bounded operators, adjoints, Hilbert–Schmidt operators)

An *operator* $A\colon H \to H$ is a linear map. It is *bounded* if $\|A\| := \sup_{\|x\|\le 1} \|Ax\| < \infty$; $\mathcal{B}(H)$ denotes the bounded operators. For $A \in \mathcal{B}(H)$ the *adjoint* $A^* \in \mathcal{B}(H)$ is defined by $(Ax,y) = (x,A^*y)$ for all x,y (for matrices: the conjugate transpose). If $H = \ell^2$, an operator can be thought of as an infinite matrix $((Ae_j,e_i))_{i,j}$ acting on sequences.

$A \in \mathcal{B}(H)$ is *Hilbert–Schmidt* if $\sum_n \|Ae_n\|^2 < \infty$ for an orthonormal basis $\{e_n\}$ (Definition 1.19). The sum does not depend on the choice of the basis: by Parseval (Theorem 1.26) applied twice, for ONBs $\{e_n\}$, $\{f_m\}$,

$$\sum_n \|Ae_n\|^2 = \sum_n \sum_m |(Ae_n, f_m)|^2 = \sum_m \sum_n |(e_n, A^* f_m)|^2 = \sum_m \|A^* f_m\|^2 ,$$

and the right side does not depend on $\{e_n\}$. We set $\|A\|_{S_2} = \left(\sum_n \|Ae_n\|^2\right)^{1/2}$ and $S_2(H) = \{A \in \mathcal{B}(H) : \sum_n \|Ae_n\|^2 < \infty\}$. The *trace* of an operator is $\operatorname{trace} T = \sum_n (Te_n, e_n)$ (when this makes sense); for $A, B \in S_2(H)$ one has $\operatorname{trace}(B^*A) = \operatorname{trace}(AB^*) = \sum_n (Ae_n, Be_n)$, and this is the inner product of $S_2(H)$.

Example 1.15 (The Fock space)**.** Polynomials, constants, sin, e^{az}, ... belong to $\mathcal{F}_2$. Indeed, in polar coordinates $z = re^{i\theta}$, $d\lambda_2 = r\, dr\, d\theta$,

$$\|z^n\|^2_{\mathcal{F}_2} = \frac{1}{\pi} \int_0^{2\pi} \int_0^\infty r^{2n} e^{-r^2}\, r\, dr\, d\theta = 2 \int_0^\infty r^{2n+1} e^{-r^2} dr \overset{s=r^2}{=} \int_0^\infty s^n e^{-s} ds = n!,$$

and $(z^n, z^m) = 0$ for $n \neq m$ (the θ-integral of $e^{i(n-m)\theta}$ vanishes). So $\{z^n/\sqrt{n!}\}_{n\geq 0}$ is an orthonormal system in $\mathcal{F}_2$ (in fact an orthonormal basis). Since $|\sin z| \leq e^{|z|}$ and $|e^{az}| \leq e^{|a||z|}$, and $\int_{\mathbb{C}} e^{2c|z|-|z|^2} d\lambda_2 < \infty$, also sin and e^{az} belong to $\mathcal{F}_2$. On the other hand $e^{z^2} \notin \mathcal{F}_2$, since $|e^{z^2}|^2 e^{-|z|^2} = e^{|z|^2}$ on the real axis.

1.3 Orthonormal bases

Definition 1.16. Let H be a Hilbert space. A family $\{e_k\}_{k\in I} \subset H$ is an *orthogonal system* if $(e_k, e_j) = 0$ for all $k, j \in I$, $k \neq j$.

Background 17. (Span, closure, density)

$\operatorname{span}\{e_k\}_{k\in I}$ is the set of all *finite* linear combinations $\sum_{k\in F} c_k e_k$, $F \subset I$ finite. The *closure* $\overline{A} = \operatorname{clos}_H A$ of a set $A \subset H$ is the set of all limits of sequences in A; A is *dense* in H if $\overline{A} = H$, i.e. every $h \in H$ can be approximated arbitrarily well by elements of A: $\forall h \in H\ \forall \varepsilon > 0\ \exists a \in A : \|h - a\| < \varepsilon$. $\operatorname{clos}\operatorname{span}\{e_k\} = \overline{\operatorname{span}}\{e_k\}$ is the *closed linear span*: the smallest closed subspace containing all e_k.

Definition 1.18. Let H be a Hilbert space. A family $\{e_k\}_{k\in I}$ is *complete* if $\operatorname{clos}_H \operatorname{span}\{e_k\}_{k\in I} = H$.

Definition 1.19. $\{e_k\}_{k\in I}$ is an *orthonormal basis* (ONB) of H if it is a complete orthogonal system in H with $\|e_k\| = 1$ for all $k \in I$.

Remark 1.20. An ONB is usually *not* a basis in the sense of linear algebra (a Hamel basis): elements of H are in general *infinite* combinations $\sum_k c_k e_k$, converging in norm (Theorem 1.26). E.g. $(1, \frac{1}{2}, \frac{1}{3}, \dots) \in \ell^2(\mathbb{N})$ is not a finite combination of the unit vectors.

Background 21. (Separable spaces)

A metric space X is *separable* if it contains a *countable dense subset*, i.e. a countable $A \subseteq X$ with $\overline{A} = X$. Example: $\mathbb{R}$ is separable since $\mathbb{Q}$ is countable and $\overline{\mathbb{Q}} = \mathbb{R}$; $\mathbb{C}^N$ is separable ($(\mathbb{Q} + i\mathbb{Q})^N$). If A is countable and dense, then for every h there is a sequence $(a_n) \subset A$ with $a_n \to h$ (pick a_n with $\|h - a_n\| < \frac{1}{n}$).

Exercise 1.22. Let H be a separable Hilbert space and $\{e_k\}_{k\in I}$ an ONB in H. Then I is countable.

Solution. For $k \neq j$, by Pythagoras, $\|e_k - e_j\|^2 = \|e_k\|^2 + \|e_j\|^2 = 2$. Hence the open balls $B_k = B(e_k, \frac{\sqrt{2}}{2})$, $k \in I$, are pairwise disjoint (a point in $B_k \cap B_j$ would give $\|e_k - e_j\| < \sqrt{2}$). Let A be a countable dense set. Every ball B_k contains some point $a_k \in A$, and $a_k \neq a_j$ for $k \neq j$ by disjointness. Thus $k \mapsto a_k$ is an injection $I \to A$, so I is countable.

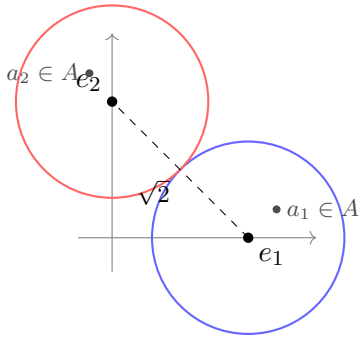


Figure 1: Orthonormal vectors are at distance $\sqrt{2}$, so balls of radius $\sqrt{2}/2$ around them are disjoint; each contains its own point of the dense set A.

Proposition 1.23. *For a Hilbert space $H \neq \{0\}$ the following are equivalent: (a) H is separable; (b) H has a countable ONB; (c) H has a countable complete system.*

Proof. (a)⇒(c): a countable dense set is a complete system. (c)⇒(b): apply the Gram–Schmidt procedure to the complete sequence (x_n): discard every x_n that lies in $\operatorname{span}\{x_1, \dots, x_{n-1}\}$ and put $e_n = \frac{x_n - \sum_{j<n} (x_n, e_j) e_j}{\|\cdots\|}$. The e_n are orthonormal and have the same span as the x_n, hence they are complete. (b)⇒(a): finite combinations of the e_n with coefficients in $\mathbb{Q} + i\mathbb{Q}$ form a countable set whose closure contains $\operatorname{span}\{e_n\}$, hence equals H. □

From now on, all Hilbert spaces will be separable. To construct a non-separable Hilbert space one needs a "wild" measure, for example the counting measure on an uncountable set: $\ell^2(\mathbb{R}) = L^2(\mathbb{R}, \#)$ consists of $f \colon \mathbb{R} \to \mathbb{C}$ vanishing outside a countable set with $\sum_x |f(x)|^2 < \infty$, and $\{\mathbb{1}_{\{x\}}\}_{x \in \mathbb{R}}$ is an uncountable ONB.

1.4 Fourier coefficients and the Riesz–Fischer theorem

Example 1.24 (Observation: the finite-dimensional case)**.** In $\mathbb{C}^N$ the standard basis is $\{e_k\}_{k=1}^N$, $e_k = (0, \dots, 0, \underbrace{1}_{k\text{-th place}}, 0, \dots, 0)^T$, and for all $p \in \mathbb{C}^N$

$$p = p_1 e_1 + \cdots + p_N e_N, \qquad p_k = (p, e_k)_{\mathbb{C}^N}$$

(p_k is the projection coefficient of p onto e_k). This holds for *every* ONB $\{e_k\}$ of $\mathbb{C}^N$: if $p = \sum_j c_j e_j$, then $(p, e_k) = \sum_j c_j (e_j, e_k) = c_k$ by orthonormality. The goal is to prove a similar formula in the infinite-dimensional case.

Definition 1.25. Let H be a Hilbert space, $\{e_k\}_{k=1}^\infty$ an ONB in H and $h \in H$. The number

$$\hat{h}(k) = (h, e_k)$$

is the *k-th Fourier coefficient* of h with respect to $\{e_n\}_{n=1}^\infty$.

Theorem 1.26 (Riesz–Fischer)**.** *Let H be a separable Hilbert space with ONB $\{e_n\}_{n=1}^\infty$. Then for every $h \in H$*

$$h = \sum_{k=1}^{\infty} \hat{h}(k)\, e_k,$$

where the series converges in the norm of H, i.e. $\|h - \sum_{k=1}^{N} \hat{h}(k)e_k\|_H \xrightarrow{N\to\infty} 0$. *Moreover*

$$\|h\|^2 = \sum_{k=1}^{\infty} \left|\hat{h}(k)\right|^2 \qquad \textit{(Parseval's identity).} \tag{3}$$

Remark 1.27. (a) "Converges in the norm" is *not* pointwise convergence: for $H = L^2$ the partial sums need not converge at any given point (this will be the main theme of Section 2).

(b) Names vary in the literature: (3) is usually called *Parseval's identity*; "Riesz–Fischer theorem" often refers to completeness of L^2, or to the converse statement Corollary 1.30.

(c) The statement also holds for non-separable H (with uncountable sums, of which only countably many terms are non-zero), but we will not need this.

We need a simple but important fact first.

Lemma 1.28 (Continuity of the inner product)**.** *If $x_N \to x$ in H, then $(x_N, y) \to (x, y)$ for all $y \in H$, and $\|x_N\| \to \|x\|$.*

Proof. By Cauchy–Schwarz, $|(x_N, y) - (x, y)| = |(x_N - x, y)| \le \|x_N - x\|\,\|y\| \to 0$. By the triangle inequality, $\|x\| \le \|x_N\| + \|x - x_N\|$ and $\|x_N\| \le \|x\| + \|x_N - x\|$, hence $\big|\|x_N\| - \|x\|\big| \le \|x_N - x\| \to 0$. □

Proof of Theorem 1.26. Step 1: Bessel's inequality. We show

$$\sum_{k=1}^{\infty} \left|\hat{h}(k)\right|^2 \le \|h\|^2 . \tag{4}$$

For this, completeness of $\{e_k\}$ is not needed – it holds for every orthonormal system. Set $h_N = \sum_{k=1}^{N} \hat{h}(k)e_k$. By orthonormality,

$$\Big\|\sum_{k=1}^{N} c_k e_k\Big\|^2 = \Big(\sum_{k} c_k e_k, \sum_{j} c_j e_j\Big) = \sum_{k,j} c_k \overline{c_j}\,(e_k, e_j) = \sum_{k=1}^{N} |c_k|^2, \qquad \text{so} \qquad \|h_N\|^2 = \sum_{k=1}^{N} \left|\hat{h}(k)\right|^2 .$$

Further, by Remark 1.3,

$$(h, h_N) = \Big(h, \sum_{k=1}^{N} \hat{h}(k)e_k\Big) = \sum_{k=1}^{N} \overline{\hat{h}(k)}\,(h, e_k) = \sum_{k=1}^{N} \overline{\hat{h}(k)}\hat{h}(k) = \sum_{k=1}^{N} \left|\hat{h}(k)\right|^2 = \|h_N\|^2 .$$

Summarizing, $(h, h_N) = \|h_N\|^2$, so by Cauchy–Schwarz $\|h_N\|^2 \le \|h\|\,\|h_N\|$, i.e. $\|h_N\| \le \|h\|$ (trivial if $h_N = 0$). Thus $\sum_{k=1}^{N} \left|\hat{h}(k)\right|^2 \le \|h\|^2$ for all N, and letting $N \to \infty$ gives (4).

Step 2: (h_N) is a Cauchy sequence. For $N > M$,

$$\|h_N - h_M\|^2 = \Big\|\sum_{k=M+1}^{N} \hat{h}(k)e_k\Big\|^2 = \sum_{k=M+1}^{N} \left|\hat{h}(k)\right|^2 \le \sum_{k=M+1}^{\infty} \left|\hat{h}(k)\right|^2 \xrightarrow{M\to\infty} 0,$$

since the tails of the convergent series (4) tend to 0. As H is complete, there exists

$$\tilde{h} = \lim_{N\to\infty} h_N = \lim_{N\to\infty} \sum_{k=1}^{N} \hat{h}(k)e_k,$$

in other words $\sum_{k=1}^{\infty} \hat{h}(k)e_k$ converges and $\tilde{h} = \sum_{k=1}^{\infty} \hat{h}(k)e_k$.

Step 3: $\tilde{h} = h$.

For fixed k and all $N \geq k$ we have $(h_N, e_k) = \sum_{j=1}^{N} \hat{h}(j)\,(e_j, e_k) = \hat{h}(k)$. Hence, by Lemma 1.28 (this is the step that was "left as an exercise"),

$$\left(\tilde{h}, e_k\right) = (\lim_{N\to\infty} h_N, e_k) = \lim_{N\to\infty} (h_N, e_k) = \hat{h}(k) = (h, e_k)\,,$$

so $\left(h - \tilde{h}, e_k\right) = 0$ for all k: h and $\tilde{h}$ have the same projections onto every e_k. By linearity $\left(h - \tilde{h}, g\right) = 0$ for all $g \in \operatorname{span}\{e_k\}$, and by Lemma 1.28 (in the second slot) for all $g \in \operatorname{clos}\operatorname{span}\{e_k\} = H$ – here completeness of the system is used. Taking $g := h - \tilde{h}$ gives $\left\|h - \tilde{h}\right\|^2 = 0$, i.e. $h = \tilde{h}$.

Step 4: Parseval. Since $h_N \to h$, Lemma 1.28 gives

$$\|h\|^2 = \lim_{N\to\infty} \|h_N\|^2 = \lim_{N\to\infty} \sum_{k=1}^{N} \left|\hat{h}(k)\right|^2 = \sum_{k=1}^{\infty} \left|\hat{h}(k)\right|^2. \qquad \square$$

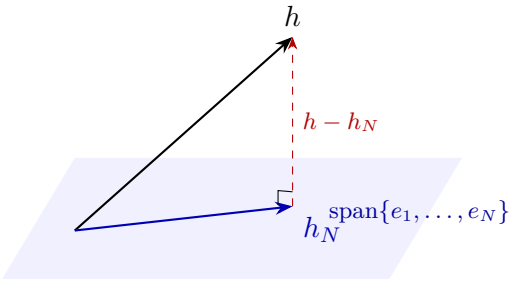


Figure 2: h_N is the orthogonal projection of h onto $\operatorname{span}\{e_1, \ldots, e_N\}$: $h - h_N \perp e_k$ for $k \leq N$. Pythagoras gives $\|h\|^2 = \|h_N\|^2 + \|h - h_N\|^2 \geq \|h_N\|^2$ – another proof of Bessel's inequality.

Corollary 1.29 (Completeness criterion)**.** *An orthonormal sequence $\{e_k\}$ in H is complete if and only if the only $h \in H$ with $(h, e_k) = 0$ for all k is $h = 0$.*

Proof. "$\Rightarrow$" is Step 3 above. "$\Leftarrow$": for any h, Steps 1–2 (which do not use completeness) give $\tilde{h} = \sum \hat{h}(k) e_k \in \operatorname{clos}\operatorname{span}\{e_k\}$, and as in Step 3, $\left(h - \tilde{h}, e_k\right) = 0$ for all k. By assumption $h = \tilde{h} \in \operatorname{clos}\operatorname{span}\{e_k\}$. $\square$

Corollary 1.30. *If $\{e_k\}$ is an ONB of H and $(c_k) \in \ell^2(\mathbb{N})$, then $\sum_k c_k e_k$ converges to some $h \in H$ with $\hat{h}(k) = c_k$. Consequently $h \mapsto (\hat{h}(k))_k$ is a bijective linear isometry $H \to \ell^2(\mathbb{N})$: every separable infinite-dimensional Hilbert space "is" ℓ^2.*

Proof. The partial sums are Cauchy exactly as in Step 2; the limit has coefficients c_k as in Step 3. $\square$

Example 1.31 (Euler's formula)**.** $\displaystyle\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}.$

Proof. Let $H = L^2([-\pi, \pi], \mu = \frac{dx}{2\pi})$ and $e_k(x) = e^{ikx}$, $k \in \mathbb{Z}$. Orthonormality is a direct computation: $(e_k, e_j) = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{i(k-j)x} dx = 1$ if $k = j$ and $= \frac{1}{2\pi} \frac{e^{i(k-j)x}}{i(k-j)}\Big|_{-\pi}^{\pi} = 0$ if $k \neq j$. The non-trivial thing is that $\{e_k\}_{k\in\mathbb{Z}}$ is *complete*: "trigonometric functions are dense in L^2". This is proved in Corollary 2.19 below (and nothing proved before it relies on this example). Since $\mathbb{Z}$ is countable, we may enumerate $\{e_k\}_{k\in\mathbb{Z}}$ as a sequence and apply Theorem 1.26.

Consider $f(x) = x$ on $[-\pi, \pi]$. Then $\|f\|_H^2 = \frac{1}{2\pi}\int_{-\pi}^{\pi} x^2 dx = \frac{1}{2\pi}\cdot\frac{2\pi^3}{3} = \frac{\pi^2}{3}$, $\hat{f}(0) = \frac{1}{2\pi}\int_{-\pi}^{\pi} x\,dx = 0$, and for $k \neq 0$, integrating by parts,

$$\hat{f}(k) = \frac{1}{2\pi}\int_{-\pi}^{\pi} x\,\overline{e^{ikx}}\,dx = \frac{1}{2\pi}\int_{-\pi}^{\pi} xe^{-ikx}dx = \frac{1}{2\pi}\Big[\frac{xe^{-ikx}}{-ik}\Big]_{-\pi}^{\pi} + \frac{1}{2\pi}\underbrace{\int_{-\pi}^{\pi}\frac{e^{-ikx}}{ik}dx}_{=0}$$

$$= \frac{1}{2\pi}\cdot\frac{\pi e^{-ik\pi} + \pi e^{ik\pi}}{-ik} = \frac{(-1)^k}{-ik} = \frac{i(-1)^k}{k}.$$

Hence $\sum_{k\in\mathbb{Z}}\left|\hat{f}(k)\right|^2 = \sum_{k\neq 0}\frac{1}{k^2} = 2\sum_{k=1}^{\infty}\frac{1}{k^2}$, and Parseval's identity $\|f\|_H^2 = \sum_k |\hat{f}(k)|^2$ – "Pythagoras in infinite dimensions" – gives $\frac{\pi^2}{3} = 2\sum_{k\geq 1}\frac{1}{k^2}$. □

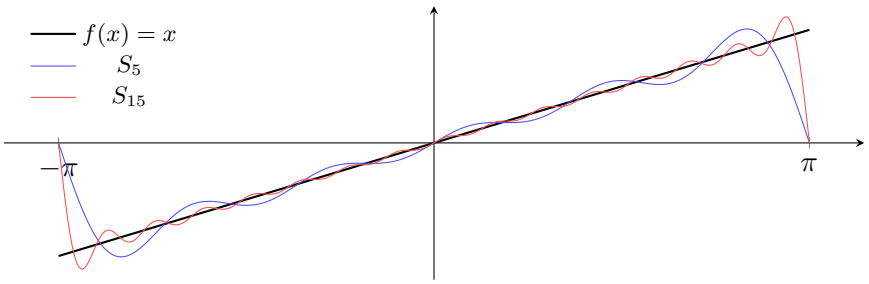


Figure 3: Partial sums $S_N(x) = \sum_{|k|\leq N}\hat{f}(k)e^{ikx} = 2\sum_{k=1}^{N}\frac{(-1)^{k+1}}{k}\sin kx$ of $f(x) = x$. They converge to f in L^2; at $\pm\pi$ (the jump of the periodic extension) they are 0, the average of the one-sided limits (cf. Theorem 2.36).

2 Classical Fourier series

2.1 Setting

Let $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$ be the unit circle and m the Lebesgue (arc-length) measure on $\mathbb{T}$ normalized by $\frac{1}{2\pi}$, so that $m(\mathbb{T}) = 1$. Concretely, $t \mapsto e^{it}$ maps $[-\pi, \pi)$ bijectively onto $\mathbb{T}$ and $m(A) = \frac{1}{2\pi}\lambda_1\{t \in [-\pi, \pi) : e^{it} \in A\}$. We work in

$$H = L^2(\mathbb{T}, m), \qquad (f, g) = \underbrace{\int_{\mathbb{T}} f\,\overline{g}\,dm}_{\text{general measure theory}} = \underbrace{\frac{1}{2\pi}\int_{-\pi}^{\pi} f(e^{it})\,\overline{g(e^{it})}\,dt}_{\text{parametrization}}.$$

Functions on $\mathbb{T}$ are the same as 2π-periodic functions on $\mathbb{R}$, via $g(t) = f(e^{it})$.

Exercise 2.1. $\{z^n\}_{n\in\mathbb{Z}}$ is an ONB in $L^2(\mathbb{T}, m)$.

Solution. Orthonormality: $\left(z^n, z^k\right) = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{i(n-k)t}dt = \delta_{nk}$, as in Example 1.31. *Completeness* needs two facts.

(1) Trigonometric polynomials $\sum_{|k|\leq N} c_k z^k$ are dense in $C(\mathbb{T})$ with the sup-norm. This follows from the Stone–Weierstrass theorem (Theorem 2.2), since they form an algebra containing 1, closed under conjugation ($\overline{z} = z^{-1}$ on $\mathbb{T}$) and separating points (already z does). We will also get a self-contained proof from Fejér's theorem (Corollary 2.18).

(2) $C(\mathbb{T})$ is dense in $L^2(\mathbb{T}, m)$ (Theorem 2.3).

Given $f \in L^2$ and $\varepsilon > 0$, choose $g \in C(\mathbb{T})$ with $\|f - g\|_{L^2} < \varepsilon$ and a trigonometric polynomial p with $\max_{\mathbb{T}}|g - p| < \varepsilon$. Since $m(\mathbb{T}) = 1$, $\|g - p\|_{L^2} \leq \max_{\mathbb{T}}|g - p|$, so $\|f - p\|_{L^2} < 2\varepsilon$. Hence $\operatorname{clos}\operatorname{span}\{z^n\} = L^2(\mathbb{T}, m)$.

Background 2. (Weierstrass and Stone–Weierstrass theorems)

Weierstrass approximation theorem. Every continuous $f\colon [a,b] \to \mathbb{C}$ is a uniform limit of polynomials. *Trigonometric version:* every $f \in C(\mathbb{T})$ is a uniform limit of trigonometric polynomials.
Stone–Weierstrass theorem. Let K be a compact metric space and $\mathcal{A} \subset C(K)$ a subalgebra (closed under sums, products and scalar multiples) which contains the constants, separates points ($\forall x \neq y\ \exists a \in \mathcal{A} : a(x) \neq a(y)$) and is closed under complex conjugation. Then $\mathcal{A}$ is dense in $C(K)$ with respect to $\|f\|_\infty = \max_K |f|$.

Background 3. (Continuous functions are dense in L^p)

If μ is a finite Borel measure on a compact metric space K and $1 \le p < \infty$, then $C(K)$ is dense in $L^p(K,\mu)$. (Idea: simple functions are dense; by regularity of μ an indicator $\mathbb{1}_A$ is approximated by $\mathbb{1}_F$ with $F \subset A$ closed, and $\mathbb{1}_F$ by the continuous functions $\max(0, 1 - n\operatorname{dist}(x,F))$.)

Definition 2.4. For $f \in L^2(\mathbb{T}, m)$ (or even $f \in L^1(\mathbb{T}, m)$) the Fourier coefficients are

$$\hat f(k) = \left(f, z^k\right) = \int_{\mathbb{T}} f(z)\,\overline{z}^k\,dm(z) = \int_{\mathbb{T}} f(z)\,z^{-k}\,dm(z) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(e^{it})e^{-ikt}dt, \qquad k \in \mathbb{Z},$$

using $\overline{z} = z^{-1}$ on $\mathbb{T}$. The *Fourier series* of f is $\sum_{k\in\mathbb{Z}} \hat f(k) z^k$ and its symmetric partial sums are $S_N(f,z) = \sum_{k=-N}^{N} \hat f(k) z^k$.

By the Riesz–Fischer theorem, for all $f \in L^2(\mathbb{T}, m)$ we have $f = \sum_{k\in\mathbb{Z}} \hat f(k) z^k$ where the series converges in $L^2(\mathbb{T}, m)$, i.e.

$$\int_{\mathbb{T}} \Big| f - \sum_{k=-N}^{N} \hat f(k) z^k \Big|^2 dm \xrightarrow{N\to\infty} 0. \tag{$**$}$$

(Here we use that $\mathbb{Z}$ can be enumerated as $0, 1, -1, 2, -2, \dots$, so the symmetric partial sums are a subsequence of the partial sums of Theorem 1.26.)

But ($**$) is not what one would naively expect: we are interested in *pointwise* convergence. In other words, we would like to have $f(z) = \sum_{k\in\mathbb{Z}} \hat f(k) z^k$ a.e. on $\mathbb{T}$ for "good" f. This is true, but very hard to prove:

Theorem 2.5 (Carleson 1966, Hunt 1968)**.** *If $f \in L^p(\mathbb{T}, m)$ for some $p \in (1, \infty]$, then $S_N(f,z) \xrightarrow{N\to\infty} f(z)$ for m-almost every $z \in \mathbb{T}$.*

For $p = 1$ this fails: Kolmogorov (1923) constructed $f \in L^1(\mathbb{T})$ whose Fourier series diverges almost everywhere (even everywhere). We will prove much simpler results, starting with convergence in a weaker sense.

Definition 2.6 (Cesàro convergence)**.** A sequence $(S_n)_{n\ge 0}$ *converges in the Cesàro sense* to L if its arithmetic means converge: $\sigma_N := \frac{1}{N+1}\sum_{n=0}^{N} S_n \xrightarrow{N\to\infty} L$. A series converges in the Cesàro sense if the sequence of its partial sums does.

Example 2.7. $S_n = -1, 1, -1, 1, \dots$ (the partial sums of $-1 + 2 - 2 + 2 - \dots$) does not converge, but $\sigma_N \to 0$: the means are 0 or $-\frac{1}{N+1}$.

Proposition 2.8 (Cesàro summation is consistent)**.** *If $S_n \to L$, then also $\sigma_N \to L$.*

Proof. Let $\varepsilon > 0$ and $|S_n - L| < \varepsilon$ for $n > n_0$. Then $|\sigma_N - L| \le \frac{1}{N+1}\sum_{n=0}^{n_0} |S_n - L| + \frac{N-n_0}{N+1}\varepsilon$, and the first term tends to 0 as $N \to \infty$. So $\limsup |\sigma_N - L| \le \varepsilon$ for every ε. □

Theorem 2.9 (Fejér). *Let $f \in C(\mathbb{T})$ (a continuous function on the unit circle). Then the Fourier series $\sum_{k=-\infty}^{\infty} \hat{f}(k) z^k$ converges to f in the Cesàro sense everywhere on $\mathbb{T}$. In other words, if $S_N(z) = \sum_{k=-N}^{N} \hat{f}(k) z^k$, then*

$$\frac{1}{N+1} \sum_{n=0}^{N} S_n(z) \longrightarrow f(z) \qquad \text{for every } z \in \mathbb{T}.$$

The proof is given in Section 2.4; first we write the Cesàro means as an integral.

2.2 The Fejér kernel

Lemma 2.10. $\dfrac{1}{N+1} \sum_{n=0}^{N} S_n(z) = \displaystyle\int_{\mathbb{T}} f(\xi)\, F_N(\xi, z)\, dm(\xi)$, *where*

$$F_N(\xi, z) := \frac{|1 + \overline{\xi} z + \cdots + (\overline{\xi} z)^N|^2}{N+1} = \frac{1}{N+1} \Big| \sum_{s=0}^{N} (\overline{\xi} z)^s \Big|^2$$

is the Fejér kernel.

Proof. Since all sums are finite, we may interchange them with the integral:

$$\begin{aligned} \frac{1}{N+1} \sum_{n=0}^{N} S_n(z) &= \frac{1}{N+1} \sum_{n=0}^{N} \sum_{k=-n}^{n} \hat{f}(k) z^k = \frac{1}{N+1} \sum_{n=0}^{N} \sum_{k=-n}^{n} \int_{\mathbb{T}} f(\xi) \overline{\xi}^k \, dm(\xi)\, z^k \\ &= \int_{\mathbb{T}} f(\xi) \Big(\frac{1}{N+1} \sum_{n=0}^{N} \sum_{k=-n}^{n} \overline{\xi}^k z^k \Big) dm(\xi). \end{aligned}$$

So the lemma reduces to the identity, for $w = \overline{\xi} z \in \mathbb{T}$,

$$\sum_{n=0}^{N} \sum_{k=-n}^{n} w^k = \Big| \sum_{s=0}^{N} w^s \Big|^2. \tag{5}$$

Direct proof. Using $\overline{w} = w^{-1}$,

$$\Big| \sum_{s=0}^{N} w^s \Big|^2 = \Big(\sum_{s=0}^{N} w^s \Big) \Big(\sum_{r=0}^{N} w^{-r} \Big) = \sum_{k=-N}^{N} \#\{(s,r) : s - r = k\}\, w^k = \sum_{k=-N}^{N} (N + 1 - |k|)\, w^k,$$

and on the left of (5) the power w^k appears once for every n with $|k| \le n \le N$, i.e. $N + 1 - |k|$ times.

Proof by induction (as in the lecture). For $N = 0$ both sides equal 1. Write $P_N = \sum_{s=0}^{N} w^s$ and suppose (5) holds for N. The left side for $N+1$ is $\sum_{k=-(N+1)}^{N+1} w^k + |P_N|^2$. Using $|a+b|^2 = |a|^2 + |b|^2 + 2\,\mathrm{Re}(a\overline{b})$ with $a = P_N$, $b = w^{N+1}$:

$$|P_{N+1}|^2 = |P_N|^2 + \underbrace{|w^{N+1}|^2}_{=1} + 2\,\mathrm{Re}\big(P_N \overline{w}^{N+1}\big), \qquad P_N \overline{w}^{N+1} = \sum_{s=0}^{N} w^{s-N-1} = \overline{w} + \overline{w}^2 + \cdots + \overline{w}^{N+1},$$

so $|P_{N+1}|^2 = |P_N|^2 + 1 + \sum_{j=1}^{N+1} (w^j + \overline{w}^j) = |P_N|^2 + \sum_{k=-(N+1)}^{N+1} w^k$, as needed. □

Remark 2.11 (Closed form; Dirichlet kernel). Write $z = e^{i\theta}$, $\xi = e^{it}$, $u = \theta - t$, so $w = \overline{\xi} z = e^{iu}$. Summing the geometric series and using $|1 - e^{ia}| = 2|\sin \frac{a}{2}|$,

$$F_N(\xi, z) = K_N(u) := \frac{1}{N+1} \frac{|1 - e^{i(N+1)u}|^2}{|1 - e^{iu}|^2} = \frac{1}{N+1} \left(\frac{\sin \frac{(N+1)u}{2}}{\sin \frac{u}{2}} \right)^2.$$

Similarly $S_N(f, e^{i\theta}) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(e^{it}) D_N(\theta - t)\, dt$ with the *Dirichlet kernel*

$$D_N(u) = \sum_{k=-N}^{N} e^{iku} = \frac{\sin\big((N+\frac{1}{2})u\big)}{\sin\frac{u}{2}}, \qquad \text{and} \qquad K_N = \frac{1}{N+1}\sum_{n=0}^{N} D_n.$$

The decisive difference: $K_N \geq 0$ while D_N changes sign, and $\frac{1}{2\pi}\int |D_N| \to \infty$ (Section 2.5). Both kernels are even functions of u.

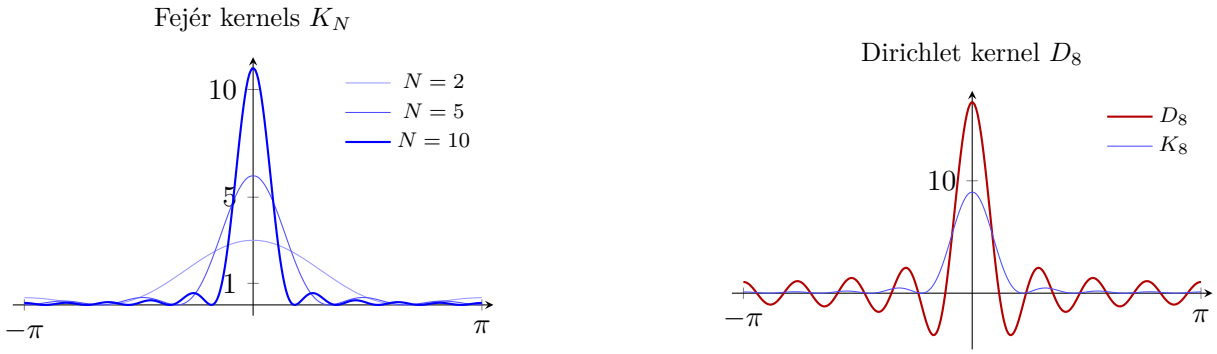


Figure 4: Left: $K_N \geq 0$, $\frac{1}{2\pi}\int K_N = 1$, and the mass concentrates at 0 – an approximate identity. Right: the Dirichlet kernel oscillates and takes negative values; its absolute integral grows like $\log N$.

2.3 Approximate identities

Definition 2.12 (Approximate identity)**.** Let X be a metric space with metric ρ, μ a Borel measure on X and $x_0 \in X$. An *approximate identity with center at* x_0 is a family of measurable functions p_n on X such that

(i) $p_n(x) \geq 0$ for all $x \in X$;

(ii) $\int_X p_n(x)\, d\mu(x) = 1$ for all n;

(iii) $\int_{X\setminus B(x_0,\delta)} p_n(x)\, d\mu(x) \xrightarrow{n\to\infty} 0$ for every $\delta > 0$, where $B(x_0,\delta) = \{y \in X : \rho(y, x_0) < \delta\}$.

Remark 2.13. (a) In the lecture (iii) was first stated as $\sup_{x\in X\setminus B(x_0,\delta)} p_n(x) \to 0$; the integral version above (suggested in the lecture as "simpler") is weaker when $\mu(X) < \infty$, since $\int_{X\setminus B} p_n\, d\mu \leq \mu(X)\sup_{X\setminus B} p_n$, and it is exactly what the proof below needs. For the Fejér kernel both hold.

(b) Whether the ball is open or closed does not matter.

(c) Intuition: the mass 1 of $p_n\, d\mu$ concentrates at x_0. In the language of distributions, $p_n \to \delta_{x_0}$ (Dirac delta): $\int f\, p_n\, d\mu \to \int f\, d\delta_{x_0} = f(x_0)$ – that is the next lemma.

Lemma 2.14. *Let (p_n) be an approximate identity with center at x_0 and let f be a bounded measurable function on X which is continuous at x_0 (e.g. f bounded and continuous). Then*

$$\int_X f(x)\, p_n(x)\, d\mu(x) \xrightarrow{n\to\infty} f(x_0).$$

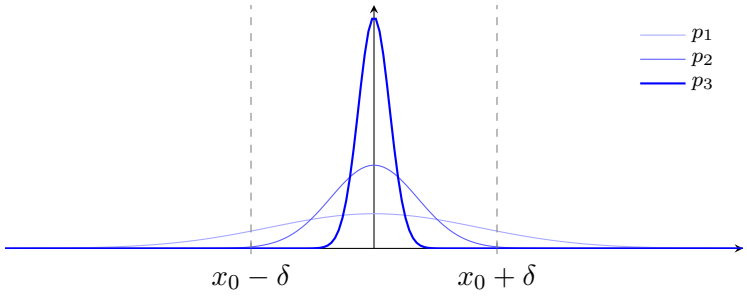


Figure 5: An approximate identity: non-negative, total mass 1, and the mass outside $B(x_0, \delta)$ tends to 0.

Proof. By (ii), $f(x_0) = \int_X f(x_0) p_n \, d\mu$, so using (i),

$$\begin{aligned}\Big|\int_X f\, p_n \, d\mu - f(x_0)\Big| &= \Big|\int_X (f(x) - f(x_0)) p_n(x) \, d\mu(x)\Big| \\ &\le \int_{B(x_0,\delta)} |f(x) - f(x_0)|\, p_n(x)\, d\mu + \int_{X\setminus B(x_0,\delta)} |f(x) - f(x_0)|\, p_n(x)\, d\mu \\ &\le \sup_{x\in B(x_0,\delta)} |f(x) - f(x_0)| \cdot \underbrace{\int_{B(x_0,\delta)} p_n \, d\mu}_{\le 1 \text{ by (i),(ii)}} + 2 \sup_{y\in X} |f(y)| \cdot \int_{X\setminus B(x_0,\delta)} p_n \, d\mu.\end{aligned}$$

Let $\varepsilon > 0$. By continuity at x_0 choose $\delta > 0$ with $|f(x) - f(x_0)| < \frac{\varepsilon}{2}$ on $B(x_0, \delta)$; then, by (iii), choose N such that the second summand is $< \frac{\varepsilon}{2}$ for $n \ge N$. □

We will see that $F_N(\xi, z)$ is an approximate identity (in ξ) at the point z.

Remark 2.15 (Dense subsets, once more). "A subset $D = \{x_1, x_2, x_3, \dots\}$ is dense in a Hilbert space H if for every $h \in H$ and every $\varepsilon > 0$ there exists some $x_n \in D$ such that $\|h - x_n\|_H < \varepsilon$." This is the definition from Theorem 1.17 specialised to countable sets; such H is separable (Theorem 1.21).

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2.4 Proof of Fejér's theorem

Recap. $H = L^2(\mathbb{T}, m)$, $\mathbb{T} = \{|z| = 1\}$, ONB $\{z^k\}_{k\in\mathbb{Z}}$,

$$S_n(f, z) := \sum_{k=-n}^{n} \hat{f}(k) z^k, \qquad \sigma_N(f, z) := \frac{1}{N+1} \sum_{n=0}^{N} S_n(f, z).$$

Theorem 2.16 (Fejér). *For every $f \in C(\mathbb{T})$ we have $\sigma_N(f, z) \xrightarrow{N\to\infty} f(z)$ for every $z \in \mathbb{T}$. Equivalently, the Fourier series $\sum_{k=-\infty}^{\infty} \hat{f}(k) z^k$ converges to $f(z)$ in the Cesàro sense at every point $z \in \mathbb{T}$.*

Proof. By Lemma 2.10, $\sigma_N(f, z) = \int_{\mathbb{T}} f(\xi) F_N(\xi, z)\, dm(\xi)$ with $F_N(\xi, z) = \frac{1}{N+1}\big|\sum_{s=0}^{N} (\bar{\xi} z)^s\big|^2$. Fix z. We check that $(F_N(\cdot, z))_N$ is an approximate identity in the variable ξ with center at z, on $X = \mathbb{T}$ with $\rho(\xi, z) = |\xi - z|$ and $\mu = m$:

(i) $F_N(\xi, z) \ge 0$ for all $\xi \in \mathbb{T}$. ✓

(ii) $\int_{\mathbb{T}} F_N(\xi, z)\, dm(\xi) = 1$. Indeed, $\xi \mapsto 1 + \bar{\xi} z + \dots + \bar{\xi}^N z^N = \sum_{s=0}^{N} z^s\, \xi^{-s}$ is a finite combination of the orthonormal functions ξ^{-s}, so by Parseval (or just orthonormality)

$$\int_{\mathbb{T}} \big|1 + \bar{\xi} z + \dots + \bar{\xi}^N z^N\big|^2 dm(\xi) = 1 + |z|^2 + |z^2|^2 + \dots + |z^N|^2 \overset{|z|=1}{=} N + 1. ✓$$

(iii) For $\xi \in \mathbb{T} \setminus B(z,\delta)$ we have $|1-\overline{\xi}z| = |\overline{\xi}|\,|\xi - z| = |\xi - z| \geq \delta$, and $|1-\overline{\xi}^{N+1}z^{N+1}| \leq 2$. Summing the geometric series,

$$\int_{\mathbb{T}\setminus B(z,\delta)} F_N(\xi,z)\,dm(\xi) \leq \underbrace{m(\mathbb{T}\setminus B(z,\delta))}_{\leq m(\mathbb{T})=1} \cdot \sup_{\xi\in\mathbb{T}\setminus B(z,\delta)} \frac{1}{N+1}\Big|\frac{1-\overline{\xi}^{N+1}z^{N+1}}{1-\overline{\xi}z}\Big|^2$$
$$\leq \frac{2^2}{\delta^2(N+1)} \xrightarrow[\delta \text{ fixed}]{N\to\infty} 0.$$

So $F_N(\cdot,z)$ is an approximate identity with center at z, and f is bounded and continuous on the compact set $\mathbb{T}$. By Lemma 2.14, $\sigma_N(f,z) \to f(z)$, for every z. □

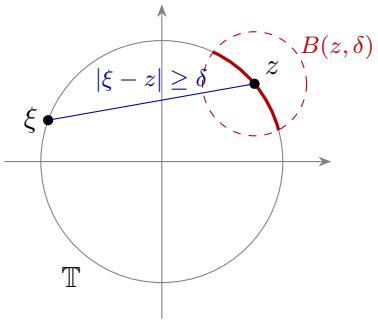


Figure 6: For ξ outside the arc $B(z,\delta)\cap\mathbb{T}$ the denominator $|1-\overline{\xi}z| = |\xi - z|$ stays $\geq \delta$, so $F_N(\xi,z) \leq \frac{4}{\delta^2(N+1)}$ there.

Remark 2.17 (Uniform convergence). In fact $\sigma_N(f,\cdot) \rightrightarrows f$ uniformly on $\mathbb{T}$, i.e. $\max_{z\in\mathbb{T}} |\sigma_N(f,z) - f(z)| \to 0$, $N\to\infty$, because all estimates were uniform in z. *Check:* f is uniformly continuous on the compact set $\mathbb{T}$, so for $\varepsilon > 0$ there is δ *not depending on* z with $|f(\xi) - f(z)| < \frac{\varepsilon}{2}$ whenever $|\xi - z| < \delta$. The proof of Lemma 2.14 then gives

$$|\sigma_N(f,z) - f(z)| \leq \frac{\varepsilon}{2} + 2\,\|f\|_\infty \cdot \frac{4}{\delta^2(N+1)} \qquad \text{for all } z \in \mathbb{T},$$

and the right side is $< \varepsilon$ for N large, independently of z.

Corollary 2.18 (Weierstrass' trigonometric approximation theorem)**.** *Trigonometric polynomials are dense in* $(C(\mathbb{T}), \|\cdot\|_\infty)$.

Proof. $\sigma_N(f,\cdot) = \sum_{|k|\leq N}(1 - \frac{|k|}{N+1})\hat{f}(k)z^k$ is a trigonometric polynomial and $\sigma_N(f,\cdot) \to f$ uniformly by Remark 2.17. □

Corollary 2.19. $\{z^n\}_{n\in\mathbb{Z}}$ *is an ONB of* $L^2(\mathbb{T},m)$*;*
equivalently $\{e^{ikx}\}_{k\in\mathbb{Z}}$ *is an ONB of* $L^2([-\pi,\pi],\frac{dx}{2\pi})$*. In particular Example 1.31 is justified.*

Proof. As in the solution of Exercise 2.1, with Corollary 2.18 in place of Stone–Weierstrass. □

Corollary 2.20. *If* $f \in C(\mathbb{T})$ *and* $S_N(f,z)$ *converges at some point* z*, then the limit is* $f(z)$ *(by Proposition 2.8 and Fejér's theorem).*

Remark 2.21 (Continuity at one point suffices). If f is bounded and measurable on $\mathbb{T}$ (not necessarily in $C(\mathbb{T})$) and $f(\xi) \to f(z_0)$ as $\xi \to z_0$, then $\sigma_N(f,z_0) \to f(z_0)$. *Check:* Lemma 2.14 only uses boundedness of f and continuity at the center $x_0 = z_0$.

Remark 2.22 (Jump discontinuities – homework). Let f be bounded and measurable, and let $z_0 = e^{it_0}$ be such that the one-sided limits

$$f(z_0 - 0) := \lim_{\varepsilon \to 0,\ \varepsilon > 0} f(e^{i(t_0 - \varepsilon)}), \qquad f(z_0 + 0) := \lim_{\varepsilon \to 0,\ \varepsilon > 0} f(e^{i(t_0 + \varepsilon)})$$

exist (f need not be continuous at z_0). Then

$$\sigma_N(f, z_0) \longrightarrow \frac{f(z_0 - 0) + f(z_0 + 0)}{2} \qquad (N \to \infty).$$

This generalizes Fejér's theorem (if f is continuous at z_0 both limits equal $f(z_0)$).

Solution. By Remark 2.11, $\sigma_N(f, z_0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(e^{i(t_0+u)}) K_N(u)\, du$ (substitute $t = t_0 + u$ and use that K_N is even and 2π-periodic). Since K_N is even and $\frac{1}{2\pi} \int_{-\pi}^{\pi} K_N = 1$, we have $\frac{1}{2\pi} \int_0^{\pi} K_N = \frac{1}{2\pi} \int_{-\pi}^{0} K_N = \frac{1}{2}$. Hence, with $L_{\pm} = f(z_0 \pm 0)$,

$$\sigma_N(f, z_0) - \frac{L_- + L_+}{2} = \frac{1}{2\pi} \int_0^{\pi} (f(e^{i(t_0+u)}) - L_+) K_N(u)\, du + \frac{1}{2\pi} \int_0^{\pi} (f(e^{i(t_0-u)}) - L_-) K_N(u)\, du.$$

Each term tends to 0 by the argument of Lemma 2.14: given ε, choose δ such that $|f(e^{i(t_0+u)}) - L_+| < \varepsilon$ for $0 < u < \delta$; the part $\int_0^{\delta}$ is $\le \frac{\varepsilon}{2}$, and the part $\int_{\delta}^{\pi}$ is $\le 2\,\|f\|_\infty \cdot \frac{1}{2\pi} \int_{\delta \le |u| \le \pi} K_N \to 0$, because $K_N(u) \le \frac{1}{(N+1)\sin^2(\delta/2)}$ there.

2.5 Divergence: the du Bois-Reymond theorem

Theorem 2.23 (du Bois-Reymond)**.** *There exists a function $f \in C(\mathbb{T})$ such that $S_N(f, z)$ does not converge at some point $z \in \mathbb{T}$.*

So for continuous functions the Cesàro means in Fejér's theorem are really necessary. The proof is based on a classical result of functional analysis.

Background 24. (Banach spaces, bounded operators and functionals)

A *Banach space* is a complete normed linear space (Theorem 1.9). Example: $C(\mathbb{T})$ with $\|f\| = \max_{z \in \mathbb{T}} |f(z)|$ – a uniform limit of continuous functions is continuous, so Cauchy sequences converge in $C(\mathbb{T})$. A linear map $T\colon X \to Y$ between normed spaces is *bounded* if $\|T\| := \sup_{\|x\| \le 1} \|Tx\| < \infty$; for linear maps, bounded $\iff$ continuous. A bounded linear map $X \to \mathbb{C}$ is a *bounded linear functional*.

Theorem 2.25 (Banach–Steinhaus, uniform boundedness principle)**.** *Let X be a Banach space, Y a normed linear space and $\{T_\alpha\}_{\alpha \in A}$ a family of bounded linear operators $X \to Y$ such that $\sup_{\alpha \in A} \|T_\alpha x\| < \infty$ for every $x \in X$. Then*

$$\sup_{\alpha \in A} \|T_\alpha\| = \sup_{\alpha \in A} \sup_{x \in X,\ \|x\| \le 1} \|T_\alpha x\| < \infty.$$

The proof was skipped in the lecture as a basic fact from functional analysis; here is the standard argument.

Background 26. (Baire category theorem)

If a complete metric space is a countable union of closed sets, then at least one of them contains an open ball.

Proof of Theorem 2.25 (sketch). The sets $E_n = \{x \in X : \sup_\alpha \|T_\alpha x\| \le n\}$ are closed (intersections of closed sets, by continuity of each T_α) and $\bigcup_n E_n = X$ by assumption. By Baire, some E_n contains a ball $B(x_0, r)$. For $\|x\| \le 1$, both x_0 and $x_0 + rx$ lie in E_n, so $\|T_\alpha x\| = \frac{1}{r} \|T_\alpha(x_0 + rx) - T_\alpha x_0\| \le \frac{2n}{r}$ for all α. □

Proof of Theorem 2.23. Let $X := C(\mathbb{T})$ with $\|f\| = \max_{z\in\mathbb{T}} |f(z)|$ (a Banach space), $Y := \mathbb{C}$ and

$$\varphi_N(f) := S_N(f,1) = \sum_{k=-N}^{N} \hat{f}(k), \qquad f \in C(\mathbb{T}).$$

φ_N is linear, $\varphi_N(\alpha f + \beta g) = \alpha\varphi_N(f) + \beta\varphi_N(g)$, because $\hat{f}(k) = \left(f, z^k\right)_{L^2}$ is linear in f. It is bounded: $|\hat{f}(k)| = \left|\int_{\mathbb{T}} f\,\overline{z}^k\,dm\right| \le \max_{\mathbb{T}} |f| \cdot m(\mathbb{T}) = \|f\|$, hence $\|\varphi_N\| \le 2N+1$.

Suppose $S_N(f,1)$ converged (to a finite limit) for every $f \in C(\mathbb{T})$. Then $\sup_N |\varphi_N(f)| < \infty$ for every f, and Banach–Steinhaus gives $\sup_{N\ge 0} \|\varphi_N\| < \infty$. We show that this is not the case, which proves that there is $f \in C(\mathbb{T})$ with $\sup_N |S_N(f,1)| = \infty$; in particular $S_N(f,1)$ diverges. (Rotating f moves the bad point 1 to any prescribed $z \in \mathbb{T}$.)

Computing $\|\varphi_N\|$. Since $\sum_{k=-N}^{N} \overline{\xi}^k = D_N(-t) = D_N(t)$ for $\xi = e^{it}$ (Remark 2.11),

$$\varphi_N(f) = \sum_{k=-N}^{N} \int_{\mathbb{T}} f(\xi)\overline{\xi}^k\,dm(\xi) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(e^{it}) D_N(t)\,dt.$$

Note D_N is real. In the lecture one takes f_N with $|f_N| = 1$ and $f_N \cdot D_N = |D_N|$, i.e. $f_N = \operatorname{sgn} D_N$; this function is not continuous (it jumps at the zeros of D_N), so we approximate it: $g_j = \frac{D_N}{|D_N| + 1/j}$ is continuous, $|g_j| \le 1$, and $g_j \to \operatorname{sgn} D_N$ wherever $D_N \neq 0$, i.e. except at finitely many points. By dominated convergence,

$$\|\varphi_N\| \ge \lim_{j\to\infty} |\varphi_N(g_j)| = \frac{1}{2\pi}\int_{-\pi}^{\pi} |D_N(t)|\,dt =: L_N \qquad \text{(the } \textit{Lebesgue constants}\text{)}.$$

So it suffices to prove $L_N \to \infty$. Multiplying by $|e^{iNt}| = 1$ and summing the geometric series,

$$\begin{aligned} L_N &= \frac{1}{2\pi}\int_{-\pi}^{\pi} |e^{-iNt} + \cdots + e^{iNt}|dt = \frac{1}{2\pi}\int_{-\pi}^{\pi} |1 + e^{it} + \cdots + e^{2iNt}|dt = \frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{|1 - e^{i(2N+1)t}|}{|1 - e^{it}|}dt \\ &\ge \frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{|\operatorname{Im}(1 - e^{i(2N+1)t})|}{|1 - e^{it}|}dt = \frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{|\sin((2N+1)t)|}{|1 - e^{it}|}dt \ge \frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{|\sin((2N+1)t)|}{|t|}dt, \end{aligned}$$

using $|a| \ge |\operatorname{Im} a|$, $\operatorname{Im}(e^{ix}) = \sin x$, and $|1 - e^{it}| \le |t|$ (a chord is shorter than the arc, Fig. 7). By evenness, the substitution $u = (2N+1)t$, and $|\sin u| \ge \sin^2 u = \frac{1-\cos 2u}{2}$,

$$\begin{aligned} L_N &\ge \frac{1}{\pi}\int_0^{(2N+1)\pi} \frac{|\sin u|}{u}\,du \ge \frac{1}{2\pi}\int_1^{(2N+1)\pi} \frac{1 - \cos(2u)}{u}\,du \\ &= \frac{1}{2\pi}\Big(\log((2N+1)\pi) - \int_1^{(2N+1)\pi} \frac{\cos(2u)}{u}\,du\Big). \end{aligned}$$

The last integral stays bounded: integrating by parts, $\int_1^R \frac{\cos 2u}{u}\,du = \left[\frac{\sin 2u}{2u}\right]_1^R + \int_1^R \frac{\sin 2u}{2u^2}du$, which is at most $\frac{1}{2} + \frac{1}{2} + \frac{1}{2}$ in absolute value (it converges to $\int_1^\infty \frac{\cos 2u}{u}\,du$). Hence $L_N \ge \frac{1}{2\pi}(\log((2N+1)\pi) - \frac{3}{2}) \to \infty$. □

2.6 Functions of bounded variation

Definition 2.27. Let $f\colon \mathbb{T} \to \mathbb{C}$. We say that f has *bounded variation*, $f \in BV(\mathbb{T})$, if

$$V_{\mathbb{T}}(f) := \sup\Big\{\sum_{j=1}^{n} |f(z_{j+1}) - f(z_j)|\Big\} < \infty,$$

the supremum taken over all finite partitions $z_1, \ldots, z_n$ of $\mathbb{T}$ (points ordered counter-clockwise, $z_{n+1} := z_1$).

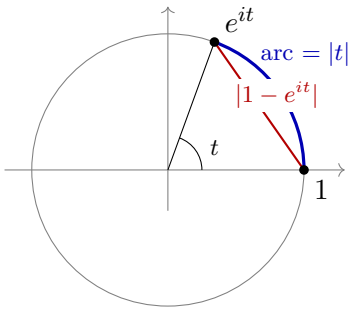


Figure 7: $|1 - e^{it}| = 2|\sin\frac{t}{2}| \le |t|$: the chord is shorter than the arc.

Definition 2.28. Let $g\colon [a,b] \to \mathbb{C}$. Then $g \in BV([a,b])$ if

$$V_a^b(g) := \sup_{a=t_1<t_2<\cdots<t_n=b} \sum_{k=1}^{n-1} |g(t_{k+1}) - g(t_k)| < \infty.$$

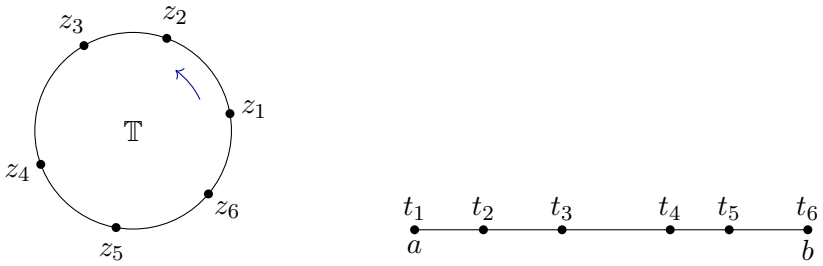


Figure 8: Partitions of $\mathbb{T}$ and of $[a,b]$ used in the definition of the total variation.

Remark 2.29. Let $g(t) = f(e^{it})$. Then $g \in BV([0,2\pi]) \iff f \in BV(\mathbb{T})$, and in fact $V_{\mathbb{T}}(f) = V_0^{2\pi}(g)$. Indeed, adding a point to a partition can only increase the sum (triangle inequality), so in both suprema we may assume the point $1 = e^{i0}$ is included; and partitions of $\mathbb{T}$ containing 1 correspond exactly to partitions $0 = t_1 < \cdots < t_{n+1} = 2\pi$ of $[0,2\pi]$, with the same sums (as $g(2\pi) = g(0)$). Also, BV functions are bounded: $|g(t)| \le |g(a)| + V_a^b(g)$.

Theorem 2.30 (Jordan decomposition)**.** $g \in BV([a,b]) \iff$ *there are bounded nondecreasing functions* $g_1, g_2, g_3, g_4\colon [a,b] \to \mathbb{R}$ *with*

$$g = g_1 - g_2 + i(g_3 - g_4).$$

Proof. "$\Leftarrow$": a nondecreasing h has $\sum_k |h(t_{k+1}) - h(t_k)| = h(b) - h(a)$ for every partition, and sums of BV functions are BV. "$\Rightarrow$": it suffices to treat real g (apply it to $\operatorname{Re} g$, $\operatorname{Im} g$, whose variation is $\le V(g)$). Let $V(x) = V_a^x(g)$. Then V is nondecreasing, and for $x < y$, $g(y) - g(x) \le V_x^y(g) = V(y) - V(x)$, so $V - g$ is nondecreasing as well; $g = V - (V - g)$. □

Background 31. (Complex measures and Lebesgue–Stieltjes measures)

A *complex measure* on $\mathcal{B}(\mathbb{R})$ is a map $\mu\colon \mathcal{B}(\mathbb{R}) \to \mathbb{C}$ with $\mu(\bigsqcup A_n) = \sum \mu(A_n)$ (absolutely convergent) for disjoint A_n. Its *total variation* is the positive measure $|\mu|(A) = \sup \sum_j |\mu(A_j)|$ (sup over finite partitions of A into Borel sets); it is always finite: $|\mu|(\mathbb{R}) < \infty$, and $\left|\int h\, d\mu\right| \le \int |h|\, d|\mu|$.
Lebesgue–Stieltjes: for every nondecreasing, right-continuous, bounded F on $[a,b]$ there is a unique finite Borel measure μ_F on $(a,b]$ with $\mu_F((a,x]) = F(x) - F(a)$.

Theorem 2.32. *($g \in BV([a,b])$ and g is continuous from the right)* $\iff$ $g(x) = g(a) + \int_{(a,x]} d\mu(s)$ *for all x, where μ is a complex valued measure on $(a,b]$ (of finite total variation).*

Idea. "⇒": write $g = g_1 - g_2 + i(g_3 - g_4)$ as in Theorem 2.30; one can choose the g_j right-continuous when g is, and then $\mu = \mu_{g_1} - \mu_{g_2} + i(\mu_{g_3} - \mu_{g_4})$. "⇐": $\sum_k |g(t_{k+1}) - g(t_k)| = \sum_k |\mu((t_k, t_{k+1}])| \le |\mu|((a,b])$, and right-continuity follows from continuity of μ along decreasing sets $(a, x + \frac{1}{n}] \downarrow (a,x]$. □

Example 2.33 (The Cantor staircase)**.** Let $C = \{\sum_{n\ge1} a_n 3^{-n} : a_n \in \{0,2\}\} \subset [0,1]$ be the Cantor set (remove middle thirds repeatedly); $\lambda(C) = 0$. The *Cantor function* $F(\sum a_n 3^{-n}) = \sum \frac{a_n}{2} 2^{-n}$ on C, extended to be constant on each removed interval, is continuous, nondecreasing, with $F(0) = 0$, $F(1) = 1$. So $F \in BV([0,1])$ and $F(x) = \mu((0,x])$ for the *Cantor measure* μ. Note $F' = 0$ a.e., yet F is not constant: μ has no atoms but lives on the null set C (it is *singular continuous*). Hence μ in Theorem 2.32 cannot in general be replaced by $g'(s)\,ds$.

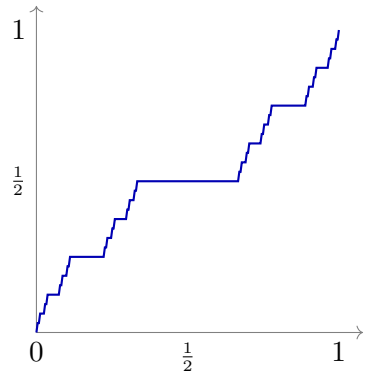


Figure 9: The Cantor staircase (5th stage of the construction): a continuous BV function whose derivative is 0 almost everywhere.

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Example 2.34. If $g \in C^1([a,b])$, then $g \in BV([a,b])$:

$$\sum_k |g(t_{k+1}) - g(t_k)| \le c \sum_k |t_{k+1} - t_k| = c\,(b-a) < \infty, \qquad c = \max_{[a,b]} |g'|,$$

by the inequality $|g(x) - g(y)| = \left|\int_y^x g'\right| \le \max_{[a,b]} |g'| \cdot |x - y|$ (for complex g use this integral form, the mean value theorem itself fails for complex valued functions). Here μ in Theorem 2.32 is $d\mu = g'(s)\,ds$: the derivative of such a function "is" a finite measure.

Remark 2.35. If $g \in BV([a,b])$, then for all $x_0 \in [a,b]$ the one-sided limits

$$g(x_0 - 0) = \lim_{x \to x_0,\, x < x_0} g(x), \qquad g(x_0 + 0) = \lim_{x \to x_0,\, x > x_0} g(x)$$

exist. Moreover g has at most countably many discontinuities.

Proof. By Theorem 2.30, $g = g_1 - g_2 + i(g_3 - g_4)$, and for bounded monotone functions $g_k(x_0 \pm 0)$ exist (e.g. $g_k(x_0 - 0) = \sup_{x<x_0} g_k(x)$ for nondecreasing g_k). A nondecreasing h is discontinuous at x iff its jump $h(x+0) - h(x-0) > 0$; since any finitely many jumps sum to at most $h(b) - h(a)$, there are at most $n(h(b) - h(a))$ jumps $> \frac{1}{n}$, so countably many jumps altogether. The discontinuities of g lie in the union of those of $g_1, \dots, g_4$. □

2.7 Hardy's theorem (Dirichlet–Jordan test)

Theorem 2.36 (Hardy)**.** *Let $f \in BV(\mathbb{T})$. Then for every $z \in \mathbb{T}$*

$$\sum_{k=-\infty}^{\infty} \hat{f}(k) z^k := \lim_{N\to\infty} S_N(f,z) = \frac{f(z-0)+f(z+0)}{2}.$$

In particular, if $f \in BV(\mathbb{T})$ is continuous at $z \in \mathbb{T}$, then $\sum_{k=-\infty}^{\infty} \hat{f}(k) z^k = f(z)$.

(This is classically known as the Dirichlet–Jordan theorem; Hardy's contribution is the Tauberian argument below.) The strategy: by Fejér the series is Cesàro summable; if in addition the terms are $O(\frac{1}{k})$, Cesàro convergence upgrades to ordinary convergence. Results of the form "summable in a weak sense + a size condition on the terms ⇒ convergent" are called *Tauberian theorems.*

Lemma 2.37 (Hardy's Tauberian theorem)**.** *Assume that $\sum_{n=1}^{\infty} a_n$ converges in the Cesàro sense and $|a_k| \le \frac{C}{k}$ for all $k \ge 1$ and some $C > 0$. Then $\sum_{k=1}^{\infty} a_k$ converges in the usual sense, to the same sum.*

Proof. Notation:

$$S_n := \sum_{k=1}^{n} a_k, \qquad \sigma_n := \frac{1}{n}\sum_{k=1}^{n} S_k \to \sigma, \qquad \sigma_{n,j} := \frac{S_{n+1}+\cdots+S_{n+j}}{j}, \qquad l_n(\varepsilon) = [\varepsilon n]$$

for $\varepsilon \in (0,1]$, $n \ge 1$, where $[\,\cdot\,]$ is the integer part. We only consider $n \ge 1/\varepsilon$, so that $j = l_n(\varepsilon) \ge 1$; note $\frac{\varepsilon n}{2} \le j \le \varepsilon n$. The proof has two steps:

- Step 1: $\lim_{n\to\infty} |\sigma_n - \sigma_{n,l_n(\varepsilon)}| = 0$;
- Step 2: $\limsup_{n\to\infty} |S_n - \sigma_{n,l_n(\varepsilon)}| \le C\varepsilon$ for every ε.

Step 1 (uses only Cesàro convergence). Since $S_1 + \cdots + S_m = m\sigma_m$,

$$\sigma_{n,j} = \frac{(S_1+\cdots+S_{n+j}) - (S_1+\cdots+S_n)}{j} = \sigma_{n+j}\frac{n+j}{j} - \sigma_n \frac{n}{j} = (\sigma_{n+j}-\sigma_n)\frac{n+j}{j} + \sigma_n.$$

Thus, with $j = l_n(\varepsilon)$,

$$|\sigma_{n,j} - \sigma_n| = |\sigma_{n+j}-\sigma_n| \cdot \frac{n+j}{j} \le \underbrace{|\sigma_{n+l_n(\varepsilon)} - \sigma_n|}_{\to 0} \cdot \underbrace{\frac{n+\varepsilon n}{\varepsilon n/2}}_{=2\frac{1+\varepsilon}{\varepsilon},\ \text{indep. of } n} \longrightarrow 0,$$

because (σ_m) converges, so $\sigma_{n+j} - \sigma_n \to 0$ for any $j = j(n) \ge 0$.

Step 2 (uses only the size condition). Since $S_n - S_{n+i} = -(a_{n+1} + \cdots + a_{n+i})$,

$$\begin{aligned}\Big|S_n - \frac{S_{n+1}+\cdots+S_{n+j}}{j}\Big| &= \frac{1}{j}\Big|\sum_{i=1}^{j}(S_n - S_{n+i})\Big| \le \frac{1}{j}\sum_{i=1}^{j}(|a_{n+1}| + \cdots + |a_{n+i}|) \\ &\le |a_{n+1}| + \cdots + |a_{n+j}| \le C\Big(\frac{1}{n+1} + \cdots + \frac{1}{n+j}\Big) \\ &\le \frac{Cj}{n+1} \le C\,\frac{\varepsilon n}{n+1} \le C\varepsilon.\end{aligned}$$

Taking the lim sup we obtain Step 2.

Conclusion. $S_n \to \sigma$ iff $\lim |S_n - \sigma| = 0$. By the triangle inequality,

$$\limsup_n |S_n - \sigma| \le \underbrace{\limsup_n |S_n - \sigma_{n,l_n(\varepsilon)}|}_{\le C\varepsilon \text{ by Step 2}} + \underbrace{\limsup_n |\sigma_{n,l_n(\varepsilon)} - \sigma_n|}_{=0 \text{ by Step 1}} + \underbrace{\limsup_n |\sigma_n - \sigma|}_{=0 \text{ by Cesàro convergence}},$$

so $\limsup_n |S_n - \sigma| \le C\varepsilon$ for every $\varepsilon > 0$, i.e. $\lim |S_n - \sigma| = 0$. □

Remark 2.38 (Indexing). Fejér's means are $\frac{1}{N+1}\sum_{n=0}^{N} S_n$ for a series $\sum_{n\ge 0} a_n$, while Lemma 2.37 uses $\frac{1}{n}\sum_{k=1}^{n} S_k$ for $\sum_{n\ge 1} a_n$. This is only a shift of index: putting $b_n = a_{n-1}$ $(n \ge 1)$, the partial sums and means of $\sum b_n$ are S_{n-1} and $\frac{1}{n}\sum_{k=0}^{n-1} S_k$, and $|a_n| \le \frac{C}{n+1}$ for $n \ge 0$ becomes $|b_n| \le \frac{C}{n}$.

Lemma 2.39. *If $f \in BV(\mathbb{T})$, then $|\hat f(k)| \le \frac{C}{|k|+1}$ for all $k \in \mathbb{Z}$, with C depending on f only.*

Proof (lecture version, assuming f continuous from the right). Let $g(t) = f(e^{it})$ for $t \in [0, 2\pi]$; then $g \in BV([0,2\pi])$ (remark after the definition of BV). Assume g is continuous from the right (an extra condition). Then, by Theorem 2.32, $g(t) = g(0) + \int_{(0,t]} d\mu(s)$ for a finite complex measure μ. For $k \ne 0$,

$$\begin{aligned}\hat f(k) &= \int_{\mathbb{T}} f(\xi)\bar\xi^{k}\, dm(\xi) = \frac{1}{2\pi}\int_0^{2\pi} g(t)e^{-ikt}dt = \frac{1}{2\pi}\int_0^{2\pi}\Big(g(0) + \int_{(0,t]} d\mu(s)\Big)e^{-ikt}dt \\ &= \underbrace{\frac{g(0)}{2\pi}\int_0^{2\pi} e^{-ikt}dt}_{=0 \text{ for } k\ne 0} + \frac{1}{2\pi}\int_0^{2\pi}\int_{(0,2\pi]} \chi_{(0,t]}(s)\, e^{-ikt}\, d\mu(s)\, dt \\ &\overset{\text{Fubini}}{=} \frac{1}{2\pi}\int_{(0,2\pi]}\Big(\int_s^{2\pi} e^{-ikt}dt\Big)d\mu(s),\end{aligned}$$

where Fubini applies since the integrand is bounded and $|\mu|$, dt are finite. Now $\int_s^{2\pi} e^{-ikt}dt = \frac{e^{-ikt}}{-ik}\Big|_s^{2\pi} = \frac{e^{-iks}-1}{ik}$ has modulus $\le \frac{2}{|k|}$, so

$$|\hat f(k)| \le \frac{1}{2\pi}\cdot\frac{2}{|k|}\cdot|\mu|((0,2\pi]) = O\Big(\frac{1}{|k|}\Big).$$

Together with $|\hat f(0)| \le \sup|f|$ this gives the claim, since $\frac{1}{|k|} \le \frac{2}{|k|+1}$ for $k \ne 0$. "In general this assumption is not important, just a longer proof" – here is a short proof without it. □

Proof (general). Extend $g(t) = f(e^{it})$ 2π-periodically; let $V = V_{\mathbb{T}}(f)$. Fix $k \ne 0$ and $h = \frac{\pi}{|k|}$, so $e^{-ik(t+h)} = -e^{-ikt}$. Shifting the variable of integration by h (periodicity), $\hat f(k) = \frac{1}{2\pi}\int_0^{2\pi} g(t+h)e^{-ik(t+h)}dt = -\frac{1}{2\pi}\int_0^{2\pi} g(t+h)e^{-ikt}dt$. Averaging with the original formula,

$$|\hat f(k)| = \Big|\frac{1}{4\pi}\int_0^{2\pi}(g(t) - g(t+h))e^{-ikt}dt\Big| \le \frac{1}{4\pi}\int_0^{2\pi}|g(t+h) - g(t)|\, dt.$$

Since $2\pi = 2|k|\cdot h$, split $[0,2\pi)$ into $2|k|$ intervals of length h:

$$\int_0^{2\pi}|g(t+h)-g(t)|\,dt = \int_0^h \sum_{j=0}^{2|k|-1}\big|g(s+(j+1)h) - g(s+jh)\big|\,ds \le h\,V,$$

because for each s the points $e^{i(s+jh)}$ form a partition of $\mathbb{T}$. Hence $|\hat f(k)| \le \frac{hV}{4\pi} = \frac{V}{4|k|}$. □

Proof of Theorem 2.36. Fix $z \in \mathbb{T}$. A BV function is bounded and has one-sided limits $f(z \pm 0)$ at every $z \in \mathbb{T}$ (Remark 2.35). By Fejér's theorem in the form of Remark 2.22, $\sum_{k=-\infty}^{\infty}\hat f(k)z^k$ converges in the Cesàro sense to $\frac{f(z-0)+f(z+0)}{2}$.

Reduction to a continuous-from-the-right f (lecture). Changing f at its points of discontinuity, e.g. replacing it by $\tilde f(e^{it}) = g(t+0)$, does not change $\sum\hat f(k)z^k$ nor $\frac{f(z-0)+f(z+0)}{2}$: the set of discontinuities is countable (Remark 2.35), hence of measure 0, so the integrals $\hat f(k) = \int f\bar\xi^{k}\,dm$ are unchanged; and one-sided limits are not affected by the values at single points. Also $V_{\mathbb{T}}(\tilde f) \le V_{\mathbb{T}}(f)$ (partition sums of $\tilde f$ are limits of partition sums of f). With the general proof of Lemma 2.39 this reduction is not even needed.

Tauberian step. By Lemma 2.39, $|\hat{f}(k)| \le \frac{C}{|k|+1}$ for all k. Group the terms symmetrically: $a_0 = \hat{f}(0)$ and $a_n = \hat{f}(n)z^n + \hat{f}(-n)z^{-n}$ for $n \ge 1$. Then $\sum_{n=0}^{N} a_n = S_N(f,z)$ and $|a_n| \le \frac{2C}{n+1}$. The series $\sum a_n$ converges in the Cesàro sense, so by Lemma 2.37 (with Remark 2.38) it converges in the usual sense to the same value:

$$\sum_{k\in\mathbb{Z}} \hat{f}(k)z^k = \lim_{N\to\infty} S_N(f,z) = \frac{f(z-0)+f(z+0)}{2}. \qquad \square$$

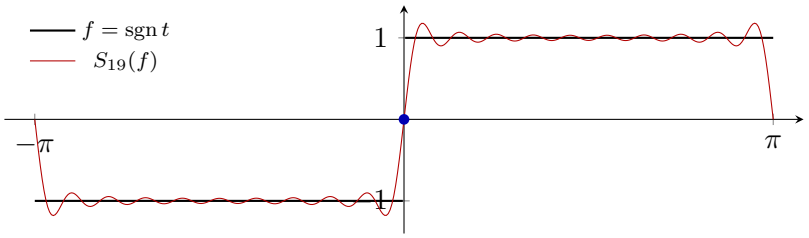


Figure 10: Hardy's theorem for the BV function $f(e^{it}) = \operatorname{sgn} t$: at the jump $t = 0$ every partial sum equals $0 = \frac{f(1-0)+f(1+0)}{2}$; at points of continuity $S_N(f) \to f$.

3 Fourier coefficients of various classes of functions

Setting. We have some class of functions on $\mathbb{T}$ (continuous, k times differentiable, integrable, of bounded variation, ...) and we look for properties of the Fourier coefficients $\hat{f}(k)$ of functions in this class, where $f = \sum_{k\in\mathbb{Z}} \hat{f}(k)z^k$. The guiding principle of this section is

the smoother f is, the faster $\hat{f}(n)$ decays as $|n| \to \infty$, and conversely.

Throughout, $\hat{f}(n) = \int_{\mathbb{T}} f(z)\bar{z}^n\, dm(z) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(e^{it})e^{-int}dt$ is defined for every $f \in L^1(\mathbb{T}, m)$ (Theorem 7.2), and $|\hat{f}(n)| \le \|f\|_{L^1}$.

Background 1. (The symbols O and o)

For sequences (x_n) and (y_n) with $y_n > 0$ we write $x_n = O(y_n)$ (as $|n| \to \infty$) if there are C, N with $|x_n| \le Cy_n$ for $|n| \ge N$, i.e. x_n/y_n is eventually bounded; and $x_n = o(y_n)$ if $x_n/y_n \to 0$. So $O(1)$ means "bounded", $o(1)$ means "tends to 0", and $x_n = o(y_n) \Rightarrow x_n = O(y_n)$. An expression like $\sum_n O(1)$ stands for $\sum_n x_n$ with some bounded sequence (x_n), which may of course diverge.

To talk about $C^k(\mathbb{T})$ we need derivatives of functions on the circle. The lecture uses the derivative along $\mathbb{T}$ in the complex sense:

Definition 3.2 (Differentiable functions on $\mathbb{T}$)**.** A function $f\colon \mathbb{T} \to \mathbb{C}$ is *differentiable* at $z_0 \in \mathbb{T}$ if the limit

$$f'(z_0) := \lim_{z\to z_0,\ z\in\mathbb{T}} \frac{f(z)-f(z_0)}{z-z_0}$$

exists and is finite. $f \in C^1(\mathbb{T})$ if f is differentiable at every point and f' is continuous on $\mathbb{T}$; recursively $f \in C^k(\mathbb{T})$ if $f \in C^1(\mathbb{T})$ and $f' \in C^{k-1}(\mathbb{T})$ (with $C^0(\mathbb{T}) = C(\mathbb{T})$), and $C^\infty(\mathbb{T}) = \bigcap_k C^k(\mathbb{T})$. For $f \in L^1(\mathbb{T})$ we say $f \in C^k(\mathbb{T})$ if f coincides a.e. with a C^k function (as elements of L^1 functions are only defined up to null sets).

For example $f(z) = z^n$ has $f'(z) = nz^{n-1}$ for every $n \in \mathbb{Z}$, exactly as in complex analysis.

Proposition 3.3. *Let $k \geq 0$. If $f \in C^k(\mathbb{T})$, then $|\hat{f}(n)| = O(\frac{1}{|n|^k})$; in fact even $|\hat{f}(n)| = o(\frac{1}{|n|^k})$.*

Proposition 3.4. *Let $k \geq 0$. If $f \in L^1(\mathbb{T})$ and $\sum_{n\in\mathbb{Z}} |n|^k |\hat{f}(n)| < \infty$, then $f \in C^k(\mathbb{T})$ (after a change on a null set). Moreover $f(z) = \sum_{n\in\mathbb{Z}} \hat{f}(n) z^n$ for every $z \in \mathbb{T}$, the series converging absolutely and uniformly.*

Remark 3.5 (The gap between Proposition 3.3 and Proposition 3.4). Proposition 3.3 and Proposition 3.4 are not converses of each other. If $f \in C^k(\mathbb{T})$, Proposition 3.3 only gives

$$\sum_{n\in\mathbb{Z}} |n|^k |\hat{f}(n)| = \sum_{n\in\mathbb{Z}} |n|^k \, O\Big(\frac{1}{|n|^k}\Big) = \sum_{n\in\mathbb{Z}} O(1),$$

which might diverge, so the hypothesis of Proposition 3.4 cannot be deduced. This is a genuine gap, not a weakness of the proofs:

(a) $f \in C(\mathbb{T})$ *does not imply* $\sum |\hat{f}(n)| < \infty$. Take $f \in C(\mathbb{T})$ from the du Bois-Reymond theorem (Theorem 2.23). If we had $\sum |\hat{f}(n)| < \infty$, then $S_N(f, z) = \sum_{|n|\leq N} \hat{f}(n) z^n$ would converge (absolutely) at every $z \in \mathbb{T}$ – but it diverges at some point. Applying this to a k-fold antiderivative shows the same for every k.

(b) *One derivative more closes the gap.* If $f \in C^{k+1}(\mathbb{T})$, then $\sum_n |n|^k |\hat{f}(n)| < \infty$. Indeed, in the proof of Proposition 3.3 below we show $|n|^{k+1}|\hat{f}(n)| = |\widehat{h^{(k+1)}}(n)|$ with $h^{(k+1)}$ continuous, so $\sum_n |n|^{2k+2}|\hat{f}(n)|^2 < \infty$ by Bessel's inequality, and by Cauchy–Schwarz in ℓ^2

$$\sum_{n\neq 0} |n|^k |\hat{f}(n)| = \sum_{n\neq 0} \frac{1}{|n|} \cdot |n|^{k+1}|\hat{f}(n)| \leq \Big(\sum_{n\neq 0} \frac{1}{n^2}\Big)^{1/2} \Big(\sum_{n\neq 0} |n|^{2k+2}|\hat{f}(n)|^2\Big)^{1/2} < \infty.$$

So $C^{k+1}(\mathbb{T}) \Rightarrow (\sum |n|^k |\hat{f}(n)| < \infty) \Rightarrow C^k(\mathbb{T})$: between "smoothness" and "summability of the coefficients" we lose exactly one derivative. A useful special case ($k = 0$): *the Fourier series of an $f \in C^1(\mathbb{T})$ converges absolutely and uniformly to f.*

Proposition 3.6. *Let $f \in L^1(\mathbb{T})$. Then $f \in C^\infty(\mathbb{T}) \iff |\hat{f}(n)| = O(\frac{1}{|n|^k})$ (as $|n| \to \infty$) for every $k \geq 0$.*

So for C^∞ the gap of Remark 3.5 disappears: smooth functions are exactly those with *rapidly decreasing* Fourier coefficients.

Proposition 3.7 (Riemann–Lebesgue lemma)**.** *If $f \in L^1(\mathbb{T})$, then $\hat{f}(n) = o(1)$, i.e. $\hat{f}(n) \to 0$ as $|n| \to \infty$.*

Remark 3.8. In the notebook the statement reads $|\hat{f}(n)| = O(1)$; this is a slip (the proof gives, and the Riemann–Lebesgue lemma states, $o(1)$). Boundedness alone is trivial: $|\hat{f}(n)| \leq \|f\|_{L^1}$. The decay $o(1)$ cannot be improved to any rate: for every sequence $\varepsilon_n \to 0$ there is $f \in L^1(\mathbb{T})$ with $\hat{f}(n) \neq O(\varepsilon_n)$ (see e.g. Katznelson, *An Introduction to Harmonic Analysis*, I.4). Integrability gives decay, but not *how fast*.

Proposition 3.9. *Let $f \in L^1(\mathbb{T})$. Then $f \in L^2(\mathbb{T}) \iff \sum_{n\in\mathbb{Z}} |\hat{f}(n)|^2 < \infty$. Moreover any sequence of numbers $\{c_n\}_{n\in\mathbb{Z}}$ with $\sum |c_n|^2 < \infty$ is the sequence of Fourier coefficients of some $f \in L^2(\mathbb{T})$.*

Proposition 3.10. *If $f \in BV(\mathbb{T})$, then $|\hat{f}(n)| = O(\frac{1}{|n|})$.*

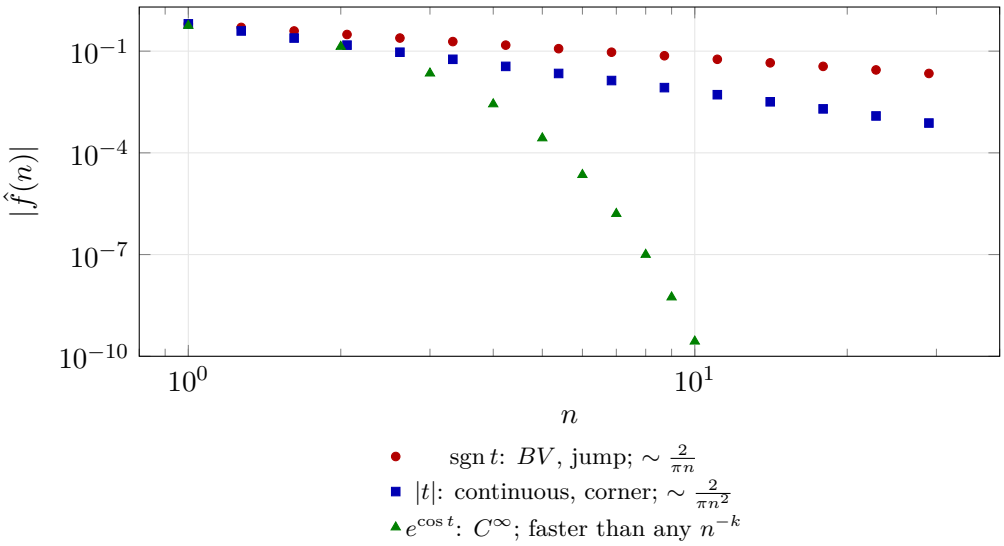


Figure 11: Decay of $|\hat{f}(n)|$ (log–log scale) for three functions $f(e^{it})$, $t \in (-\pi, \pi]$. For $\operatorname{sgn} t$ and $|t|$ only odd n are shown (the even coefficients vanish, $n \neq 0$). On a log–log scale n^{-k} is a line of slope $-k$: the jump gives slope -1 (Proposition 3.10), the corner slope -2, and the smooth function falls off every line (Proposition 3.6).

3.1 Proofs

Derivatives on $\mathbb{T}$ versus derivatives of periodic functions. Write $h(t) := f(e^{it})$, $t \in \mathbb{R}$; h is 2π-periodic and every 2π-periodic function arises in this way.

Lemma 3.11. *Let $f\colon \mathbb{T} \to \mathbb{C}$ and $h(t) = f(e^{it})$. Then $f \in C^k(\mathbb{T}) \iff h \in C^k(\mathbb{R})$, and in this case $h'(t) = ie^{it} f'(e^{it})$.*

Proof. Pointwise. Fix $t_0 \in \mathbb{R}$, $z_0 = e^{it_0}$. The map $t \mapsto e^{it}$ is a homeomorphism of $(t_0 - \pi, t_0 + \pi)$ onto $\mathbb{T} \setminus \{-z_0\}$, so "$z \to z_0$ in $\mathbb{T}$" is the same as "$t \to t_0$" with $z = e^{it}$. For $0 < |t - t_0| < \pi$,

$$\frac{h(t) - h(t_0)}{t - t_0} = \frac{f(e^{it}) - f(e^{it_0})}{e^{it} - e^{it_0}} \cdot \underbrace{\frac{e^{it} - e^{it_0}}{t - t_0}}_{\to\, ie^{it_0} \neq 0}. \tag{6}$$

Hence the difference quotient of h at t_0 converges iff that of f at z_0 does (if one converges, multiply or divide by the second factor, which has a non-zero limit), and $h'(t_0) = ie^{it_0} f'(z_0)$. (In the notebook the reciprocal factor $\frac{t - t_0}{e^{it} - e^{it_0}} \to \frac{1}{ie^{it_0}}$ is circled: it is the reason the two notions agree.)

$k = 1$. f' is continuous on $\mathbb{T}$ iff $t \mapsto f'(e^{it})$ is continuous on $\mathbb{R}$ iff $h'(t) = ie^{it} f'(e^{it})$ is continuous (multiplication by the continuous non-vanishing $ie^{\pm it}$).

Induction. Suppose the claim holds for $k - 1$ and all functions. The parametrization of f' is $t \mapsto f'(e^{it}) = -ie^{-it} h'(t)$. Therefore

$$\begin{aligned} f \in C^k(\mathbb{T}) &\iff f \in C^1(\mathbb{T}),\ f' \in C^{k-1}(\mathbb{T}) \iff h \in C^1(\mathbb{R}),\ e^{-it} h'(t) \in C^{k-1}(\mathbb{R}) \\ &\iff h \in C^1(\mathbb{R}),\ h' \in C^{k-1}(\mathbb{R}) \iff h \in C^k(\mathbb{R}), \end{aligned}$$

where in the third step we used that multiplying by the C^∞ functions $e^{\mp it}$ preserves C^{k-1} (Leibniz rule). □

Remark 3.12. The notebook writes $h \in C^k([-\pi, \pi])$. What is really needed is $h \in C^k(\mathbb{R})$ for the periodic extension, i.e. $h \in C^k([-\pi, \pi])$ *and* $h^{(j)}(-\pi) = h^{(j)}(\pi)$ for $j \leq k$. For instance $f(e^{it}) = t$ on $(-\pi, \pi]$ is C^∞ on $[-\pi, \pi]$ but jumps at $z = -1$ as a function on $\mathbb{T}$; its coefficients are only $O(\frac{1}{|n|})$ (Example 1.31). This periodicity is exactly what kills the boundary term in the integration by parts below.

Lemma 3.13 (Integration by parts)**.** *Let $h \in C^1(\mathbb{R})$ be 2π-periodic and put*

$$\hat{h}(n) := \frac{1}{2\pi}\int_{-\pi}^{\pi} h(t)e^{-int}dt.$$

Then $\widehat{h'}(n) = in\,\hat{h}(n)$ for all $n \in \mathbb{Z}$.

Proof.

$$\widehat{h'}(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} h'(t)e^{-int}dt = \underbrace{\frac{1}{2\pi}\Big[h(t)e^{-int}\Big]_{-\pi}^{\pi}}_{=0} - \frac{1}{2\pi}\int_{-\pi}^{\pi} h(t)(-in)e^{-int}dt = in\,\hat{h}(n).$$

The boundary term vanishes because $h(\pi) = h(-\pi)$ and $e^{-in\pi} = e^{in\pi} = (-1)^n$: "$e^{int}$ is periodic as well as h". □

If $h(t) = f(e^{it})$ then $\hat{h}(n) = \hat{f}(n)$ (this is the parametrization of the integral over $\mathbb{T}$). Iterating the lemma, for $h \in C^k(\mathbb{R})$ periodic,

$$\widehat{h^{(k)}}(n) = (in)^k\,\hat{f}(n), \qquad n \in \mathbb{Z}. \tag{7}$$

Proof of Proposition 3.3. Let $f \in C^k(\mathbb{T})$ and $h(t) = f(e^{it})$. By Lemma 3.11, $h \in C^k(\mathbb{R})$, so $h^{(k)}$ is continuous and 2π-periodic; in particular it is bounded, hence $h^{(k)} \in L^2([-\pi,\pi], \frac{dt}{2\pi})$. By Bessel's inequality (or Parseval, Theorem 1.26 with the ONB $\{e^{int}\}$, Corollary 2.19),

$$\sum_{n\in\mathbb{Z}} \big|\widehat{h^{(k)}}(n)\big|^2 \le \big\|h^{(k)}\big\|^2 < \infty, \qquad \text{in particular} \qquad \widehat{h^{(k)}}(n) \xrightarrow{|n|\to\infty} 0.$$

By (7), for $n \neq 0$,

$$\hat{f}(n) = \frac{\widehat{h^{(k)}}(n)}{(in)^k} = o\Big(\frac{1}{|n|^k}\Big). \qquad \square$$

Remark 3.14 (Normalization in the lecture). In the lecture the computation was done with the ONB $\{\frac{e^{int}}{\sqrt{2\pi}}\}$ of $L^2([-\pi,\pi], dt)$, i.e. $\widehat{h^{(k)}}(n)$ meant $\frac{1}{\sqrt{2\pi}}\int h^{(k)}(t)e^{-int}dt$; then $\widehat{h^{(k)}}(n) = (in)^k\sqrt{2\pi}\hat{f}(n)$ and $\hat{f}(n) = \frac{\widehat{h^{(k)}}(n)}{\sqrt{2\pi}(in)^k}$. The factors $\sqrt{2\pi}$ are harmless constants; with the normalized measure $\frac{dt}{2\pi}$ they disappear.

A side note in the notebook asks whether $\widehat{h^{(k)}}(n) \to 0$ holds "because $h^{(k)} \in BV$, being continuous on a compact interval". That reason would be wrong: a continuous function need not be of bounded variation (e.g. $t\sin\frac{1}{t}$ near 0). The correct reason is the one given: the coefficients of an L^2 function form an ℓ^2 sequence and so tend to 0 (alternatively: Proposition 3.7).

Proof of Proposition 3.4. We first need two auxiliary facts. The first is the "observation" of the lecture: an absolutely summable combination of uniformly bounded continuous functions is continuous.

Lemma 3.15. *Let $\sum_{n\in\mathbb{Z}} |c_n| < \infty$ and let $g_n\colon \mathbb{T} \to \mathbb{C}$ satisfy $M := \sup_{n\in\mathbb{Z},\,t\in\mathbb{T}} |g_n(t)| < \infty$ and $g_n(t) \to g_n(t_0)$ as $t \to t_0$, for every n. Then*

$$\sum_{n\in\mathbb{Z}} c_n g_n(t) \xrightarrow{t\to t_0} \sum_{n\in\mathbb{Z}} c_n g_n(t_0).$$

(The same proof works with $\mathbb{T}$ replaced by any metric space.)

Proof. All series converge absolutely since $|c_n g_n(t)| \le M|c_n|$. For every N,

$$\Big|\sum_n c_n g_n(t) - \sum_n c_n g_n(t_0)\Big| \le \sum_{|n|\le N} |c_n|\,|g_n(t) - g_n(t_0)| + \sum_{|n|>N} |c_n| \cdot 2M.$$

Given $\varepsilon > 0$, choose $N = N(\varepsilon)$ with $2M \sum_{|n|>N} |c_n| < \frac{\varepsilon}{2}$ (tail of a convergent series). For this N the first sum is a *finite* sum of terms tending to 0, so there is $\delta > 0$ with $\sum_{|n|\le N} |c_n||g_n(t) - g_n(t_0)| < \frac{\varepsilon}{2}$ for $|t - t_0| < \delta$. For such t the left side is $< \varepsilon$. □

This is a variant of the Weierstrass M-test (Theorem 7.3). The second fact is that an integrable function is determined by its Fourier coefficients. The lecture uses this silently in the case $k = 0$: there f is only in L^1, and since $L^1(\mathbb{T}) \not\subset L^2(\mathbb{T})$, the Riesz–Fischer expansion $f = \sum \hat{f}(n) z^n$ is not available.

Lemma 3.16 (Uniqueness theorem)**.** *If $f \in L^1(\mathbb{T})$ and $\hat{f}(n) = 0$ for all $n \in \mathbb{Z}$, then $f = 0$ a.e.*

Proof. Step 1: $\int_{\mathbb{T}} fg\,dm = 0$ for all $g \in C(\mathbb{T})$. By assumption $\int_{\mathbb{T}} fp\,dm = 0$ for every trigonometric polynomial p (p is a combination of $\overline{z}^n = z^{-n}$). Given $g \in C(\mathbb{T})$ choose trigonometric polynomials $p_j \to g$ uniformly (Corollary 2.18); then $\left|\int fg\,dm\right| = \left|\int f(g - p_j)\,dm\right| \le \|f\|_{L^1} \max_{\mathbb{T}} |g - p_j| \to 0$.

Step 2. Let $u = \overline{f}/|f|$ where $f \neq 0$ and $u = 0$ elsewhere; u is measurable, $|u| \le 1$ and $fu = |f|$. Since $C(\mathbb{T})$ is dense in $L^1(\mathbb{T})$ (Theorem 2.3) there are $g_j \in C(\mathbb{T})$ with $\|g_j - u\|_{L^1} \to 0$. Replacing g_j by $r \circ g_j$, where $r(w) = w$ for $|w| \le 1$ and $r(w) = w/|w|$ otherwise (r is continuous and $|r(w) - r(w')| \le |w - w'|$, and $r \circ u = u$), we may assume $|g_j| \le 1$. Passing to a subsequence, $g_j \to u$ a.e. Then $|fg_j| \le |f| \in L^1$ and by dominated convergence (Theorem 7.2) and Step 1,

$$\int_{\mathbb{T}} |f|\,dm = \int_{\mathbb{T}} fu\,dm = \lim_j \int_{\mathbb{T}} fg_j\,dm = 0.$$

Hence $f = 0$ a.e. □

(The subsequence step uses the standard fact that L^1-convergence implies a.e. convergence of a subsequence.)

Proof of Proposition 3.4. By induction on k; the statement is proved for all f simultaneously.

$k = 0$. Assume $\sum_n |\hat{f}(n)| < \infty$ and put $F(z) := \sum_{n\in\mathbb{Z}} \hat{f}(n) z^n$. By Lemma 3.15 with $c_n = \hat{f}(n)$, $g_n(z) = z^n$ ($|g_n| = 1$), F is continuous on $\mathbb{T}$; the series converges absolutely and uniformly (Theorem 7.3). Uniform convergence allows us to integrate term by term: $\hat{F}(m) = \sum_n \hat{f}(n) \int_{\mathbb{T}} z^n \overline{z}^m\,dm = \hat{f}(m)$. So $f - F \in L^1(\mathbb{T})$ has all Fourier coefficients 0, and by Lemma 3.16 $f = F$ a.e. Thus $f \in C(\mathbb{T})$ and $f(z) = \sum_n \hat{f}(n) z^n$ everywhere (after the change on a null set).

$k - 1 \to k$ *($k \ge 1$).* Assume $\sum_n |n|^k |\hat{f}(n)| < \infty$. Since $|n|^k \ge 1$ for $n \neq 0$, also $\sum_n |\hat{f}(n)| < \infty$, so by the case $k = 0$ we may take $f(z) = \sum_n \hat{f}(n) z^n$, continuous. Fix $z_0 \in \mathbb{T}$. For $z \in \mathbb{T} \setminus \{z_0\}$,

$$\frac{f(z) - f(z_0)}{z - z_0} = \sum_{n\in\mathbb{Z}} \hat{f}(n) \frac{z^n - z_0^n}{z - z_0} = \sum_{n\neq 0} \underbrace{n\hat{f}(n)}_{=:c_n} \cdot \underbrace{\frac{z^n - z_0^n}{n(z - z_0)}}_{=:g_n(z)}.$$

We check the hypotheses of Lemma 3.15 (on the metric space $\mathbb{T}$, with $g_n(z_0) := z_0^{n-1}$):

- $\sum_n |c_n| = \sum_n |n||\hat{f}(n)| \le \sum_n |n|^k |\hat{f}(n)| < \infty$.
- For $n \ge 1$, $z^n - z_0^n = (z - z_0) \sum_{j=0}^{n-1} z^j z_0^{n-1-j}$, so $g_n(z) = \frac{1}{n} \sum_{j=0}^{n-1} z^j z_0^{n-1-j}$. This is a polynomial in z, hence continuous, with value z_0^{n-1} at $z = z_0$, and $|g_n(z)| \le \frac{1}{n} \cdot n = 1$. (The notebook writes the same thing as $|g_n(z)| = \frac{1}{n}|\sum_{j=0}^{n-1} (\overline{z} z_0)^j| \le 1$.)
- For $n = -m \le -1$: $z^n - z_0^n = \frac{z_0^m - z^m}{z^m z_0^m}$, so $g_n(z) = \frac{1}{z^m z_0^m} \cdot \frac{z^m - z_0^m}{m(z - z_0)}$ is again continuous, bounded by 1, with value $z_0^{-2m} z_0^{m-1} = z_0^{n-1}$ at z_0.

By Lemma 3.15,

$$\frac{f(z)-f(z_0)}{z-z_0} \xrightarrow{z\to z_0} \sum_{n\in\mathbb{Z}} n\hat{f}(n) z_0^{n-1}.$$

So f is differentiable at every point, with $f'(z)=\sum_n n\hat{f}(n)z^{n-1}$; this series is again of the form of Lemma 3.15 ($c_n = n\hat{f}(n)$, $g_n = z^{n-1}$), so f' is continuous and $f \in C^1(\mathbb{T})$. Integrating term by term (uniform convergence) gives the Fourier coefficients of f':

$$\widehat{f'}(n-1) = n\hat{f}(n), \qquad \text{i.e.} \qquad \widehat{f'}(m) = (m+1)\hat{f}(m+1).$$

Now we repeat the argument for $(f', k-1)$ in place of (f,k). Using $|m| = |n-1| \le 2|n|$ for $n \neq 0$,

$$\sum_{m\in\mathbb{Z}} |m|^{k-1}|\widehat{f'}(m)| = \sum_{n\neq 0} |n-1|^{k-1}|n||\hat{f}(n)| \le 2^{k-1}\sum_{n\in\mathbb{Z}} |n|^k|\hat{f}(n)| < \infty.$$

By the induction hypothesis f' coincides a.e. with a C^{k-1} function; both are continuous, and two continuous functions that agree a.e. on $\mathbb{T}$ agree everywhere (a non-empty open arc has positive measure). Hence $f' \in C^{k-1}(\mathbb{T})$, i.e. $f \in C^k(\mathbb{T})$. □

Remark 3.17. The formula $\widehat{f'}(n-1) = n\hat{f}(n)$ is consistent with (7): by Lemma 3.11, $h'(t) = ie^{it}f'(e^{it})$, and multiplying by $z = e^{it}$ shifts coefficients by one, so $\widehat{h'}(n) = i\,\widehat{f'}(n-1) = in\hat{f}(n)$. The complex derivative along $\mathbb{T}$ and the derivative in t differ only by the factor iz.

Proof of Proposition 3.6. ($\Rightarrow$) $C^\infty(\mathbb{T}) \subset C^k(\mathbb{T})$ for every k, so Proposition 3.3 gives $\hat{f}(n) = O(|n|^{-k})$ for every k.

($\Leftarrow$) Fix k. By assumption (applied with $k+2$) $|\hat{f}(n)| \le C|n|^{-k-2}$ for $|n| \ge N$, hence

$$\sum_{|n|\ge N} |n|^k|\hat{f}(n)| \le C\sum_{|n|\ge N}\frac{1}{n^2} < \infty,$$

and the finitely many terms with $|n| < N$ do not matter. By Proposition 3.4, $f \in C^k(\mathbb{T})$. All the modifications on null sets give the same continuous function (namely $\sum \hat{f}(n)z^n$), so $f \in C^k$ for all k, i.e. $f \in C^\infty(\mathbb{T})$. □

Proof of Proposition 3.7 (Riemann–Lebesgue). Let $f \in L^1(\mathbb{T})$ and $\varepsilon > 0$. *Trigonometric polynomials are dense in* $L^1(\mathbb{T})$: choose $g \in C(\mathbb{T})$ with $\|f-g\|_{L^1} < \frac{\varepsilon}{2}$ (Theorem 2.3) and a trigonometric polynomial $p = \sum_{k=-n}^n c_k z^k$ with $\max_{\mathbb{T}}|g-p| < \frac{\varepsilon}{2}$ (Corollary 2.18); since $m(\mathbb{T}) = 1$, $\|g-p\|_{L^1} \le \max_{\mathbb{T}}|g-p|$, so $\|f-p\|_{L^1} < \varepsilon$. Now split, for any $j \in \mathbb{Z}$,

$$\hat{f}(j) = \int_{\mathbb{T}} f(z)\overline{z}^j\,dm = \underbrace{\int_{\mathbb{T}} (f(z)-p(z))\overline{z}^j\,dm}_{A_j} + \underbrace{\int_{\mathbb{T}} p(z)\overline{z}^j\,dm}_{B_j}.$$

- $|A_j| \le \int_{\mathbb{T}} |f-p|\,|z|^{-j}\,dm = \|f-p\|_{L^1} < \varepsilon$ for *every* j (because $|z| = 1$ on $\mathbb{T}$).
- $B_j = \sum_{k=-n}^n c_k \left(z^k, z^j\right) = 0$ for $|j| \ge n+1$, by orthogonality of $\{z^k\}$.

Hence $|\hat{f}(j)| < \varepsilon$ for all $|j| \ge n+1$. As ε was arbitrary, $\hat{f}(j) \to 0$ as $|j| \to \infty$. □

The proof is a prototype of a standard technique: prove the statement for a dense class where it is obvious (trigonometric polynomials: their coefficients are eventually 0) and pass to the closure using a uniform estimate (here $|\hat{f}(j) - \hat{p}(j)| \le \|f-p\|_{L^1}$ for all j).

Propositions 3.9 and 3.10 we already know from the Riesz–Fischer theorem and from the proof of Hardy's theorem:

Proof of Proposition 3.9. ($\Rightarrow$) If $f \in L^2(\mathbb{T})$, Parseval (3) for the ONB $\{z^n\}$ gives $\sum_n |\hat f(n)|^2 = \|f\|_{L^2}^2 < \infty$.

The "moreover" part is Corollary 1.30: for $(c_n) \in \ell^2(\mathbb{Z})$ the series $g = \sum c_n z^n$ converges in $L^2(\mathbb{T})$ and $\hat g(n) = c_n$.

($\Leftarrow$) Let $f \in L^1(\mathbb{T})$ with $\sum |\hat f(n)|^2 < \infty$. Let $g \in L^2(\mathbb{T})$ have $\hat g(n) = \hat f(n)$ (previous paragraph). Then $f - g \in L^1(\mathbb{T})$ (as $L^2(\mathbb{T}) \subset L^1(\mathbb{T})$, Problem 2) has all coefficients 0, so $f = g$ a.e. by Lemma 3.16; thus $f \in L^2(\mathbb{T})$. □

Proof of Proposition 3.10. This is Lemma 2.39: $|\hat f(n)| \le \frac{V_{\mathbb{T}}(f)}{4|n|}$ for $n \neq 0$. □

Example 3.18 (The three functions of Fig. 11)**.** (a) $f(e^{it}) = \operatorname{sgn} t$ on $(-\pi, \pi]$ is in $BV(\mathbb{T})$ but not continuous. For $n \neq 0$: $\hat f(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} \operatorname{sgn} t\, e^{-int} dt = \frac{-i}{\pi}\int_0^{\pi} \sin(nt)\, dt = \frac{-i(1-(-1)^n)}{\pi n}$, so $|\hat f(n)| = \frac{2}{\pi|n|}$ for odd n. This is $O(\frac{1}{|n|})$ as Proposition 3.10 predicts, and not $o(\frac{1}{|n|})$: Proposition 3.10 is sharp. Also $\sum |\hat f(n)| = \infty$, consistent with Proposition 3.4 (f is not continuous).

(b) $f(e^{it}) = |t|$ is continuous (also at $z = -1$, as $|-\pi| = |\pi|$) with a corner, so $f \in C(\mathbb{T}) \setminus C^1(\mathbb{T})$. Here $\hat f(n) = \frac{1}{\pi}\int_0^{\pi} t\cos(nt)\, dt = \frac{(-1)^n - 1}{\pi n^2}$, i.e. $|\hat f(n)| = \frac{2}{\pi n^2}$ for odd n. Then $\sum |\hat f(n)| < \infty$, so by Proposition 3.4 the Fourier series converges absolutely and uniformly to f, even though $f \notin C^1$. But $\sum |n||\hat f(n)| = \infty$, so Proposition 3.4 does not give C^1, as it must not.

(c) $f(e^{it}) = e^{\cos t} = e^{(z+\bar z)/2} = \sum_{j,l\ge 0} \frac{z^j \bar z^l}{2^{j+l} j!\, l!}$ is C^∞; collecting powers z^n gives

$$\hat f(n) = \sum_{l \ge 0} \frac{1}{2^{2l+|n|}\, l!\, (l+|n|)!} \le \frac{e^{1/4}}{2^{|n|}|n|!},$$

which decays faster than every $|n|^{-k}$ (Proposition 3.6). (These are the values $I_n(1)$ of the modified Bessel function.)

4 The classical form of Fourier series

Before complex exponentials became standard, Fourier series were written with sines and cosines. This is the same expansion, only grouped differently.

Theorem 4.1. *Let $f\colon \mathbb{R} \to \mathbb{C}$ be a 2π-periodic function with $f \in L^2([-\pi, \pi])$. Then*

$$f(x) = a_0 + \sum_{k=1}^{\infty} a_k \cos(kx) + \sum_{k=1}^{\infty} b_k \sin(kx), \tag{$*$}$$

where

$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\, dx, \qquad a_k = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(kx)\, dx,$$
$$b_k = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin(kx)\, dx \qquad (k \ge 1),$$

and the series converges in $L^2([-\pi, \pi])$, in the sense that the partial sums $a_0 + \sum_{k=1}^{N}(a_k\cos(kx) + b_k\sin(kx))$ converge to f in L^2 as $N \to \infty$.

Note the different constant in a_0: it is the *mean value* of f, while a_k, b_k carry the factor $\frac{1}{\pi}$ (see Remark 4.2). Some books write $\frac{a_0}{2}$ with $a_0 = \frac{1}{\pi}\int f$ to unify the formulas.

Proof (lecture). Let $H := L^2([-\pi,\pi], \frac{dx}{2\pi})$ with the ONB $\{e^{inx}\}_{n\in\mathbb{Z}}$ (Corollary 2.19). The norms of $L^2([-\pi,\pi], dx)$ and H differ by the constant $\sqrt{2\pi}$, so convergence in one is convergence in the other.

Reduction to real f. Write $f = u + iv$ with $u = \operatorname{Re} f$, $v = \operatorname{Im} f$ real, 2π-periodic and in L^2 (as $|u|, |v| \le |f|$). The coefficients are linear in f: $a_k(f) = a_k(u) + ia_k(v)$, $b_k(f) = b_k(u) + ib_k(v)$, so the partial sums of $(*)$ for f are those for u plus i times those for v. If both converge in L^2 to u and v, they converge to f. By linearity it therefore suffices to prove $(*)$ for $f\colon \mathbb{R} \to \mathbb{R}$.

Real f. By Riesz–Fischer, $f = \sum_{n\in\mathbb{Z}} \hat f(n) e^{inx}$ in H, with symmetric partial sums $S_N = \sum_{|n|\le N} \hat f(n) e^{inx}$ (Eq. $(**)$). The map $g \mapsto \operatorname{Re} g$ is real linear and continuous on H ($|\operatorname{Re} g_1 - \operatorname{Re} g_2| \le |g_1 - g_2|$ pointwise), so taking real parts of $S_N \to f$ gives $\operatorname{Re} S_N \to \operatorname{Re} f = f$, since f is real:

$$f \overset{f \text{ real}}{=} \sum_{n\in\mathbb{Z}} \operatorname{Re}(\hat f(n) e^{inx}) = \sum_{n\in\mathbb{Z}} \operatorname{Re} \hat f(n) \cdot \cos(nx) - \sum_{n\in\mathbb{Z}} \operatorname{Im} \hat f(n) \cdot \sin(nx),$$

using $\operatorname{Re}(\alpha\beta) = \operatorname{Re}\alpha \operatorname{Re}\beta - \operatorname{Im}\alpha \operatorname{Im}\beta$ and $e^{inx} = \cos(nx) + i\sin(nx)$ (all sums are symmetric limits $\lim_N \sum_{|n|\le N}$). Since f is real,

$$\operatorname{Re} \hat f(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} \operatorname{Re}(f(s)e^{-ins}) ds = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(s)\cos(ns)\, ds =: \tilde a_n,$$

$$\operatorname{Im} \hat f(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} \operatorname{Im}(f(s)e^{-ins}) ds = -\frac{1}{2\pi}\int_{-\pi}^{\pi} f(s)\sin(ns)\, ds =: -\tilde b_n.$$

Hence $f = \sum_{n\in\mathbb{Z}} \tilde a_n \cos(nx) + \sum_{n\in\mathbb{Z}} \tilde b_n \sin(nx)$. Since cosine is even and sine is odd, $\tilde a_{-n} = \tilde a_n$, $\tilde b_{-n} = -\tilde b_n$ and $\tilde b_0 = 0$, so the terms with n and $-n$ are equal: $\tilde a_{-n}\cos(-nx) = \tilde a_n \cos(nx)$ and $\tilde b_{-n}\sin(-nx) = \tilde b_n \sin(nx)$. Grouping them, the symmetric partial sum is

$$\sum_{|n|\le N} (\tilde a_n \cos(nx) + \tilde b_n \sin(nx)) = \tilde a_0 + \sum_{n=1}^{N} 2\tilde a_n \cos(nx) + \sum_{n=1}^{N} 2\tilde b_n \sin(nx),$$

and $\tilde a_0 = a_0$, $2\tilde a_n = a_n$, $2\tilde b_n = b_n$. This finishes the proof. □

Remark 4.2 (A shorter route, valid for complex f). For any f (real or complex) and $k \ge 1$, using $e^{\pm ikx} = \cos(kx) \pm i\sin(kx)$,

$$\hat f(k)e^{ikx} + \hat f(-k)e^{-ikx} = (\hat f(k) + \hat f(-k))\cos(kx) + i(\hat f(k) - \hat f(-k))\sin(kx),$$

and $\hat f(k) + \hat f(-k) = \frac{1}{2\pi}\int f(x)(e^{-ikx} + e^{ikx})dx = a_k$, $i(\hat f(k) - \hat f(-k)) = \frac{i}{2\pi}\int f(x)(-2i\sin kx)\, dx = b_k$. So the partial sum of $(*)$ is *exactly* $S_N(f)$, and $(*)$ is just $(**)$. This also explains the constants: $a_0 = \hat f(0)$ appears once, while $\cos(kx)$ collects the two coefficients $\hat f(\pm k)$, which doubles $\frac{1}{2\pi}$ to $\frac{1}{\pi}$.

Corollary 4.3 (Trigonometric ONB and classical Parseval)**.** *The functions $\frac{1}{\sqrt{2\pi}}$, $\frac{\cos(kx)}{\sqrt{\pi}}$, $\frac{\sin(kx)}{\sqrt{\pi}}$ ($k \ge 1$) form an ONB of $L^2([-\pi,\pi], dx)$, and for $f \in L^2([-\pi,\pi])$*

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} |f(x)|^2 dx = |a_0|^2 + \frac{1}{2}\sum_{k=1}^{\infty} (|a_k|^2 + |b_k|^2).$$

Proof. Orthonormality is the computation

$$\int_{-\pi}^{\pi} \cos(kx)\cos(lx)\, dx = \int_{-\pi}^{\pi} \sin(kx)\sin(lx)\, dx = \pi\delta_{kl}, \qquad \int_{-\pi}^{\pi} \sin(kx)\cos(lx)\, dx = 0$$

($k, l \ge 1$), e.g. from $\cos(kx) = \frac{e^{ikx}+e^{-ikx}}{2}$ and orthogonality of exponentials. Completeness: by Theorem 4.1 every f is an L^2-limit of their linear combinations. For Parseval, by Remark 4.2 and the parallelogram law,

$$|a_k|^2 + |b_k|^2 = |\hat f(k) + \hat f(-k)|^2 + |\hat f(k) - \hat f(-k)|^2 = 2(|\hat f(k)|^2 + |\hat f(-k)|^2),$$

and $\sum_n |\hat f(n)|^2 = \frac{1}{2\pi}\int |f|^2$ by (3). □

Example 4.4. For $f(x) = x$ on $(-\pi, \pi]$ (extended periodically) f is odd, so $a_k = 0$, and $b_k = \frac{1}{\pi}\int_{-\pi}^{\pi} x \sin(kx)\,dx = \frac{2(-1)^{k+1}}{k}$ (integrate by parts, or use $\hat f(k) = \frac{e^{-ik\pi}}{-ik}$ from Example 1.31 and $b_k = i(\hat f(k) - \hat f(-k))$). Thus

$$x = 2\Big(\sin x - \frac{\sin 2x}{2} + \frac{\sin 3x}{3} - \dots\Big) \quad \text{in } L^2([-\pi,\pi]),$$

and the classical Parseval identity gives $\frac{\pi^2}{3} = \frac12 \sum_k \frac{4}{k^2}$, which is again Euler's formula $\sum \frac{1}{k^2} = \frac{\pi^2}{6}$. Since f is BV on $\mathbb{T}$, Hardy's theorem (Theorem 2.36) even gives pointwise convergence for $|x| < \pi$, and convergence to $\frac{\pi+(-\pi)}{2} = 0$ at $x = \pm\pi$ (Fig. 12).

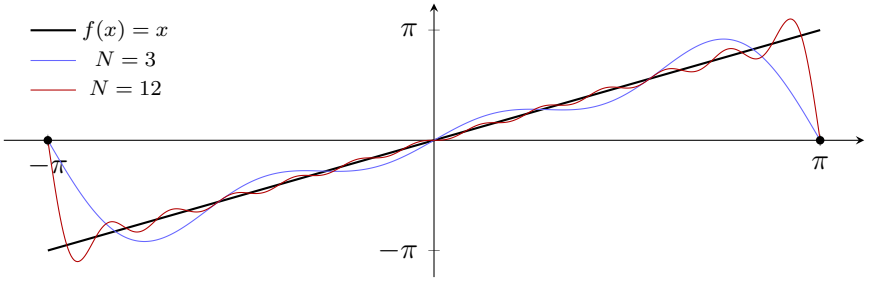


Figure 12: Partial sums $2\sum_{k=1}^{N} \frac{(-1)^{k+1}}{k} \sin(kx)$ of the classical Fourier series of $f(x) = x$. All of them vanish at $x = \pm\pi$, the midpoint of the jump of the periodic extension.

5 Gram–Schmidt orthonormalization and orthogonal polynomials

5.1 Gram–Schmidt orthonormalization

Theorem 5.1 (Gram–Schmidt)**.** *Let H be a Hilbert space and $\{h_k\}_{k=0}^{\infty}$ a linearly independent sequence in H (every finite subfamily is linearly independent). Then there exists a sequence $\{e_k\}_{k=0}^{\infty} \subset H$ such that*

(i) $\|e_k\| = 1$ *for all k;*

(ii) $e_k \perp e_j$ *for all $k \neq j$;*

(iii) $\operatorname{span}(e_0, \dots, e_k) = \operatorname{span}(h_0, \dots, h_k)$ *for all k.*

Moreover one can achieve (iv) $(h_k, e_k) > 0$ for all k, and (i)–(iv) determine $\{e_k\}$ uniquely.

(Property (iv) and the uniqueness were not stated in the lecture; they are needed in the proof of Favard's theorem.)

Proof. Construction. Put $e_0 := \frac{h_0}{\|h_0\|}$; here $\|h_0\| \neq 0$ because a linearly independent family cannot contain 0. Recursively, for $n \geq 1$,

$$u_n := h_n - \sum_{k=0}^{n-1} (h_n, e_k)\, e_k, \qquad e_n := \frac{u_n}{\|u_n\|};$$

in particular $e_1 = \frac{h_1 - (h_1,e_0)e_0}{\|h_1 - (h_1,e_0)e_0\|}$. The vector $\sum_{k<n} (h_n, e_k)\, e_k$ is the orthogonal projection of h_n onto $\operatorname{span}(e_0, \dots, e_{n-1})$ (cf. Problem 1), so u_n is the component of h_n orthogonal to the previous vectors (Fig. 13).

We prove (i)–(iv) by induction on n, assuming them for indices $< n$.

- $u_n \neq 0$: otherwise $h_n = \sum_{k<n} (h_n, e_k)\, e_k \in \operatorname{span}(e_0, \dots, e_{n-1}) = \operatorname{span}(h_0, \dots, h_{n-1})$, contradicting linear independence. So e_n is well defined and (i) holds.

- (ii): for $j < n$, $(u_n, e_j) = (h_n, e_j) - \sum_{k<n} (h_n, e_k)\,(e_k, e_j) = (h_n, e_j) - (h_n, e_j) = 0$.
- (iii): $e_n \in \operatorname{span}(h_n, e_0, \dots, e_{n-1}) = \operatorname{span}(h_0, \dots, h_n)$, and conversely $h_n = \|u_n\|\, e_n + \sum_{k<n} (h_n, e_k)\, e_k \in \operatorname{span}(e_0, \dots, e_n)$.
- (iv): $(h_n, e_n) = (u_n, e_n) = \|u_n\| > 0$, since $h_n - u_n \perp e_n$.

Uniqueness. If $\{e'_k\}$ also satisfies (i)–(iii), then $e'_n \in V_n := \operatorname{span}(h_0, \dots, h_n)$ and $e'_n \perp V_{n-1}$. The orthogonal complement of V_{n-1} in V_n has dimension 1 and is spanned by e_n, so $e'_n = \lambda e_n$ with $|\lambda| = 1$. If both satisfy (iv), then $0 < (h_n, e'_n) = \overline{\lambda}\,(h_n, e_n)$ with $(h_n, e_n) > 0$ forces $\overline{\lambda} > 0$, so $\lambda = 1$. □

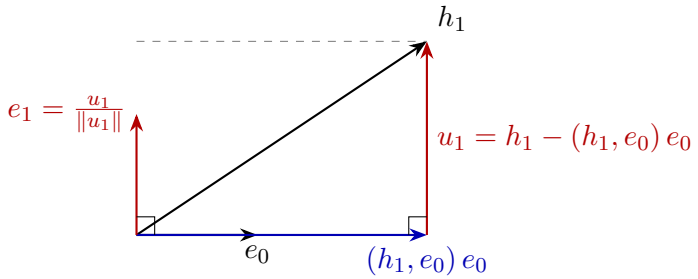


Figure 13: One step of the Gram–Schmidt process: subtract from h_1 its projection onto e_0 and normalize the rest.

Corollary 5.2. *Every separable Hilbert space has an ONB.*

Proof. If $H = \{0\}$ the empty family is an ONB. Otherwise take a dense sequence $(d_m)_{m\geq 1}$ in H (Theorem 1.21) and remove some of its elements to get a linearly independent sequence: go through $d_1, d_2, \dots$ and keep d_m iff $d_m \notin \operatorname{span}(d_1, \dots, d_{m-1})$. Call the kept vectors $h_0, h_1, \dots$. They are linearly independent (each is not in the span of the earlier ones), and every discarded d_m lies in the span of earlier kept ones, so $\operatorname{span}\{h_k\} = \operatorname{span}\{d_m\}$. Apply Gram–Schmidt (if only finitely many h_k are kept, the same finite construction works). The resulting $\{e_k\}$ is orthonormal and by (iii)

$$\operatorname{span}\{e_k\} = \operatorname{span}\{h_k\} = \operatorname{span}\{d_m\} \supset \{d_m\},$$

which is dense in H. So $\operatorname{clos}\operatorname{span}\{e_k\} = H$, i.e. $\{e_k\}$ is complete, hence an ONB (Definition 1.19). □

Remark 5.3. Together with Exercise 1.22 this gives: a Hilbert space is separable iff it has a countable (finite or countably infinite) ONB. Non-separable Hilbert spaces also have ONBs, but the proof needs Zorn's lemma instead of an explicit construction.

5.2 Orthogonal polynomials and Favard's theorem

Background 4. (Probability measures on $\mathbb{R}$, support, Dirac masses)

A *probability measure* on $\mathbb{R}$ is a (positive) Borel measure μ with $\mu(\mathbb{R}) = 1$. The *Dirac mass* at x_0 is $\delta_{x_0}(A) = 1$ if $x_0 \in A$ and 0 otherwise, so $\int g\, d\delta_{x_0} = g(x_0)$. A *finite linear combination of δ-masses* is $\mu = \sum_{k=1}^{n} c_k \delta_{x_k}$ (for a probability measure: $c_k > 0$, $\sum c_k = 1$); then $\int g\, d\mu = \sum_k c_k g(x_k)$. The *support* $\operatorname{supp}\mu$ is the set of x such that every neighbourhood of x has positive measure; it is the smallest closed set of full measure. "μ is supported on a compact set" means $\operatorname{supp}\mu \subset [-R, R]$ for some R, i.e. $\mu(\mathbb{R} \setminus [-R, R]) = 0$. Example: $\frac{dx}{2}$ on $[-1, 1]$.

Setting. Let μ be a probability measure on $\mathbb{R}$ with $\operatorname{supp}\mu \subset [-R,R]$, and $H = L^2(\mu) = L^2(\mathbb{R},\mu)$, $(f,g) = \int_{\mathbb{R}} f\overline{g}\,d\mu$. Every polynomial q is bounded on $[-R,R]$, so $\int |q|^2 d\mu \le \max_{[-R,R]} |q|^2 < \infty$: polynomials belong to H. We would like to define

$$\{p_n\}_{n=0}^{\infty} := \text{Gram–Schmidt of the sequence } \{x^k\}_{k\in\mathbb{N}_0} = \{1, x, x^2, \dots\},$$

the *orthonormal polynomials* of μ. For this we need that $\{x^k\}$ is linearly independent *in* $L^2(\mu)$ (where functions equal μ-a.e. are identified!).

Lemma 5.5. *$\{x^k\}_{k=0}^{\infty}$ is linearly independent in $L^2(\mu)$ $\iff$ μ is not a finite linear combination of Dirac masses $\iff$ $\operatorname{supp}\mu$ is an infinite set.*

Proof. The monomials are dependent iff there is a non-zero polynomial q with $q = 0$ in $L^2(\mu)$, i.e. $\int |q|^2 d\mu = 0$, i.e. $\mu(\{q \neq 0\}) = 0$. If $\mu = \sum_{k=1}^{n} c_k \delta_{x_k}$, then $q(x) = \prod_k (x - x_k) \neq 0$ has $\int |q|^2 d\mu = \sum_k c_k |q(x_k)|^2 = 0$. Conversely, if $\mu(\{q \neq 0\}) = 0$ for some $q \neq 0$, then μ is concentrated on the finite zero set $Z = \{x_1, \dots, x_n\}$ of q, so $\mu = \sum_k \mu(\{x_k\})\delta_{x_k}$. Finally, μ is a finite combination of Dirac masses iff its support (the closed set carrying it) is finite. □

In the notebook the condition is also written as "$\int f\,d\mu \neq \sum_{k=1}^{n} f(x_k)c_k$", meaning: for every choice of n, x_k, c_k there is a polynomial f violating the equality, i.e. μ cannot be replaced by a discrete measure on polynomials. From now on we exclude the discrete case, and thus have independence.

Lemma 5.6. *Let μ be a probability measure with compact infinite support and p_n its orthonormal polynomials. Then*

(a) $\deg p_n = n$ *and* $p_n = \varepsilon_n x^n + (\text{lower order terms})$ *with* $\varepsilon_n > 0$*;*

(b) $\operatorname{span}(p_0, \dots, p_n) = \mathcal{P}_n := \{\text{polynomials of degree} \le n\}$*, and* $p_n \perp \mathcal{P}_{n-1}$*;*

(c) the p_n have real coefficients, $p_0 = 1$*;*

(d) $\{p_n\}$ is an ONB of $L^2(\mu)$.

Proof. (b) is property (iii) of Theorem 5.1, and $p_n \perp p_0, \dots, p_{n-1}$ gives $p_n \perp \mathcal{P}_{n-1}$. (a): in the construction, $u_n = x^n - (\text{element of } \mathcal{P}_{n-1})$, so $p_n = u_n / \|u_n\|$ has leading coefficient $\varepsilon_n = 1/\|u_n\| > 0$. (c): $p_0 = 1/\|1\| = 1$ since $\mu(\mathbb{R}) = 1$. If $p_0, \dots, p_{n-1}$ are real, the numbers $(x^n, p_k) = \int x^n p_k\,d\mu$ are real, so u_n and p_n are real. (d): it remains to show completeness, i.e. that $\mathcal{P} = \bigcup_n \mathcal{P}_n = \operatorname{span}\{p_n\}$ is dense in $L^2(\mu)$. Since μ lives on $[-R,R]$, $L^2(\mu) = L^2([-R,R],\mu)$, and $C([-R,R])$ is dense in it (Theorem 2.3). Given $g \in C([-R,R])$, the Weierstrass theorem (Theorem 2.2) gives polynomials $q_j \to g$ uniformly on $[-R,R]$, and $\|g - q_j\|_{L^2(\mu)} \le \max_{[-R,R]} |g - q_j| \to 0$ because μ is a probability measure. □

By (c), $\overline{p_k} = p_k$ on $\mathbb{R}$, so $(f, p_k) = \int f p_k\,d\mu$. A key (trivial but crucial) observation is that multiplication by x is *symmetric*: since x is real,

$$(xf, g) = \int_{\mathbb{R}} x f \overline{g}\,d\mu = (f, xg) \qquad (f, g \in L^2(\mu)), \qquad \text{and} \qquad \|xf\| \le R\,\|f\|\,. \tag{8}$$

Theorem 5.7 (Favard)**.** *Let μ be a probability measure on a compact subset of $\mathbb{R}$ which is not a finite linear combination of δ-masses, and let $\{p_n\}_{n=0}^{\infty}$ be its orthonormal polynomials. Then there are $a_n \in \mathbb{R}$ $(n \ge 0)$ and $b_n > 0$ $(n \ge 1)$ with*

$$x p_n = b_{n+1} p_{n+1} + a_n p_n + b_n p_{n-1}, \qquad n \ge 0, \qquad p_{-1} := 0,\ p_0 = 1, \tag{RR}$$

(a three-term recurrence relation*), and* $\sup_n |a_n| + \sup_n b_n < \infty$.

Conversely, for all bounded sequences $\{a_n\}_{n\ge0} \subset \mathbb{R}$ and $\{b_n\}_{n\ge1} \subset (0,\infty)$, the polynomials defined by (RR) are orthonormal with respect to

some unique probability measure μ, supported on a compact subset of $\mathbb{R}$, which is not a finite linear combination of δ-masses.

Remark 5.8 (Differences from the notebook). (a) The notebook writes $b_n > 0$ for all $n \geq 0$; but b_0 only multiplies $p_{-1} = 0$, so only b_n, $n \geq 1$, matters.

(b) In the converse, the notebook asks only $a_n \in \mathbb{R}$, $b_n > 0$. Boundedness must be added: the proof of "⇒" shows that compact support forces $|a_n|, b_n \leq R$, so unbounded coefficients can never come from a compactly supported μ. (For unbounded coefficients there is still an orthogonality measure on $\mathbb{R}$, but it need not be compactly supported nor unique – this is the *moment problem*.)

(c) The notebook says "orthogonal" in the converse; the proof gives ortho*normal*.

(d) "A sequence of orthonormal polynomials" means the Gram–Schmidt sequence, with $\deg p_n = n$ and positive leading coefficients. (Replacing p_n by $-p_n$ keeps orthonormality but flips the sign of b_n and b_{n+1}.)

Remark 5.9 (Jacobi matrices). (RR) says that the operator $f \mapsto xf$ on $L^2(\mu)$, written in the ONB $\{p_n\}$, is the tridiagonal matrix

$$J = \begin{pmatrix} a_0 & b_1 & & & \\ b_1 & a_1 & b_2 & & \\ & b_2 & a_2 & b_3 & \\ & & b_3 & a_3 & \ddots \\ & & & \ddots & \ddots \end{pmatrix},$$

a *Jacobi matrix*: the n-th column holds the coefficients of xp_n. Favard's theorem says that $\mu \mapsto J$ is a bijection between probability measures with compact infinite support and bounded Jacobi matrices with $b_n > 0$. So questions about measures can be translated into spectral theory of J and back; this is the starting point of the theory of orthogonal polynomials.

Proof of "⇒". Let R with $\operatorname{supp}\mu \subset [-R, R]$. Since $\{p_n\}$ is an ONB (Lemma 5.6), by Riesz–Fischer

$$xp_n = \sum_{k=0}^{\infty} c_k p_k, \qquad c_k = (xp_n, p_k)\,, \qquad \sum_k |c_k|^2 < \infty.$$

Only three terms survive.

- $k \geq n+2$: $xp_n \in \mathcal{P}_{n+1}$ (degree $\leq n+1$) and $p_k \perp \mathcal{P}_{k-1} \supset \mathcal{P}_{n+1}$, so $c_k = 0$.
- $k \leq n-2$: by (8), $c_k = (xp_n, p_k) = (p_n, xp_k)$, and $xp_k \in \mathcal{P}_{k+1} \subset \mathcal{P}_{n-1} \perp p_n$, so $c_k = 0$.

Hence $xp_n = c_{n,1}p_{n+1} + c_{n,2}p_n + c_{n,3}p_{n-1}$ with

$$c_{n,1} = (xp_n, p_{n+1})\,, \qquad c_{n,2} = (xp_n, p_n)\,, \qquad c_{n,3} = (xp_n, p_{n-1}) \quad (n \geq 1),$$

and the p_{n-1} term is absent for $n = 0$. (In fact the expansion is finite from the start, as $xp_n \in \mathcal{P}_{n+1} = \operatorname{span}(p_0, \ldots, p_{n+1})$.)

All coefficients are real. $c_{n,2} = \int x|p_n|^2\, d\mu \in \mathbb{R}$; and $c_{n,1}, c_{n,3}$ are integrals of real functions (p_k real). Set $a_n := c_{n,2} \in \mathbb{R}$.

The matrix is symmetric: $c_{n,3} = c_{n-1,1}$. By (8) and realness,

$$c_{n,3} = (xp_n, p_{n-1}) = (p_n, xp_{n-1}) = \overline{(xp_{n-1}, p_n)} = c_{n-1,1}.$$

So with $b_n := c_{n,3} = (xp_n, p_{n-1})$ for $n \geq 1$ we have $c_{n,1} = b_{n+1}$, which is (RR).

Positivity: $b_n > 0$. $b_n = (p_n, xp_{n-1})$. From the Gram–Schmidt process, $p_k = \varepsilon_k x^k + \dots$ with $\varepsilon_k > 0$ (Lemma 5.6(a)). So $xp_{n-1} = \varepsilon_{n-1}x^n + \dots$, and comparing leading coefficients,

$$xp_{n-1} = \delta_n p_n + (\text{element of } \mathcal{P}_{n-1}), \qquad \delta_n = \frac{\varepsilon_{n-1}}{\varepsilon_n} > 0.$$

As $p_n \perp \mathcal{P}_{n-1}$, $b_n = (p_n, \delta_n p_n) = \delta_n \|p_n\|^2 = \delta_n > 0$.

Boundedness. With $\|p_n\| = 1$ and (8),

$$|a_n| = \Big| \int_{\mathbb{R}} x|p_n|^2 \, d\mu \Big| \le \sup_{x \in \operatorname{supp}\mu} |x| \cdot \|p_n\|^2 \le R,$$
$$|b_n| = |\,(xp_n, p_{n-1})\,| \le \|xp_n\| \, \|p_{n-1}\| \le R \, \|p_n\| \, \|p_{n-1}\| = R,$$

by Cauchy–Schwarz. (The notebook has $\|xp_n\| \cdot \|p_n\|$ in the middle; it should be $\|p_{n-1}\|$, which equals 1 anyway.) □

The converse was only stated in the lecture. Its proof needs the spectral theorem.

Background 10. (Self-adjoint operators and spectral measures)

A bounded operator A on a Hilbert space H (Theorem 2.24) is *self-adjoint* if $(Au, v) = (u, Av)$ for all u, v. For a polynomial $q(x) = \sum_j \gamma_j x^j$ put $q(A) = \sum_j \gamma_j A^j$; if q has real coefficients, $q(A)$ is again self-adjoint, and $(q_1 q_2)(A) = q_1(A)q_2(A)$.
Spectral theorem (form needed here). If A is bounded and self-adjoint and $v \in H$, there is a unique finite positive Borel measure μ_v on $\mathbb{R}$, supported in $[-\|A\|, \|A\|]$, with

$$(q(A)v, v) = \int_{\mathbb{R}} q \, d\mu_v \qquad \text{for every polynomial } q;$$

in particular $\mu_v(\mathbb{R}) = \|v\|^2$. *Idea:* if $q \ge 0$ on $[-\|A\|, \|A\|]$ then $q(A)$ is a positive operator, so $q \mapsto (q(A)v, v)$ is a positive linear functional on polynomials, bounded by $\|v\|^2 \max_{[-\|A\|,\|A\|]} |q|$. It extends by Weierstrass to $C([-\|A\|, \|A\|])$, and the Riesz–Markov representation theorem turns it into a measure. See e.g. Reed–Simon, *Methods of Modern Mathematical Physics I*, Thm. VII.3.
Uniqueness of measures. Two finite Borel measures on $[-R, R]$ with $\int g \, d\mu = \int g \, d\nu$ for all $g \in C([-R, R])$ coincide (uniqueness in Riesz–Markov); by Weierstrass it suffices to check this for all polynomials g.

Proof of "⇐". Let $\{a_n\}_{n\ge 0} \subset \mathbb{R}$ and $\{b_n\}_{n \ge 1} \subset (0, \infty)$ be bounded, $|a_n| \le A_0$, $b_n \le B_0$.

The polynomials. (RR) defines recursively $p_{n+1} = \frac{1}{b_{n+1}}((x - a_n)p_n - b_n p_{n-1})$ with $p_{-1} = 0$, $p_0 = 1$ (and $b_0 p_{-1} = 0$). By induction p_n has real coefficients, $\deg p_n = n$ and leading coefficient $\frac{1}{b_1 \cdots b_n} > 0$.

The Jacobi operator. On $\ell^2(\mathbb{N}_0)$ with standard basis $\{e_n\}_{n \ge 0}$ define

$$(Jv)_n = b_n v_{n-1} + a_n v_n + b_{n+1} v_{n+1} \qquad (v_{-1} := 0), \qquad \text{so} \qquad Je_n = b_{n+1}e_{n+1} + a_n e_n + b_n e_{n-1}$$

$(e_{-1} := 0)$. J is bounded: it is the sum of a diagonal operator and two weighted shifts, so $\|Jv\| \le (A_0 + 2B_0)\|v\|$. It is self-adjoint: $(Je_n, e_m) = (e_n, Je_m)$ for all n, m (both equal the real symmetric matrix entry J_{mn}), hence by linearity and continuity $(Ju, v) = (u, Jv)$ for all u, v.

Key identity: $p_n(J)e_0 = e_n$. True for $n = 0$ ($p_0 = 1$) and, with $p_{-1}(J)e_0 = 0 = e_{-1}$, for $n = -1$. If it holds for $n - 1$ and n, then by (RR)

$$p_{n+1}(J)e_0 = \frac{1}{b_{n+1}}\big((J - a_n)p_n(J)e_0 - b_n p_{n-1}(J)e_0\big) = \frac{1}{b_{n+1}}(Je_n - a_n e_n - b_n e_{n-1}) = e_{n+1}.$$

Existence. Let $\mu := \mu_{e_0}$ be the spectral measure of J and e_0 (Theorem 5.10). It is supported in $[-\|J\|, \|J\|]$, compact, and $\mu(\mathbb{R}) = \|e_0\|^2 = 1$. Since p_n has real coefficients, $p_n(J)$ is self-adjoint, and

$$\int_{\mathbb{R}} p_n p_m \, d\mu = ((p_n p_m)(J)e_0, e_0) = (p_m(J)e_0, p_n(J)e_0) = (e_m, e_n) = \delta_{mn}.$$

So $\{p_n\}$ is orthonormal in $L^2(\mu)$. In particular $L^2(\mu)$ contains infinitely many orthonormal vectors, so it is infinite-dimensional; if $\mu = \sum_{k=1}^{N} c_k \delta_{x_k}$, then $L^2(\mu)$ would have dimension $\le N$. Hence μ is not a finite combination of δ-masses. Moreover, by uniqueness in Theorem 5.1 (orthonormal, $\deg p_n = n$, positive leading coefficient), $\{p_n\}$ is exactly the Gram–Schmidt sequence of μ.

Uniqueness. Let ν be another compactly supported probability measure for which $\{p_n\}$ is orthonormal. Then, as $p_0 = 1$,

$$\int p_n \, d\nu = (p_n, p_0)_{L^2(\nu)} = \delta_{n0} = \int p_n \, d\mu \qquad \text{for all } n.$$

The p_n span all polynomials, so $\int q \, d\nu = \int q \, d\mu$ for every polynomial q. Both measures live on some common $[-R, R]$, so $\nu = \mu$ by the uniqueness statement in Theorem 5.10. □

Example 5.11 (Chebyshev and Legendre polynomials)**.** (a) $d\mu = \frac{dx}{\pi\sqrt{1-x^2}}$ on $[-1, 1]$ (the *arc-sine* distribution). With $x = \cos\theta$, $T_n(x) = \cos(n\theta)$ is a polynomial of degree n (Chebyshev polynomial) and $\int T_n T_m \, d\mu = \frac{1}{\pi}\int_0^\pi \cos(n\theta)\cos(m\theta)\, d\theta$, which is 1 for $n = m = 0$, $\frac{1}{2}$ for $n = m \ge 1$, and 0 otherwise. So $p_0 = 1$, $p_n = \sqrt{2}\, T_n$. The identity $2\cos\theta\cos(n\theta) = \cos((n+1)\theta) + \cos((n-1)\theta)$ gives $xT_n = \frac{1}{2}(T_{n+1} + T_{n-1})$, hence

$$a_n = 0, \qquad b_1 = \frac{1}{\sqrt{2}}, \qquad b_n = \frac{1}{2} \ (n \ge 2).$$

(For $n = 0$: $x = T_1 = \frac{1}{\sqrt{2}} p_1$; for $n = 1$: $xp_1 = \sqrt{2}\,\frac{T_2 + T_0}{2} = \frac{1}{2}p_2 + \frac{1}{\sqrt{2}}p_0$.)

(b) $d\mu = \frac{dx}{2}$ on $[-1, 1]$: $p_n = \sqrt{2n+1}\, P_n$ with the Legendre polynomials P_n, and from Bonnet's recursion $(n+1)P_{n+1} = (2n+1)xP_n - nP_{n-1}$ one gets $a_n = 0$, $b_n = \frac{n}{\sqrt{4n^2-1}}$. E.g. $b_1 = \frac{1}{\sqrt{3}}$: indeed $x = \frac{1}{\sqrt{3}} p_1$ with $p_1 = \sqrt{3}x$.

In both cases $a_n = 0$ because μ is symmetric (p_n is even or odd, so $x|p_n|^2$ is odd), and $b_n \to \frac{1}{2}$. This is no accident: if $\operatorname{supp}\mu = [-1, 1]$ and the density of μ is positive a.e. on $[-1, 1]$, then $a_n \to 0$, $b_n \to \frac{1}{2}$ (Rakhmanov's theorem).

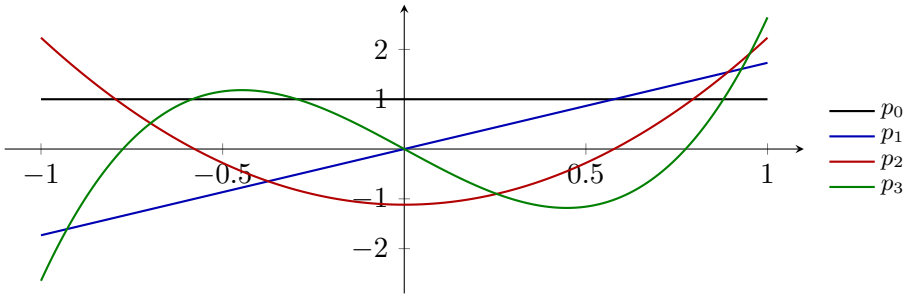


Figure 14: The first orthonormal polynomials of $d\mu = \frac{dx}{2}$ on $[-1, 1]$ (normalized Legendre polynomials). p_n has n simple zeros in $(-1, 1)$, and the zeros of p_n and p_{n+1} interlace. Both facts follow from (RR).

6 Summary of theorems

This chapter lists, in the order of the course, the results that were stated in the lectures or proved on the problem sheets (nothing added only in these notes). Each one is stated completely, with all its hypotheses, followed by a short note on why it matters; the link on the right points to the proof. The *key results*, the ones to know by heart, are in blue boxes.

Notation. $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$, m is the normalized arc length on $\mathbb{T}$ ($m(\mathbb{T}) = 1$), $\int_{\mathbb{T}} f\,dm = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(e^{it})\,dt$. For $f \in L^1(\mathbb{T})$: $\hat{f}(k) = \int_{\mathbb{T}} f(z)\overline{z}^k\,dm(z)$, $S_N(f,z) = \sum_{k=-N}^{N} \hat{f}(k)z^k$, $\sigma_N(f,z) = \frac{1}{N+1}\sum_{n=0}^{N} S_n(f,z)$. For a Hilbert space H with an orthonormal system $\{e_k\}$: $\hat{h}(k) = (h, e_k)$.

Hilbert spaces

T1 $\|\cdot\|$ is a norm. If $(\cdot,\cdot)$ is an inner product on a complex vector space H, then $\|x\| = \sqrt{(x,x)}$ is a norm on H; in particular $\|x+y\| \le \|x\| + \|y\|$. [→ Proposition 1.6]

Why it matters. Every inner product space is automatically a normed (metric) space, so we can talk about convergence, Cauchy sequences and completeness, i.e. about Hilbert spaces.

T2 Cauchy–Schwarz inequality key result

Let H be an inner product space. For all $x, y \in H$

$$|(x,y)| \le \|x\|\,\|y\|\,,$$

with equality if and only if x and y are linearly dependent. [→ Theorem 1.7]

Why it matters. The basic inequality of the whole course: it gives the triangle inequality, continuity of the inner product, Bessel's inequality, $L^2 \subset L^1$ on finite measure spaces, and countless estimates of Fourier coefficients.

T3 Continuity of the inner product. If $x_N \to x$ in a Hilbert space H, then $(x_N, y) \to (x, y)$ for every $y \in H$ and $\|x_N\| \to \|x\|$. [→ Lemma 1.28]

Why it matters. It allows us to exchange limits (and infinite sums) with inner products, e.g. to compute $(\sum_k c_k e_k, e_j) = c_j$ for convergent series.

T4 Orthonormal systems in separable spaces are countable. Let H be a separable Hilbert space and $(e_i)_{i\in I}$ an orthonormal system in H. Then I is at most countable. [→ Problem 1]

Why it matters. Justifies indexing every ONB by $\mathbb{N}$ (or $\mathbb{Z}$) and working with ordinary series; all spaces in this course (ℓ^2, $L^2(\mathbb{T})$, $L^2(\mu)$) are separable.

T5 Bessel's inequality. Let $\{e_k\}_{k=1}^{\infty}$ be an orthonormal system (not necessarily complete) in a Hilbert space H. Then for every $h \in H$

$$\sum_{k=1}^{\infty} |(h, e_k)|^2 \le \|h\|^2\,.$$

Moreover, $P_N h = \sum_{k=1}^{N} (h, e_k)\,e_k$ is the orthogonal projection of h onto $\operatorname{span}\{e_1, \dots, e_N\}$. [→ Eq. (4)]

Why it matters. The coefficients of any vector form an ℓ^2 sequence; in particular $(h, e_k) \to 0$. It is the first step of Riesz–Fischer and is used to prove $\hat{f}(n) \to 0$.

T6 Riesz–Fischer theorem and Parseval's identity key result

Let H be a separable Hilbert space with an ONB $\{e_k\}_{k=1}^{\infty}$. Then for every $h \in H$

$$h = \sum_{k=1}^{\infty} \hat{h}(k) e_k \quad \text{(convergence in the norm of } H\text{)}, \qquad \|h\|^2 = \sum_{k=1}^{\infty} \left|\hat{h}(k)\right|^2.$$

Conversely, for every $(c_k) \in \ell^2(\mathbb{N})$ the series $\sum_k c_k e_k$ converges in H to some h with $\hat{h}(k) = c_k$. Hence $h \mapsto (\hat{h}(k))_k$ is a bijective linear isometry $H \to \ell^2(\mathbb{N})$. [→ Theorem 1.26]

Why it matters. This is the abstract Fourier expansion: every vector is the sum of its projections, and the norm is "Pythagoras in infinite dimensions". For $H = L^2(\mathbb{T})$ it says $f = \sum \hat{f}(k) z^k$ in L^2 and $\|f\|_2^2 = \sum |\hat{f}(k)|^2$. It is the reason L^2 is the easiest space for Fourier series – convergence in L^2 is free, pointwise convergence is not.

T7 Characterizations of an ONB. Let $(e_n)_{n=1}^{\infty}$ be an orthonormal system in a Hilbert space H. The following are equivalent: (i) (e_n) is an ONB (i.e. complete: $\operatorname{clos}\operatorname{span}\{e_n\} = H$); (ii) if $(h, e_k) = 0$ for all k, then $h = 0$; (iii) $h = \sum_k (h, e_k)\, e_k$ for every $h \in H$; (iv) $\|h\|^2 = \sum_k |(h, e_k)|^2$ for every $h \in H$. [→ Problem 1]

Why it matters. Gives several practical ways to check completeness; (ii) is usually the easiest to verify, (iv) is the equality case of Bessel's inequality.

T8 Euler's formula and its relatives. $\displaystyle\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}$ and $\displaystyle\sum_{k=1}^{\infty} \frac{1}{k^4} = \frac{\pi^4}{90}$ (Parseval applied to $f(x) = x$ and $f(x) = x^2$ on $[-\pi, \pi]$). [→ Example 1.31]

Why it matters. The model application of Parseval's identity: computing a norm in two ways evaluates a numerical series.

Classical Fourier series

T9 $L^2 \subset L^1$ on finite measure spaces. If (X, μ) is a measure space with $\mu(X) < \infty$, then $L^2(X, \mu) \subseteq L^1(X, \mu)$ and $\|f\|_{L^1} \le \mu(X)^{1/2} \|f\|_{L^2}$. In particular $\hat{f}(k)$ is well defined for $f \in L^2(\mathbb{T}, m)$ and $|\hat{f}(k)| \le \|f\|_{L^1} \le \|f\|_{L^2}$. [→ Problem 2]

Why it matters. Fourier coefficients make sense for every $f \in L^1(\mathbb{T})$, a larger class than $L^2(\mathbb{T})$; the inclusion is strict (e.g. $|x|^{-1/2}$).

T10 Density of continuous functions. $C(\mathbb{T})$ is dense in $L^2(\mathbb{T}, m)$: for every $f \in L^2(\mathbb{T})$ and $\varepsilon > 0$ there is $g \in C(\mathbb{T})$ with $\|f - g\|_{L^2} < \varepsilon$. [→ Problem 3]

Why it matters. Lets us prove statements for nice (continuous) functions and extend them to all of L^2 by approximation.

T11 Trigonometric polynomials; the exponential ONB. Trigonometric polynomials $\sum_{k=-N}^{N} a_k z^k$ are dense in $C(\mathbb{T})$ with respect to the sup-norm (Stone–Weierstrass, or Fejér), and $\{z^k\}_{k\in\mathbb{Z}}$ is an ONB of $L^2(\mathbb{T}, m)$; equivalently $\{e^{ikx}\}_{k\in\mathbb{Z}}$ is an ONB of $L^2([-\pi, \pi], \frac{dx}{2\pi})$. [→ Corollary 2.19]

Why it matters. This makes Riesz–Fischer applicable to $L^2(\mathbb{T})$ and so starts classical Fourier analysis: every $f \in L^2(\mathbb{T})$ equals its Fourier series in the L^2 sense.

T12 Carleson–Hunt theorem (stated without proof). If $f \in L^p(\mathbb{T}, m)$ for some $p \in (1, \infty]$, then $S_N(f, z) \to f(z)$ for m-almost every $z \in \mathbb{T}$. [→ Theorem 2.5]

Why it matters. The deep answer to the question of pointwise convergence. For $p = 1$ it fails (Kolmogorov), and for continuous functions "almost everywhere" cannot be replaced by "everywhere" (du Bois-Reymond).

T13 Fejér kernel representation. For $f \in L^1(\mathbb{T})$, $z \in \mathbb{T}$ and $N \geq 0$,

$$\sigma_N(f,z) = \int_{\mathbb{T}} f(\xi) F_N(\xi, z)\, dm(\xi), \qquad F_N(\xi,z) = \frac{1}{N+1}\Big|\sum_{s=0}^{N} (\bar{\xi} z)^s\Big|^2.$$

[→ Lemma 2.10]

Why it matters. Turns the Cesàro means into an integral against a *non-negative* kernel; positivity is exactly what the Dirichlet kernel (for S_N) lacks, and why Cesàro means behave better.

T14 Approximate identities. Let (X, ρ) be a metric space with a Borel measure μ and (p_n) an approximate identity centred at x_0 (i.e. $p_n \geq 0$, $\int_X p_n\, d\mu = 1$, and $\int_{X \setminus B(x_0,\delta)} p_n\, d\mu \to 0$ for every $\delta > 0$). Then for every bounded measurable f on X that is continuous at x_0,

$$\int_X f\, p_n\, d\mu \xrightarrow{n\to\infty} f(x_0).$$

[→ Lemma 2.14]

Why it matters. The general mechanism behind Fejér's theorem: kernels that concentrate their mass at a point act like the Dirac delta δ_{x_0}. The same scheme reappears for the Poisson kernel, heat kernel, mollifiers, ...

T15 Fejér's theorem key result

For every $f \in C(\mathbb{T})$, $\sigma_N(f,z) \to f(z)$ as $N \to \infty$, for every $z \in \mathbb{T}$, and even uniformly on $\mathbb{T}$. In other words, the Fourier series of a continuous function converges to it in the Cesàro sense everywhere. [→ Theorem 2.16]

Why it matters. The first positive pointwise result: a continuous function is recovered from its Fourier coefficients at *every* point. Consequences: trigonometric polynomials are uniformly dense in $C(\mathbb{T})$ (Weierstrass), $\{z^k\}$ is an ONB of $L^2(\mathbb{T})$, and if $S_N(f,z)$ converges at all, its limit is $f(z)$.

T16 Fejér's theorem: refinements. Let f be bounded and measurable on $\mathbb{T}$ and $z_0 = e^{it_0} \in \mathbb{T}$. (a) If f is continuous at z_0, then $\sigma_N(f,z_0) \to f(z_0)$. (b) If the one-sided limits $f(z_0 \pm 0) = \lim_{\varepsilon \downarrow 0} f(e^{i(t_0 \pm \varepsilon)})$ exist, then $\sigma_N(f,z_0) \to \frac{f(z_0-0)+f(z_0+0)}{2}$. [→ Remark 2.22]

Why it matters. At a jump, the Fourier series "chooses the midpoint". This is the Cesàro half of Hardy's theorem.

T17 Banach–Steinhaus theorem (uniform boundedness principle). Let X be a Banach space, Y a normed space and $\{T_\alpha\}_{\alpha \in A}$ a family of bounded linear operators $X \to Y$ with $\sup_\alpha \|T_\alpha x\| < \infty$ for every $x \in X$. Then $\sup_\alpha \|T_\alpha\| < \infty$. [→ Theorem 2.25]

Why it matters. Pointwise boundedness implies uniform boundedness. Contrapositive use: if the norms $\|T_\alpha\|$ are unbounded, then some single x makes $T_\alpha x$ unbounded – a non-constructive way to produce counterexamples.

T18 du Bois-Reymond theorem. There exists $f \in C(\mathbb{T})$ such that $S_N(f,z)$ does not converge at some point $z \in \mathbb{T}$. (The functionals $f \mapsto S_N(f,1)$ on $C(\mathbb{T})$ have norms $\geq c \log N \to \infty$.) [→ Theorem 2.23]

Why it matters. Continuity is *not* enough for pointwise convergence of Fourier series, so the Cesàro means in Fejér's theorem cannot be dropped; extra regularity (e.g. bounded variation) is needed.

T19 Basic properties of BV. $BV([a,b])$ is a vector space closed under products; every $g \in BV([a,b])$ is bounded and has one-sided limits $g(x_0 \pm 0)$ at every point; $C^1([a,b]) \subset BV([a,b])$; and $f \in BV(\mathbb{T}) \iff t \mapsto f(e^{it}) \in BV([0,2\pi])$. [→ Problem 6]

Why it matters. BV is the natural class for Hardy's theorem: functions with jumps are allowed, but they are "tame".

T20 Jordan decomposition. $g\colon [a,b] \to \mathbb{C}$ is of bounded variation if and only if $g = g_1 - g_2 + i(g_3 - g_4)$ with bounded nondecreasing $g_1, \dots, g_4\colon [a,b] \to \mathbb{R}$. For real g one can take $g_1 = V$, $g_2 = V - g$ with $V(x) = \mathrm{Var}_{[a,x]}(g)$. [→ Theorem 2.30]

Why it matters. Reduces questions about BV functions to monotone functions (one-sided limits, countably many jumps).

T21 BV functions and measures. $g \in BV([a,b])$ and g is right-continuous if and only if $g(x) = g(a) + \mu((a,x])$ for all $x \in [a,b]$, for some complex Borel measure μ on $(a,b]$ of finite total variation. [→ Theorem 2.32]

Why it matters. "The derivative of a BV function is a finite measure", which is what allows the integration by parts giving $\hat{f}(n) = O(\frac{1}{|n|})$. The Cantor function shows that μ need not have a density.

T22 Hardy's Tauberian theorem. If $\sum_{n\geq 1} a_n$ converges in the Cesàro sense and $|a_k| \leq \frac{C}{k}$ for all $k \geq 1$, then $\sum_{n\geq 1} a_n$ converges in the usual sense, to the same sum. [→ Lemma 2.37]

Why it matters. A model Tauberian theorem: weak (averaged) convergence plus a size condition on the terms gives genuine convergence. It converts Fejér-type results into results about S_N.

T23 Hardy's theorem (Dirichlet–Jordan test) key result

Let $f \in BV(\mathbb{T})$. Then for every $z \in \mathbb{T}$

$$\lim_{N\to\infty} S_N(f,z) = \frac{f(z-0) + f(z+0)}{2};$$

in particular $S_N(f,z) \to f(z)$ at every point z where f is continuous. [→ Theorem 2.36]

Why it matters. The main pointwise convergence theorem of the course: for functions of bounded variation the Fourier series converges *everywhere*, without Cesàro means, to the midpoint of the jump. Proof scheme: Fejér + ($\hat{f}(n) = O(\frac{1}{|n|})$) + Hardy's Tauberian theorem.

T24 Fourier coefficients of a product. If $f, g \in L^2(\mathbb{T}, m)$, then $\widehat{fg}(n) = \sum_{k\in\mathbb{Z}} \hat{f}(k)\hat{g}(n-k)$ for every $n \in \mathbb{Z}$, the series converging absolutely. [→ Problem 5]

Why it matters. Multiplication of functions corresponds to convolution of coefficient sequences – the first instance of the duality "product ↔ convolution" of Fourier analysis.

Fourier coefficients of various classes of functions

T25 Smoothness gives decay (Proposition 1) key result

Let $k \geq 0$ and $f \in C^k(\mathbb{T})$. Then $\hat{f}(n) = O(\frac{1}{|n|^k})$, in fact $\hat{f}(n) = o(\frac{1}{|n|^k})$, as $|n| \to \infty$. More precisely, for $h(t) = f(e^{it})$: $\widehat{h^{(k)}}(n) = (in)^k \hat{f}(n)$. [→ Proposition 3.3]

Why it matters. Differentiation becomes multiplication by in on the Fourier side; hence every derivative of f buys one power of $\frac{1}{|n|}$ of decay.

T26 Decay gives smoothness (Proposition 2) key result

Let $k \geq 0$, $f \in L^1(\mathbb{T})$ and $\sum_{n\in\mathbb{Z}} |n|^k |\hat{f}(n)| < \infty$. Then f coincides a.e. with a function in $C^k(\mathbb{T})$, and $f(z) = \sum_{n\in\mathbb{Z}} \hat{f}(n) z^n$ for every z, the series converging absolutely and uniformly. [→ Proposition 3.4]

Why it matters. The converse direction. There is a gap of one derivative: $C^{k+1} \Rightarrow \sum |n|^k |\hat{f}(n)| < \infty \Rightarrow C^k$. In particular the Fourier series of an $f \in C^1(\mathbb{T})$ converges absolutely and uniformly to f.

T27 Characterization of C^∞ (Proposition 3). Let $f \in L^1(\mathbb{T})$. Then $f \in C^\infty(\mathbb{T})$ (after a change on a null set) if and only if $\hat{f}(n) = O(\frac{1}{|n|^k})$ as $|n| \to \infty$ for every $k \geq 0$. [→ Proposition 3.6]

Why it matters. For infinitely smooth functions the gap disappears: smooth functions are exactly those with rapidly decreasing coefficients.

T28 Riemann–Lebesgue lemma (Proposition 4) key result

If $f \in L^1(\mathbb{T})$, then $\hat{f}(n) \to 0$ as $|n| \to \infty$. [→ Proposition 3.7]

Why it matters. The minimal information about Fourier coefficients of an arbitrary integrable function. No rate of decay can be guaranteed. The proof (approximate by a trigonometric polynomial, use $|\hat{f}(n) - \hat{p}(n)| \leq \|f - p\|_1$) is a template for density arguments.

T29 L^2 and coefficients (Proposition 5). Let $f \in L^1(\mathbb{T})$. Then $f \in L^2(\mathbb{T})$ if and only if $\sum_{n\in\mathbb{Z}} |\hat{f}(n)|^2 < \infty$, and then $\|f\|_2^2 = \sum |\hat{f}(n)|^2$. Every $(c_n) \in \ell^2(\mathbb{Z})$ is the coefficient sequence of some $f \in L^2(\mathbb{T})$. [→ Proposition 3.9]

Why it matters. $L^2(\mathbb{T})$ is exactly characterized on the Fourier side, unlike $C(\mathbb{T})$ or $L^1(\mathbb{T})$: $f \mapsto \hat{f}$ is an isometric isomorphism $L^2(\mathbb{T}) \to \ell^2(\mathbb{Z})$.

T30 BV and coefficients (Proposition 6). If $f \in BV(\mathbb{T})$, then $|\hat{f}(n)| \leq \frac{\mathrm{Var}_{\mathbb{T}}(f)}{4|n|}$ for $n \neq 0$; in particular $\hat{f}(n) = O(\frac{1}{|n|})$. [→ Lemma 2.39]

Why it matters. This is the size condition needed in Hardy's Tauberian theorem; it is sharp, as $\operatorname{sgn} t$ shows.

The classical form of Fourier series

T31 Fourier series with sines and cosines. Let $f\colon \mathbb{R} \to \mathbb{C}$ be 2π-periodic with $f \in L^2([-\pi,\pi])$. Then, with convergence in $L^2([-\pi,\pi])$,

$$f(x) = a_0 + \sum_{k=1}^{\infty} (a_k \cos(kx) + b_k \sin(kx)),$$

$$a_0 = \tfrac{1}{2\pi}\int_{-\pi}^{\pi} f, \quad a_k = \tfrac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(kx)\,dx, \quad b_k = \tfrac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin(kx)\,dx.$$

[→ Theorem 4.1]

Why it matters. The real form of the same expansion ($a_k = \hat{f}(k) + \hat{f}(-k)$, $b_k = i(\hat{f}(k) - \hat{f}(-k))$), convenient for real, even or odd functions: even f has only cosines, odd f only sines.

Gram–Schmidt and orthogonal polynomials

T32 Gram–Schmidt orthonormalization. Let $\{h_k\}_{k=0}^{\infty}$ be a linearly independent sequence in a Hilbert space H. Then there is a sequence $\{e_k\}_{k=0}^{\infty}$ with $\|e_k\| = 1$, $e_k \perp e_j$ for $k \neq j$, and $\operatorname{span}(e_0,\dots,e_k) = \operatorname{span}(h_0,\dots,h_k)$ for all k; explicitly $e_n = u_n/\|u_n\|$ with $u_n = h_n - \sum_{k<n} (h_n, e_k)\, e_k$. [→ Theorem 5.1]

Why it matters. Turns any independent sequence into an orthonormal one without changing the spans; applied to $1, x, x^2, \dots$ it produces orthogonal polynomials.

T33 Existence of ONB. Every separable Hilbert space has an ONB (apply Gram–Schmidt to a dense sequence with dependent vectors removed). [→ Corollary 5.2]

Why it matters. Together with Riesz–Fischer: every separable infinite-dimensional Hilbert space is isometrically isomorphic to ℓ^2.

T34 Independence of monomials. Let μ be a compactly supported probability measure on $\mathbb{R}$. The monomials $\{x^k\}_{k\geq 0}$ are linearly independent in $L^2(\mu)$ if and only if μ is not a finite linear combination of Dirac masses. [→ Lemma 5.5]

Why it matters. Exactly the condition under which the orthonormal polynomials p_n (Gram–Schmidt of the monomials x^n) exist for all n.

T35 Favard's theorem key result

Let μ be a probability measure on a compact subset of $\mathbb{R}$ which is not a finite linear combination of δ-masses, and let $\{p_n\}_{n\geq 0}$ be its orthonormal polynomials (Gram–Schmidt of $1, x, x^2, \dots$ in $L^2(\mu)$). Then there are $a_n \in \mathbb{R}$ and $b_n > 0$, both bounded sequences, with

$$xp_n = b_{n+1}p_{n+1} + a_n p_n + b_n p_{n-1} \quad (n \geq 0), \qquad p_{-1} = 0,\ p_0 = 1.$$

Conversely, for all bounded sequences $\{a_n\}_{n\geq 0} \subset \mathbb{R}$, $\{b_n\}_{n\geq 1} \subset (0, \infty)$ the polynomials defined by this recurrence are orthonormal with respect to a unique probability measure μ with compact support which is not a finite linear combination of δ-masses. [→ Theorem 5.7]

Why it matters. Orthogonal polynomials are characterized by a three-term recurrence, so measures correspond one-to-one to bounded Jacobi matrices. Questions about a measure become questions about the spectral theory of an operator, and back.

7 Exercises

The problem sheets of the course, with detailed solutions. The sheets write $\langle \cdot, \cdot \rangle$ for the inner product and χ_A for the indicator function of a set A; we keep their notation here, so $\langle h, e \rangle = (h, e)$ in the notation of the lectures.

7.1 List 1 (13. 10. 2026)

Problem 1. Let H be a (complex, separable) Hilbert space.

(a) Let $(e_i)_{i \in I}$ be an ONS (orthonormal system). Using separability, prove that I is at most countable. *Hint.* Show that $\|e_i - e_j\| = \sqrt{2}$ for $i \neq j$, and use a countable dense subset of H.

(b) Let $e_1, \dots, e_N$ be orthonormal, and put

$$V_N = \operatorname{span}\{e_1, \dots, e_N\}, \qquad P_N h = \sum_{k=1}^{N} \langle h, e_k \rangle e_k.$$

Verify that $P_N h$ is the orthogonal projection of h onto V_N, and then use the Pythagorean theorem to deduce Bessel's inequality: $\sum_{k=1}^{\infty} |\langle h, e_k \rangle|^2 \leq \|h\|^2$.

(c) Let $(e_n)_{n=1}^{\infty}$ be an ONS. Prove that the following are equivalent:

(i) $(e_n)_{n=1}^{\infty}$ is an ONB;
(ii) $\langle h, e_k \rangle = 0$ for all $k \geq 1$ implies $h = 0$;
(iii) for every $h \in H$, $h = \sum_{k=1}^{\infty} \langle h, e_k \rangle e_k$ in H;
(iv) for every $h \in H$, $\|h\|^2 = \sum_{k=1}^{\infty} |\langle h, e_k \rangle|^2$.

Hint. For (iv)⇒(i), you may use the Pythagorean theorem.

Solution. **(a)** This is Exercise 1.22; we repeat the argument for an arbitrary ONS (completeness is not needed). For $i \neq j$, by orthogonality (Pythagoras),

$$\|e_i - e_j\|^2 = \langle e_i - e_j, e_i - e_j \rangle = \|e_i\|^2 - \langle e_i, e_j \rangle - \langle e_j, e_i \rangle + \|e_j\|^2 = 1 - 0 - 0 + 1 = 2.$$

Let $D \subset H$ be countable and dense (it exists since H is separable, Theorem 1.21). For each $i \in I$ the open ball $B_i = B(e_i, \frac{\sqrt{2}}{2})$ contains some point of D (density); choose one and call it d_i. If $d_i = d_j$ for $i \neq j$, then $\|e_i - e_j\| \leq \|e_i - d_i\| + \|d_j - e_j\| < \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} = \sqrt{2}$, a contradiction. So $i \mapsto d_i$ is an injective map $I \to D$, and I is at most countable.

(b) *Orthogonal projection.* Recall that the orthogonal projection of h onto a subspace V is the (unique) vector $v \in V$ with $h - v \perp V$, i.e. $\langle h - v, w \rangle = 0$ for all $w \in V$. Uniqueness: if v, v' both have this property then $v - v' \in V$ and $v - v' = (h - v') - (h - v) \perp V$, so $\langle v - v', v - v' \rangle = 0$.

Clearly $P_N h \in V_N$. For $j \leq N$, by orthonormality,

$$\langle h - P_N h, e_j \rangle = \langle h, e_j \rangle - \sum_{k=1}^{N} \langle h, e_k \rangle \langle e_k, e_j \rangle = \langle h, e_j \rangle - \langle h, e_j \rangle = 0,$$

and by linearity (and conjugate linearity in the second slot) $\langle h - P_N h, w \rangle = 0$ for every $w \in V_N$. Hence $P_N h$ is the orthogonal projection of h onto V_N. It is also the point of V_N closest to h: for $v \in V_N$, $P_N h - v \in V_N$, so by Pythagoras $\|h - v\|^2 = \|h - P_N h\|^2 + \|P_N h - v\|^2 \geq \|h - P_N h\|^2$.

Bessel. Since $h = P_N h + (h - P_N h)$ with $P_N h \perp h - P_N h$, the Pythagorean theorem gives

$$\|h\|^2 = \|P_N h\|^2 + \|h - P_N h\|^2 \geq \|P_N h\|^2 = \Big\|\sum_{k=1}^{N} \langle h, e_k\rangle e_k\Big\|^2 = \sum_{k=1}^{N} |\langle h, e_k\rangle|^2,$$

the last equality again by orthonormality. Now let $e_1, e_2, \ldots$ be an infinite ONS: the inequality holds for every N, the partial sums of $\sum_k |\langle h, e_k\rangle|^2$ are nondecreasing and bounded by $\|h\|^2$, so the series converges and $\sum_{k=1}^{\infty} |\langle h, e_k\rangle|^2 \leq \|h\|^2$. (Compare with the proof of Theorem 1.26, Step 1, where the same inequality was obtained from Cauchy–Schwarz.)

(c) Throughout, $P_N h = \sum_{k=1}^{N} \langle h, e_k\rangle e_k$ as in (b), and we use $\|h\|^2 = \|P_N h\|^2 + \|h - P_N h\|^2$ and $\|P_N h\|^2 = \sum_{k \leq N} |\langle h, e_k\rangle|^2$.

(i)⇒(ii). Let $\langle h, e_k\rangle = 0$ for all k. Then $\langle h, w\rangle = 0$ for all $w \in \operatorname{span}\{e_k\}$. If $w_n \to w$, then $\langle h, w\rangle = \lim \langle h, w_n\rangle = 0$ (continuity of the inner product, Lemma 1.28); so $\langle h, w\rangle = 0$ for all $w \in \operatorname{clos}\operatorname{span}\{e_k\} = H$. Take $w = h$: $\|h\|^2 = 0$, so $h = 0$.

(ii)⇒(iii). Let $h \in H$. For $N > M$, $\|P_N h - P_M h\|^2 = \sum_{k=M+1}^{N} |\langle h, e_k\rangle|^2$, which tends to 0 as $M \to \infty$ by Bessel's inequality (tails of a convergent series). So $(P_N h)$ is Cauchy, and since H is complete it has a limit $\tilde{h} = \sum_{k=1}^{\infty} \langle h, e_k\rangle e_k$. For fixed k and $N \geq k$, $\langle P_N h, e_k\rangle = \langle h, e_k\rangle$; letting $N \to \infty$, $\langle \tilde{h}, e_k\rangle = \langle h, e_k\rangle$. Thus $\langle h - \tilde{h}, e_k\rangle = 0$ for all k, and (ii) gives $h = \tilde{h}$.

(iii)⇒(iv). By (iii), $P_N h \to h$, hence $\|P_N h\| \to \|h\|$ (Lemma 1.28), i.e. $\sum_{k=1}^{N} |\langle h, e_k\rangle|^2 = \|P_N h\|^2 \to \|h\|^2$.

(iv)⇒(i). The system is orthonormal by assumption; we must show completeness. By Pythagoras and (iv),

$$\|h - P_N h\|^2 = \|h\|^2 - \|P_N h\|^2 = \|h\|^2 - \sum_{k=1}^{N} |\langle h, e_k\rangle|^2 \xrightarrow{N\to\infty} 0.$$

So $h = \lim P_N h$ is a limit of elements of $\operatorname{span}\{e_k\}$, i.e. $h \in \operatorname{clos}\operatorname{span}\{e_k\}$. As h was arbitrary, $\operatorname{clos}\operatorname{span}\{e_k\} = H$ and (e_n) is an ONB.

Remark 7.1. Problem 1(c) is a reformulation of the Riesz–Fischer theorem (Theorem 1.26) together with Corollary 1.29: (i)⇒(iii),(iv) is Theorem 1.26, and (ii) is the practical test for completeness.

Problem 2. Let (X, μ) be a finite measure space.

(a) Prove that $L^2(X, \mu) \subseteq L^1(X, \mu)$ and that $\|f\|_{L^1} \leq \mu(X)^{1/2} \|f\|_{L^2}$. In particular, explain why the Fourier coefficients of functions in $L^2(\mathbb{T}, m)$ are well defined, where m is the normalized arc length measure defined in the lectures, given by $\int_{\mathbb{T}} f(z)\, dm(z) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(e^{ix})\, dx$.

(b) For $\alpha \geq 0$ consider $f_\alpha(x) = |x|^{-\alpha}$, $x \in (-1, 1) \setminus \{0\}$. Determine the values of α for which $f_\alpha \in L^1(-1, 1)$.

(c) Let $f \in L^2(X, \mu)$ and set $f_M = f \cdot \chi_{\{|f| \leq M\}}$, $M > 0$. Prove that $f_M \to f$ in $L^2(X, \mu)$ as $M \to \infty$. *Hint.* Apply DCT (the dominated convergence theorem) to $|f - f_M|^2$.

(d) On $(0, 1)$ with Lebesgue measure, consider $f_n = \sqrt{n}\, \chi_{(0,1/n)}$. Show that $f_n \to 0$ almost everywhere but not in L^2. Explain why DCT cannot be used here to conclude L^2 convergence.

Background 2. (L^1 and the convergence theorems)

$L^1(X, \mu)$ is the space of (classes of a.e. equal) measurable f with $\|f\|_{L^1} = \int_X |f|\, d\mu < \infty$; for such f the integral $\int_X f\, d\mu \in \mathbb{C}$ is defined and $\left|\int f\, d\mu\right| \leq \int |f|\, d\mu$.
Monotone convergence theorem (MCT). If $0 \leq g_1 \leq g_2 \leq \ldots$ are measurable and $g_n \to g$ pointwise, then $\int g_n\, d\mu \to \int g\, d\mu$ (possibly $= \infty$).
Dominated convergence theorem (DCT). If $g_n \to g$ μ-a.e. and there is an integrable $G \geq 0$ (independent of n) with $|g_n| \leq G$ a.e. for all n, then $g \in L^1$ and $\int |g_n - g|\, d\mu \to 0$; in particular

$\int g_n \to \int g$.

Solution. **(a)** Let $f \in L^2(X,\mu)$. The constant function 1 is in $L^2(X,\mu)$ since $\int_X 1^2\,d\mu = \mu(X) < \infty$. By Cauchy–Schwarz in L^2 (Theorem 1.7) applied to $|f|$ and 1,

$$\|f\|_{L^1} = \int_X |f| \cdot 1\,d\mu = \langle |f|, 1\rangle_{L^2} \le \|f\|_{L^2}\,\|1\|_{L^2} = \mu(X)^{1/2}\,\|f\|_{L^2} < \infty.$$

So $f \in L^1$. (Functions equal a.e. have the same L^1 and L^2 norms, so this is a statement about classes, and the inclusion map $L^2 \to L^1$ is a bounded linear map.)

Fourier coefficients. For $f \in L^2(\mathbb{T}, m)$ and $k \in \mathbb{Z}$, the function $z \mapsto f(z)\overline{z}^k$ is measurable and $|f(z)\overline{z}^k| = |f(z)|$ since $|z| = 1$. As $m(\mathbb{T}) = 1$, $\int_{\mathbb{T}} |f\overline{z}^k|\,dm = \|f\|_{L^1} \le \|f\|_{L^2} < \infty$, so

$$\hat{f}(k) = \int_{\mathbb{T}} f(z)\overline{z}^k\,dm(z) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(e^{ix})e^{-ikx}dx$$

is the integral of an integrable function, hence a well defined complex number, with $|\hat{f}(k)| \le \|f\|_{L^1} \le \|f\|_{L^2}$. It does not depend on the representative of f (changing f on a null set does not change the integral). Note also $\hat{f}(k) = \langle f, z^k\rangle_{L^2}$, which is finite by Cauchy–Schwarz directly. (Since $m(\mathbb{T}) = 1$ the coefficients make sense even for $f \in L^1(\mathbb{T}, m)$, a larger space by (b).)

(b) $f_\alpha \ge 0$ is even, so $\int_{-1}^1 f_\alpha = 2\int_0^1 x^{-\alpha}dx$. By MCT, $\int_0^1 x^{-\alpha}dx = \lim_{\varepsilon\downarrow 0}\int_\varepsilon^1 x^{-\alpha}dx$, and

$$\int_\varepsilon^1 x^{-\alpha}dx = \begin{cases} \dfrac{1-\varepsilon^{1-\alpha}}{1-\alpha} \xrightarrow{\varepsilon\to 0} \dfrac{1}{1-\alpha}, & \alpha < 1, \\ -\log\varepsilon \xrightarrow{\varepsilon\to 0} \infty, & \alpha = 1, \\ \dfrac{\varepsilon^{1-\alpha}-1}{\alpha-1} \xrightarrow{\varepsilon\to 0} \infty, & \alpha > 1. \end{cases}$$

Hence $f_\alpha \in L^1(-1,1)$ if and only if $0 \le \alpha < 1$, and then $\|f_\alpha\|_{L^1} = \frac{2}{1-\alpha}$.

Comment. Since $f_\alpha^2 = f_{2\alpha}$, we get $f_\alpha \in L^2(-1,1) \iff \alpha < \frac{1}{2}$. So for $\frac{1}{2} \le \alpha < 1$, $f_\alpha \in L^1 \setminus L^2$: the inclusion in (a) is strict. On spaces of infinite measure there is no inclusion at all, e.g. $\frac{1}{1+|x|} \in L^2(\mathbb{R}) \setminus L^1(\mathbb{R})$.

(c) f_M is measurable ($\{|f| \le M\}$ is a measurable set) and

$$|f - f_M|^2 = |f|^2\,\chi_{\{|f|>M\}}.$$

Pointwise convergence. Since $\int |f|^2 d\mu < \infty$, $|f(x)| < \infty$ for a.e. x. For such x, as soon as $M \ge |f(x)|$ we have $\chi_{\{|f|>M\}}(x) = 0$; hence $|f - f_M|^2(x) \to 0$ as $M \to \infty$, for a.e. x.

Domination. $|f - f_M|^2 \le |f|^2$, and $|f|^2 \in L^1$ because $f \in L^2$.

By DCT (along any sequence $M_n \to \infty$), $\|f - f_M\|_{L^2}^2 = \int_X |f - f_M|^2\,d\mu \to 0$. Since $M \mapsto \|f - f_M\|_{L^2}$ is nonincreasing, convergence along sequences implies convergence as $M \to \infty$ through all reals. (Finiteness of μ is not needed here.)

Meaning. Bounded functions are dense in L^2: every f is approximated by its truncations f_M, $|f_M| \le M$.

(d) *Pointwise convergence.* Fix $x \in (0,1)$. For $n > \frac{1}{x}$ we have $x \notin (0, \frac{1}{n})$, so $f_n(x) = 0$. Hence $f_n(x) \to 0$ for *every* $x \in (0,1)$, in particular almost everywhere.

No L^2 convergence. $\|f_n\|_{L^2}^2 = \int_0^{1/n}(\sqrt{n})^2 dx = n \cdot \frac{1}{n} = 1$, so $\|f_n - 0\|_{L^2} = 1 \not\to 0$. Moreover (f_n) has no L^2-limit at all: if $f_n \to g$ in L^2, a subsequence would converge to g a.e. (a standard fact), so $g = 0$ a.e., contradicting $\|f_n\| = 1$.

Why DCT fails. To apply DCT to $|f_n - 0|^2 = n\chi_{(0,1/n)}$ we would need an integrable G with $n\chi_{(0,1/n)} \le G$ for all n. The smallest candidate is $G = \sup_n n\chi_{(0,1/n)}$. For $x \in [\frac{1}{n+1}, \frac{1}{n})$ we have $x \in (0, \frac{1}{k})$ exactly for $k \le n$, so $G(x) = n > \frac{1}{x} - 1$. Thus

$$\int_0^1 G\,dx \ge \int_0^1 \Big(\frac{1}{x} - 1\Big)dx = \infty,$$

and no integrable dominating function exists. The mass $\int f_n^2 = 1$ does not disappear; it concentrates in the shrinking interval $(0, \frac{1}{n})$ (it "escapes to a point"), which pointwise convergence cannot see.

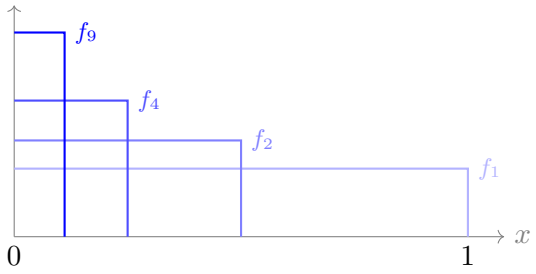


Figure 15: $f_n = \sqrt{n}\,\chi_{(0,1/n)}$: higher and narrower bumps, each with $\int f_n^2 = 1$.

Problem 3. Let $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$.

(a) Show that $\{z^k : k \in \mathbb{Z}\}$ is an ONS in $L^2(\mathbb{T}, m)$, where m is the normalized arc length measure defined above.

(b) Let $I \subset \mathbb{T}$ be a closed arc of positive length strictly less than 2π, i.e. $I = \{e^{it} : x_0 \le t \le x_1\}$, $0 < x_1 - x_0 < 2\pi$. Construct continuous functions φ_ε on $\mathbb{T}$ such that $0 \le \varphi_\varepsilon \le 1$ and $\varphi_\varepsilon \to \chi_I$ in $L^2(\mathbb{T}, m)$ $(\varepsilon \to 0^+)$. *Hint.* We only need to modify χ_I near the endpoints of I.

(c) (optional) Prove that for every measurable set $A \subseteq \mathbb{T}$ and every $\varepsilon > 0$ there is a finite union of arcs $U \subseteq \mathbb{T}$ such that $m(A \triangle U) < \varepsilon$. In particular, $\|\chi_U - \chi_A\|_{L^2}^2 < \varepsilon$.

(d) Deduce that $C(\mathbb{T})$ is dense in $L^2(\mathbb{T}, m)$.

(e) Let $\mathcal{P} = \left\{\sum_{k=-N}^{N} a_k z^k : N \in \mathbb{N},\ a_k \in \mathbb{C}\right\}$. Apply the complex Stone–Weierstrass theorem to show that $\mathcal{P}$ is uniformly dense in $C(\mathbb{T})$. Conclude that $\{z^k : k \in \mathbb{Z}\}$ is an ONB of $L^2(\mathbb{T}, m)$.

This problem is a detailed version of Exercise 2.1, where the two density facts were only quoted.

Solution. **(a)** For $k, j \in \mathbb{Z}$, using the parametrization $z = e^{ix}$,

$$\langle z^k, z^j \rangle = \int_{\mathbb{T}} z^k \bar{z}^j \, dm = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{i(k-j)x} dx = \begin{cases} 1, & k = j, \\ \dfrac{1}{2\pi}\left[\dfrac{e^{i(k-j)x}}{i(k-j)}\right]_{-\pi}^{\pi} = 0, & k \neq j, \end{cases}$$

since $e^{i(k-j)x}$ is 2π-periodic. Each z^k is bounded, hence in L^2. So $\{z^k\}$ is an ONS.

(b) Write $\ell = x_1 - x_0 \in (0, 2\pi)$ and let $0 < \varepsilon < \frac{2\pi - \ell}{2}$, so that the two "ramps" added below do not overlap on the other side of the circle. Define a function ψ_ε on $[x_0 - \varepsilon,\ x_0 - \varepsilon + 2\pi)$ (one full turn) by

$$\psi_\varepsilon(t) = \begin{cases} \dfrac{t - (x_0 - \varepsilon)}{\varepsilon}, & x_0 - \varepsilon \le t \le x_0, \\ 1, & x_0 \le t \le x_1, \\ \dfrac{(x_1 + \varepsilon) - t}{\varepsilon}, & x_1 \le t \le x_1 + \varepsilon, \\ 0, & x_1 + \varepsilon \le t < x_0 - \varepsilon + 2\pi, \end{cases}$$

and set $\varphi_\varepsilon(e^{it}) = \psi_\varepsilon(t)$. The pieces agree at the break points, and ψ_ε tends to 0 at both ends of the period interval, so φ_ε is a well-defined continuous function on $\mathbb{T}$ with $0 \le \varphi_\varepsilon \le 1$ and $\varphi_\varepsilon = \chi_I$

outside the two arcs $J_\varepsilon^- = \{e^{it} : x_0 - \varepsilon < t < x_0\}$ and $J_\varepsilon^+ = \{e^{it} : x_1 < t < x_1 + \varepsilon\}$. On these arcs $|\varphi_\varepsilon - \chi_I| \le 1$. Hence

$$\|\varphi_\varepsilon - \chi_I\|_{L^2}^2 = \int_{J_\varepsilon^- \cup J_\varepsilon^+} |\varphi_\varepsilon - \chi_I|^2 \, dm \le m(J_\varepsilon^-) + m(J_\varepsilon^+) = \frac{2\varepsilon}{2\pi} = \frac{\varepsilon}{\pi} \xrightarrow{\varepsilon \to 0^+} 0.$$

(The same construction works for open or half-open arcs, since endpoints have measure 0; for the full circle $\chi_{\mathbb{T}} = 1$ is already continuous.)

(c) We use the definition of Lebesgue measure through the outer measure: in the parametrization $t \in [-\pi, \pi)$,

$$m(A) = \inf\Big\{\sum_{j=1}^{\infty} m(J_j) : \ J_j \text{ open arcs}, \ A \subseteq \bigcup_j J_j\Big\} \qquad \text{for measurable } A \subseteq \mathbb{T}.$$

Let $\varepsilon > 0$. Choose open arcs $J_1, J_2, \dots$ covering A with $\sum_j m(J_j) < m(A) + \frac{\varepsilon}{2}$, and put $G = \bigcup_j J_j \supseteq A$. Then

$$m(G \setminus A) = m(G) - m(A) \le \sum_j m(J_j) - m(A) < \tfrac{\varepsilon}{2}.$$

As $\sum_j m(J_j) < \infty$, choose n with $\sum_{j>n} m(J_j) < \frac{\varepsilon}{2}$ and let $U = J_1 \cup \cdots \cup J_n$, a finite union of arcs. Since $A \triangle U = (A \setminus U) \cup (U \setminus A)$ is a disjoint union,

$$m(A \triangle U) = m(A \setminus U) + m(U \setminus A) \le m\Big(\bigcup_{j>n} J_j\Big) + m(G \setminus A) < \tfrac{\varepsilon}{2} + \tfrac{\varepsilon}{2} = \varepsilon,$$

because $A \setminus U \subseteq G \setminus U \subseteq \bigcup_{j>n} J_j$ and $U \setminus A \subseteq G \setminus A$. Finally, $|\chi_U - \chi_A| = \chi_{A \triangle U}$ (it equals 1 exactly at points lying in one of the sets but not the other), so

$$\|\chi_U - \chi_A\|_{L^2}^2 = \int_{\mathbb{T}} \chi_{A \triangle U} \, dm = m(A \triangle U) < \varepsilon.$$

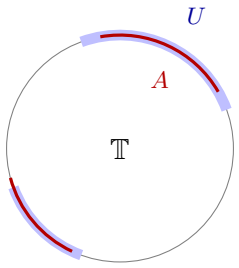


Figure 16: A measurable set A (here two arcs, but in general arbitrary) and a finite union of arcs U with small symmetric difference $A \triangle U$.

(d) Let $\overline{C}$ denote the closure of $C(\mathbb{T})$ in $L^2(\mathbb{T}, m)$; it is a closed linear subspace (closure of a subspace). We show $\overline{C} = L^2$.

Step 1: indicators of arcs. If I is an arc of length $< 2\pi$, then $\chi_I \in \overline{C}$ by (b) (open, closed or half-open arcs differ by at most two points, a null set). For $I = \mathbb{T}$, $\chi_{\mathbb{T}} = 1 \in C(\mathbb{T})$.

Step 2: indicators of finite unions of arcs. A finite union of arcs U can be written as a disjoint union of finitely many arcs $I_1, \dots, I_r$ (up to finitely many points): cut $\mathbb{T}$ at all endpoints and keep the pieces lying in U. Then $\chi_U = \sum_i \chi_{I_i}$ a.e., which lies in $\overline{C}$ by Step 1, as $\overline{C}$ is a linear subspace.

Step 3: indicators of measurable sets. By (c), χ_A is an L^2-limit of functions χ_U from Step 2, so $\chi_A \in \overline{C}$ since $\overline{C}$ is closed.

Step 4: simple functions. A simple function $s = \sum_{j=1}^{r} c_j \chi_{A_j}$ ($c_j \in \mathbb{C}$, A_j measurable) lies in $\overline{C}$ by linearity.

Step 5: all of L^2. Let $f \in L^2$. Writing $f = (u^+ - u^-) + i(v^+ - v^-)$ with $u = \operatorname{Re} f$, $v = \operatorname{Im} f$, each of the four non-negative parts is the pointwise limit of a nondecreasing sequence of simple functions (e.g. $s_n = \min(n, 2^{-n}\lfloor 2^n u^+ \rfloor)$). This gives simple functions $s_n \to f$ pointwise a.e. with $|s_n| \le |f|$ (the standard construction can be arranged so that each part is bounded by the corresponding part of f). Then $|f - s_n|^2 \le (2|f|)^2 = 4|f|^2 \in L^1$ and $|f - s_n|^2 \to 0$ a.e., so by DCT $\|f - s_n\|_{L^2} \to 0$. Hence $f \in \overline{C}$.

Therefore $C(\mathbb{T})$ is dense in $L^2(\mathbb{T}, m)$.

(e) *Stone–Weierstrass* (Theorem 2.2): $\mathbb{T}$ is a compact metric space, and $\mathcal{P} \subset C(\mathbb{T})$ satisfies:

- $\mathcal{P}$ is a subalgebra: it is clearly a linear subspace, and the product of two trigonometric polynomials is one, since $z^k z^j = z^{k+j}$;
- $\mathcal{P}$ contains the constants ($N = 0$);
- $\mathcal{P}$ separates points: $p(z) = z$ belongs to $\mathcal{P}$ and is injective;
- $\mathcal{P}$ is closed under complex conjugation: on $\mathbb{T}$ we have $\overline{z} = z^{-1}$, so $\overline{\sum_{k=-N}^{N} a_k z^k} = \sum_{k=-N}^{N} \overline{a_k} z^{-k} = \sum_{k=-N}^{N} \overline{a_{-k}} z^k \in \mathcal{P}$.

(The last condition is essential: the algebra of polynomials $\sum_{k\ge 0} a_k z^k$ satisfies the first three but is not dense in $C(\mathbb{T})$ – e.g. $\overline{z}$ is not a uniform limit of them, since $\int_{\mathbb{T}} p(z)\, z\, dm = 0$ for such p, while $\int_{\mathbb{T}} \overline{z}\, z\, dm = 1$.) Hence $\mathcal{P}$ is dense in $(C(\mathbb{T}), \|\cdot\|_\infty)$.

ONB. By (a), $\{z^k\}$ is an ONS. Let $f \in L^2(\mathbb{T}, m)$ and $\varepsilon > 0$. By (d) there is $g \in C(\mathbb{T})$ with $\|f - g\|_{L^2} < \varepsilon$, and by the above there is $p \in \mathcal{P}$ with $\|g - p\|_\infty < \varepsilon$. As $m(\mathbb{T}) = 1$, $\|g - p\|_{L^2} = \left(\int |g - p|^2 dm\right)^{1/2} \le \|g - p\|_\infty$, hence

$$\|f - p\|_{L^2} \le \|f - g\|_{L^2} + \|g - p\|_{L^2} < 2\varepsilon.$$

Since $p \in \operatorname{span}\{z^k : k \in \mathbb{Z}\}$, this shows $\operatorname{clos}\operatorname{span}\{z^k\} = L^2(\mathbb{T}, m)$, i.e. $\{z^k : k \in \mathbb{Z}\}$ is an ONB of $L^2(\mathbb{T}, m)$. (Alternatively, use Fejér's theorem instead of Stone–Weierstrass, Corollaries 2.18 and 2.19.)

7.2 List 2 (20. 10. 2026)

The problems of this sheet are numbered as on the sheet itself (continuing from List 1).

Problem 4. Consider

$$H = L^2([-\pi, \pi], dm), \qquad f(x) = x^2,$$

where m is as defined previously.

(a) Compute $\hat{f}(k)$ for every $k \in \mathbb{Z}$.

(b) Compute $\|f\|_{L^2}^2$, and apply Parseval's identity to establish $\displaystyle\sum_{k=1}^{\infty} \frac{1}{k^4} = \frac{\pi^4}{90}$.

(c) Prove that the Fourier series of f converges absolutely and uniformly on $[-\pi, \pi]$, and show that its sum equals f everywhere on this interval. *Hint.* Use Parseval's identity and the continuity of f.

Here $dm = \frac{dx}{2\pi}$ is the normalized measure, i.e. we identify $[-\pi, \pi)$ with $\mathbb{T}$ via $x \mapsto e^{ix}$ as in Section 2; so $\langle g, h\rangle = \frac{1}{2\pi}\int_{-\pi}^{\pi} g\,\overline{h}\,dx$, the ONB is $\{e^{ikx}\}_{k\in\mathbb{Z}}$ (Problem 3) and $\hat{g}(k) = \frac{1}{2\pi}\int_{-\pi}^{\pi} g(x)e^{-ikx}dx$. This is the same setting as in Example 1.31, where $f(x) = x$ gave $\sum 1/k^2$; one more power of x gives $\sum 1/k^4$.

Solution. **(a)** For $k=0$: $\hat f(0)=\frac{1}{2\pi}\int_{-\pi}^{\pi}x^2dx=\frac{1}{2\pi}\cdot\frac{2\pi^3}{3}=\frac{\pi^2}{3}$.

For $k\neq 0$: since x^2 is even and $\sin kx$ is odd, $\int_{-\pi}^{\pi}x^2e^{-ikx}dx=\int_{-\pi}^{\pi}x^2\cos kx\,dx$. Integrate by parts twice, using $\sin k\pi=0$ and $\cos(\pm k\pi)=(-1)^k$:

$$\int_{-\pi}^{\pi}x^2\cos kx\,dx=\Big[x^2\frac{\sin kx}{k}\Big]_{-\pi}^{\pi}-\frac{2}{k}\int_{-\pi}^{\pi}x\sin kx\,dx=-\frac{2}{k}\int_{-\pi}^{\pi}x\sin kx\,dx,$$

$$\int_{-\pi}^{\pi}x\sin kx\,dx=\Big[-x\frac{\cos kx}{k}\Big]_{-\pi}^{\pi}+\frac{1}{k}\int_{-\pi}^{\pi}\cos kx\,dx=-\frac{\pi(-1)^k+\pi(-1)^k}{k}+0=-\frac{2\pi(-1)^k}{k}.$$

Hence $\int_{-\pi}^{\pi}x^2\cos kx\,dx=\frac{4\pi(-1)^k}{k^2}$ and

$$\boxed{\hat f(0)=\frac{\pi^2}{3},\qquad \hat f(k)=\frac{2(-1)^k}{k^2}\quad(k\neq 0).}$$

(Check: $\hat f(-k)=\hat f(k)\in\mathbb{R}$, as it must be for a real even function.)

(b) $\|f\|_{L^2}^2=\frac{1}{2\pi}\int_{-\pi}^{\pi}x^4dx=\frac{1}{2\pi}\cdot\frac{2\pi^5}{5}=\frac{\pi^4}{5}$. By Parseval's identity (3) for the ONB $\{e^{ikx}\}_{k\in\mathbb{Z}}$ (enumerated as a sequence, as in Example 1.31),

$$\frac{\pi^4}{5}=\sum_{k\in\mathbb{Z}}|\hat f(k)|^2=\frac{\pi^4}{9}+\sum_{k\neq 0}\frac{4}{k^4}=\frac{\pi^4}{9}+8\sum_{k=1}^{\infty}\frac{1}{k^4}.$$

Therefore

$$\sum_{k=1}^{\infty}\frac{1}{k^4}=\frac{1}{8}\Big(\frac{\pi^4}{5}-\frac{\pi^4}{9}\Big)=\frac{1}{8}\cdot\frac{4\pi^4}{45}=\frac{\pi^4}{90}.$$

(c) The Fourier series of f is

$$\sum_{k\in\mathbb{Z}}\hat f(k)e^{ikx}=\frac{\pi^2}{3}+\sum_{k\neq 0}\frac{2(-1)^k}{k^2}e^{ikx}=\frac{\pi^2}{3}+4\sum_{k=1}^{\infty}\frac{(-1)^k}{k^2}\cos kx,$$

pairing the terms k and $-k$ ($e^{ikx}+e^{-ikx}=2\cos kx$).

Absolute and uniform convergence. For every x, $|\hat f(k)e^{ikx}|=|\hat f(k)|=M_k$, where $\sum_k M_k=\frac{\pi^2}{3}+4\sum_{k\geq 1}\frac{1}{k^2}=\frac{\pi^2}{3}+\frac{4\pi^2}{6}=\pi^2<\infty$ (Example 1.31). By the Weierstrass M-test (Theorem 7.3) the series converges absolutely and uniformly on $[-\pi,\pi]$ (in fact on $\mathbb{R}$), and its sum

$$g(x)=\sum_{k\in\mathbb{Z}}\hat f(k)e^{ikx}$$

does not depend on the order of summation (in particular $g=\lim_N S_N(f)$ uniformly) and is continuous, as a uniform limit of the continuous functions $S_N(f)$. It is also 2π-periodic, so $g(-\pi)=g(\pi)$.

The sum is f almost everywhere. Since $m([-\pi,\pi])=1$, $\|h\|_{L^2}\leq\sup|h|$, so $S_N(f)\to g$ also in L^2. On the other hand $S_N(f)\to f$ in L^2 by Riesz–Fischer (**). Limits in L^2 are unique, so $f=g$ in L^2, i.e. $\|f-g\|_{L^2}=0$. In the language of the hint: by continuity of the inner product, $\widehat{g}(k)=\lim_N\langle S_N(f),e^{ikx}\rangle=\hat f(k)$ for all k, so Parseval's identity applied to $f-g$ gives $\|f-g\|_{L^2}^2=\sum_k|\hat f(k)-\hat g(k)|^2=0$. Either way, $f=g$ almost everywhere.

Everywhere. Both f and g are continuous on $[-\pi,\pi]$. If $f(x_0)\neq g(x_0)$ for some x_0, then by continuity $|f-g|>\delta>0$ on some interval $J\ni x_0$ of positive length (relative to $[-\pi,\pi]$), and $\|f-g\|_{L^2}^2\geq\delta^2 m(J)>0$, a contradiction. Hence for *every* $x\in[-\pi,\pi]$

$$x^2=\frac{\pi^2}{3}+4\sum_{k=1}^{\infty}\frac{(-1)^k}{k^2}\cos kx.\tag{9}$$

This works at the endpoints too precisely because the periodic extension of x^2 is continuous: $f(-\pi)=f(\pi)$. Compare with $f(x)=x$ in Example 1.31, whose periodic extension jumps at $\pm\pi$, and whose Fourier series there converges to $0\neq\pm\pi$.

Background 3. (Weierstrass M-test)

Let $u_k\colon E\to\mathbb{C}$ be functions on a set E with $\sup_E|u_k|\le M_k$ and $\sum_k M_k<\infty$. Then $\sum_k u_k$ converges absolutely at every point and uniformly on E: the partial sums are uniformly Cauchy, since $\sup_E|\sum_{k=n+1}^{n'}u_k|\le\sum_{k>n}M_k\to 0$. If the u_k are continuous, so is the sum (a uniform limit of continuous functions is continuous). Absolute convergence also implies that every rearrangement has the same sum.

Remark 7.4. Evaluating (9) at special points gives more series for free:

- $x=\pi$: $\pi^2=\frac{\pi^2}{3}+4\sum_k\frac{1}{k^2}$, i.e. $\sum_{k\ge1}\frac{1}{k^2}=\frac{\pi^2}{6}$ again (now from pointwise convergence instead of Parseval);
- $x=0$: $0=\frac{\pi^2}{3}+4\sum_k\frac{(-1)^k}{k^2}$, i.e. $\sum_{k\ge1}\frac{(-1)^{k+1}}{k^2}=\frac{\pi^2}{12}$.

The general principle behind (c): *if $\sum_k|\hat f(k)|<\infty$ and f is continuous, then the Fourier series converges to f uniformly.* Smoothness of f (here: f is continuous and piecewise C^1 as a function on $\mathbb{T}$) gives decay of $\hat f(k)$ (here $\sim k^{-2}$), and decay gives good convergence. For $f(x)=x$ the coefficients only decay like k^{-1}, which is not summable, and uniform convergence fails.

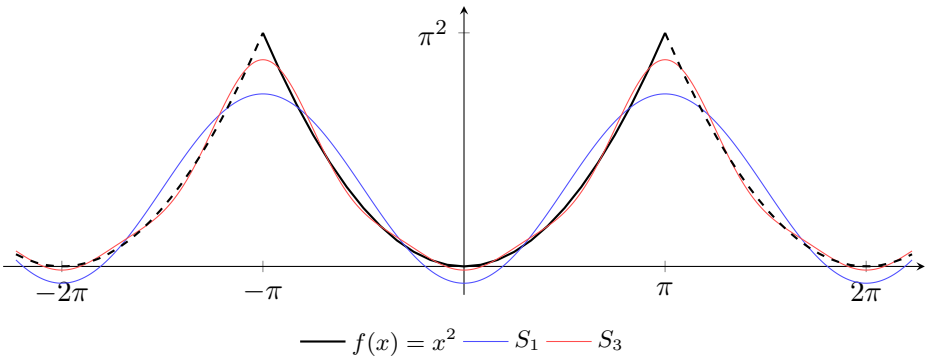


Figure 17: $f(x)=x^2$ on $[-\pi,\pi]$ (solid), its 2π-periodic extension (dashed) and the partial sums $S_N=\frac{\pi^2}{3}+4\sum_{k=1}^N\frac{(-1)^k}{k^2}\cos kx$. The extension is continuous, and already S_3 is uniformly close to f, in contrast with $f(x)=x$ (Example 1.31).

Problem 5. Let $f,g\in L^2(\mathbb{T},m)$. Prove that, for every $n\in\mathbb{Z}$,

$$\widehat{fg}(n)=\sum_{k\in\mathbb{Z}}\hat f(k)\hat g(n-k),$$

and that the series converges absolutely.

Solution. Step 0: everything is well defined. By Cauchy–Schwarz, $\int_{\mathbb{T}}|fg|\,dm\le\|f\|_{L^2}\|g\|_{L^2}<\infty$, so $fg\in L^1(\mathbb{T},m)$ and $\widehat{fg}(n)=\int_{\mathbb{T}}fg\,\overline{z}^n\,dm$ is defined (although fg need not be in L^2; cf. Problem 2(a)).

Step 1: absolute convergence. By Parseval, $\sum_k|\hat f(k)|^2=\|f\|^2<\infty$ and, substituting $j=n-k$ (a bijection of $\mathbb{Z}$), $\sum_k|\hat g(n-k)|^2=\sum_j|\hat g(j)|^2=\|g\|^2$. Cauchy–Schwarz in $\ell^2(\mathbb{Z})$ gives, for every finite $F\subset\mathbb{Z}$,

$$\sum_{k\in F}|\hat f(k)||\hat g(n-k)|\le\Big(\sum_{k\in F}|\hat f(k)|^2\Big)^{1/2}\Big(\sum_{k\in F}|\hat g(n-k)|^2\Big)^{1/2}\le\|f\|_{L^2}\|g\|_{L^2}.$$

So $\sum_{k\in\mathbb{Z}}|\hat f(k)\hat g(n-k)|\le\|f\|\,\|g\|<\infty$: the series converges absolutely, hence its sum does not depend on the order of summation, and it equals $\lim_{N\to\infty}\sum_{|k|\le N}\hat f(k)\hat g(n-k)$.

Step 2: write $\widehat{fg}(n)$ as an inner product. Put $h(z) = \overline{g(z)}z^n$. Since $|z| = 1$, $|h| = |g|$, so $h \in L^2(\mathbb{T}, m)$, and

$$\widehat{fg}(n) = \int_{\mathbb{T}} f\, g\, \overline{z}^n \, dm = \int_{\mathbb{T}} f\, \overline{\overline{g}\, z^n}\, dm = \langle f, h\rangle.$$

The Fourier coefficients of h are, using $\overline{z} = z^{-1}$ on $\mathbb{T}$,

$$\hat{h}(k) = \int_{\mathbb{T}} \overline{g}\, z^n \overline{z}^k \, dm = \overline{\int_{\mathbb{T}} g\, \overline{z}^n z^k \, dm} = \overline{\int_{\mathbb{T}} g\, \overline{z}^{\,n-k}\, dm} = \overline{\hat{g}(n-k)}.$$

Step 3: Parseval for inner products. For $u, v \in L^2(\mathbb{T}, m)$,

$$\langle u, v\rangle = \sum_{k\in\mathbb{Z}} \hat{u}(k)\overline{\hat{v}(k)}. \tag{10}$$

Indeed, by Riesz–Fischer $S_N(u) \to u$ and $S_N(v) \to v$ in L^2, so by continuity of the inner product (Lemma 1.28) and orthonormality of $\{z^k\}$,

$$\langle u, v\rangle = \lim_{N\to\infty} \langle S_N(u), S_N(v)\rangle = \lim_{N\to\infty} \sum_{|k|,|j|\le N} \hat{u}(k)\overline{\hat{v}(j)}\langle z^k, z^j\rangle = \lim_{N\to\infty} \sum_{|k|\le N} \hat{u}(k)\overline{\hat{v}(k)}.$$

(Alternatively, (10) follows from (3) by polarization, $\langle u, v\rangle = \frac{1}{4}\sum_{r=0}^{3} i^r \, \|u + i^r v\|^2$.)

Step 4: conclusion. Applying (10) with $u = f$, $v = h$:

$$\widehat{fg}(n) = \langle f, h\rangle = \sum_{k\in\mathbb{Z}} \hat{f}(k)\,\overline{\hat{h}(k)} = \sum_{k\in\mathbb{Z}} \hat{f}(k)\hat{g}(n-k),$$

the series being absolutely convergent by Step 1.

Remark 7.5. The right-hand side is the *convolution* of the sequences $\hat{f}, \hat{g} \in \ell^2(\mathbb{Z})$: $(\hat{f} * \hat{g})(n) = \sum_k \hat{f}(k)\hat{g}(n-k)$. So the problem says $\widehat{fg} = \hat{f} * \hat{g}$: *the Fourier transform turns products into convolutions.* The dual statement, $\widehat{f * g} = \hat{f}\,\hat{g}$ for the convolution of functions on $\mathbb{T}$, is the one that appeared with the Fejér kernel ($\sigma_N(f) = f * F_N$). For trigonometric polynomials the identity is just the rule for multiplying polynomials: the coefficient of z^n in $\big(\sum_k a_k z^k\big)\big(\sum_j b_j z^j\big)$ is $\sum_{k+j=n} a_k b_j$. Step 1 also shows $|\widehat{fg}(n)| \le \|f\|_{L^2}\,\|g\|_{L^2}$, consistent with $|\widehat{fg}(n)| \le \|fg\|_{L^1}$.

Problem 6. Let $BV([a,b])$ denote the set of $\mathbb{C}$-valued functions of bounded variation on $[a,b]$, where

$$\mathrm{Var}_{[a,b]}(f) := \sup_{a=t_0<\cdots<t_N=b} \sum_{j=0}^{N-1} |f(t_{j+1}) - f(t_j)|.$$

(a) Verify that $BV([a,b])$ is a vector space.

(b) Show that every $f \in BV([a,b])$ is bounded. *Hint.* Fix $x \in [a,b]$ and estimate $|f(x) - f(a)|$.

(c) Prove that if $f, g \in BV([a,b])$, then $fg \in BV([a,b])$. *Hint.* Use

$$f(t_{j+1})g(t_{j+1}) - f(t_j)g(t_j) = f(t_{j+1})(g(t_{j+1}) - g(t_j)) + g(t_j)(f(t_{j+1}) - f(t_j)).$$

(d) Let $f \in BV([a,b])$ be real-valued. Define

$$V(x) := \mathrm{Var}_{[a,x]}(f), \qquad x \in [a,b].$$

Show that both V and $V - f$ are nondecreasing on $[a,b]$. Deduce the Jordan decomposition theorem mentioned in the lectures.

In the lectures the same quantity was written $V_a^b(f)$; we write $\mathrm{Var}_{[a,b]}(f)$ as on the sheet. For a partition $P = \{a = t_0 < \cdots < t_N = b\}$ write $S(f,P) = \sum_{j=0}^{N-1} |f(t_{j+1}) - f(t_j)|$, so $\mathrm{Var}_{[a,b]}(f) = \sup_P S(f,P)$ and $f \in BV([a,b])$ means $\mathrm{Var}_{[a,b]}(f) < \infty$. We set $\mathrm{Var}_{[a,a]}(f) = 0$.

Solution. **(a)** $BV([a,b])$ is a subset of the vector space of all functions $[a,b]\to\mathbb{C}$; we check that it is a linear subspace. The zero function has $S(0,P)=0$ for all P, so $0\in BV$. Let $f,g\in BV$, $\alpha,\beta\in\mathbb{C}$, and P any partition. By the triangle inequality,

$$\begin{aligned}S(\alpha f+\beta g,P)&=\sum_j|\alpha(f(t_{j+1})-f(t_j))+\beta(g(t_{j+1})-g(t_j))|\\&\le|\alpha|S(f,P)+|\beta|S(g,P)\le|\alpha|\operatorname{Var}(f)+|\beta|\operatorname{Var}(g).\end{aligned}$$

Taking the supremum over P: $\operatorname{Var}(\alpha f+\beta g)\le|\alpha|\operatorname{Var}(f)+|\beta|\operatorname{Var}(g)<\infty$, so $\alpha f+\beta g\in BV$. (In particular Var is a seminorm on BV; it vanishes exactly on constants, and $\|f\|_{BV}=|f(a)|+\operatorname{Var}_{[a,b]}(f)$ is a norm.)

(b) Let $x\in(a,b)$. Then $a<x<b$ is a partition, so

$$|f(x)-f(a)|\le|f(x)-f(a)|+|f(b)-f(x)|=S(f,\{a,x,b\})\le\operatorname{Var}_{[a,b]}(f).$$

For $x=b$ use the partition $a<b$, and for $x=a$ the estimate is trivial. Hence for all $x\in[a,b]$

$$|f(x)|\le|f(a)|+|f(x)-f(a)|\le|f(a)|+\operatorname{Var}_{[a,b]}(f)<\infty,$$

so $\|f\|_\infty:=\sup_{[a,b]}|f|\le|f(a)|+\operatorname{Var}_{[a,b]}(f)$.

(c) Let P be a partition. By the identity in the hint and the triangle inequality,

$$\begin{aligned}S(fg,P)&\le\sum_j|f(t_{j+1})|\,|g(t_{j+1})-g(t_j)|+\sum_j|g(t_j)|\,|f(t_{j+1})-f(t_j)|\\&\le\|f\|_\infty S(g,P)+\|g\|_\infty S(f,P).\end{aligned}$$

Taking the supremum over P and using (b) (which guarantees $\|f\|_\infty,\|g\|_\infty<\infty$),

$$\operatorname{Var}(fg)\le\|f\|_\infty\operatorname{Var}(g)+\|g\|_\infty\operatorname{Var}(f)<\infty.$$

So $BV([a,b])$ is not only a vector space but an *algebra*. (The boundedness in (b) is essential here: this is the only place where it is used.)

(d) *Key inequality.* For $a\le x<y\le b$:

$$V(y)\ge V(x)+|f(y)-f(x)|.\tag{11}$$

Proof: if $x=a$, then $V(a)=0$ and $\{a<y\}$ is a partition of $[a,y]$, so $V(y)\ge|f(y)-f(a)|$. If $x>a$, let $P=\{a=t_0<\cdots<t_N=x\}$ be any partition of $[a,x]$. Then $P'=P\cup\{y\}$ is a partition of $[a,y]$ and

$$S(f,P)+|f(y)-f(x)|=S(f,P')\le V(y).$$

Taking the supremum over P gives (11). In particular (with $y=b$) $V(x)\le V(b)=\operatorname{Var}_{[a,b]}(f)<\infty$, so V is a finite, real-valued function.

V is nondecreasing: by (11), $V(y)-V(x)\ge|f(y)-f(x)|\ge0$ for $x<y$.

$V-f$ is nondecreasing: since f is real, $|f(y)-f(x)|\ge f(y)-f(x)$, so by (11)

$$\big(V(y)-f(y)\big)-\big(V(x)-f(x)\big)=\big(V(y)-V(x)\big)-\big(f(y)-f(x)\big)\ge|f(y)-f(x)|-\big(f(y)-f(x)\big)\ge0.$$

(Similarly $V+f$ is nondecreasing.)

Jordan decomposition (Theorem 2.30): $g\in BV([a,b])$ if and only if $g=g_1-g_2+i(g_3-g_4)$ with $g_1,\dots,g_4\colon[a,b]\to\mathbb{R}$ bounded and nondecreasing.

"⇒": Let $g\in BV([a,b])$ and $u=\operatorname{Re}g$, $v=\operatorname{Im}g$. Since $|\operatorname{Re}w|\le|w|$, $S(u,P)\le S(g,P)$ for every P, so $u\in BV$, and likewise $v\in BV$. By the above, with $V_u(x)=\operatorname{Var}_{[a,x]}(u)$,

$$u=V_u-(V_u-u),$$

a difference of two nondecreasing functions; they are bounded, since a nondecreasing h on $[a,b]$ satisfies $h(a) \le h \le h(b)$. Do the same for v and set $g_1 = V_u$, $g_2 = V_u - u$, $g_3 = V_v$, $g_4 = V_v - v$.

"$\Leftarrow$": A nondecreasing h is in BV: for every partition the sum telescopes, $S(h,P) = \sum_j (h(t_{j+1}) - h(t_j)) = h(b) - h(a)$, so $\mathrm{Var}(h) = h(b) - h(a) < \infty$. By (a), the linear combination $g_1 - g_2 + ig_3 - ig_4$ is in BV.

Remark 7.6. (a) The decomposition is far from unique (add the same nondecreasing function to g_1 and g_2), but V is in a sense optimal: one can show that $\mathrm{Var}_{[x,y]}(f) = V(y) - V(x)$ (additivity of the variation), so for real f the two nondecreasing functions $\frac{V+f}{2}$ and $\frac{V-f}{2}$ are the "positive and negative variation" of f.

(b) Using (11) instead of additivity makes the proof shorter than the one in the lectures, which used $f(y) - f(x) \le V_x^y(f) = V(y) - V(x)$.

(c) Consequences: by the Jordan decomposition and the fact that monotone functions have at most countably many jumps and are differentiable a.e. (Lebesgue), the same holds for BV functions. This is the input for Theorem 2.32 and Hardy's theorem (Theorem 2.36).

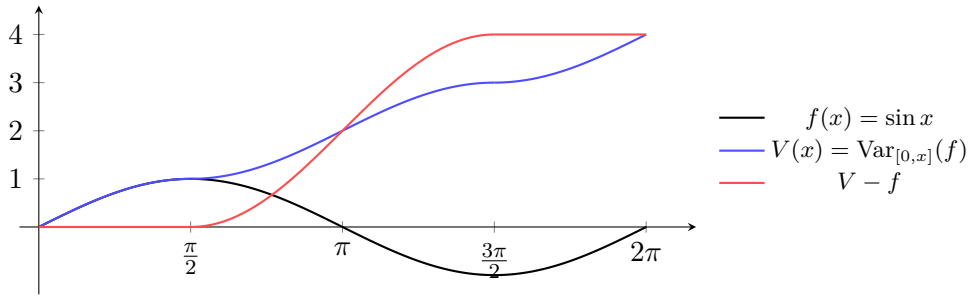


Figure 18: Jordan decomposition of $f(x) = \sin x$ on $[0, 2\pi]$: V grows wherever f moves (up or down), $V - f$ grows only where f decreases, and $f = V - (V - f)$. Here $\mathrm{Var}_{[0,2\pi]}(\sin) = 4$.